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Taxing Income Splitting Within Business-Owning Families*

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Abstract

Business owners can reduce their family tax liability by paying dividends to lower-income spouses. Governments have long used tax rules to restrict this practice, but little is known about how families respond to these policies. Using linked personal, corporate, and payroll tax records, I study Canada's 2018 expansion of the Tax on Split Income rules. Relative to controls, treated spouses report 53% less dividend income. Higher owner dividends offset 57% of the dollar decline in spouse dividends, while the increase in owner employment income offsets the remainder. The reform therefore primarily reallocates reported income within families. Tax liability rises proportionally more among moderate-income families than among the highest-income families, suggesting that income splitting played a relatively larger role in reducing their tax burden. Among treated families, the elasticity of spouse dividends with respect to the pre-reform tax advantage of splitting income is similar across income groups. With access to the same income-shifting channel, families respond similarly to tax incentives. This finding suggests that differences in access to tax planning may help explain the larger taxable-income elasticities often estimated for higher-income taxpayers.

JEL classification: D13, G35, H24, H25, H26

Keywords: dividend taxation, income shifting, tax avoidance, tax-planning strategies, family businesses, intrahousehold income allocation

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1 Introduction

Business owners have strong incentives to allocate income to family members to minimize their tax liabilities.[1] Governments have therefore adopted a variety of policies to limit tax-motivated income splitting within families. Joint taxation, rules that attribute income to the person who generated it, and higher tax rates on income allocated to family members all seek to reduce the tax advantage from within-family income splitting.[2] Despite the longstanding use of this class of policies and extensive concern about family income splitting, there is little empirical evidence on whether these policies reduce income splitting or how business-owning families adjust when they are introduced. Because these policies alter the tax consequences of allocating income within the family, understanding how families respond is necessary to assess both their equity and efficiency effects.

I address this gap by studying Canada's 2018 expansion of the Tax on Split Income (TOSI) rules. Canada provides a useful setting because linked longitudinal personal, corporate, and payroll tax records from the Canadian Employer-Employee Dynamics Database (CEEDD) allow me to follow both spouses and their family business over time. Canada's progressive personal income tax is levied at the individual rather than family level. Before the reform, business owners could reduce their family's tax liability by issuing dividends to lower-income family members.[3] The federal government announced the proposed expansion of the TOSI rules in July 2017, and the new rules took effect on January 1, 2018. Dividends received from a related private corporation became subject to the top marginal tax rate unless the recipient qualified for an exemption based on the age of the business owner, ownership of the corporation, participation in the business, or a reasonable return on contributions of labour, capital, or risk. These exemptions provide a basis for comparing families affected by the reform with those that are not.

My first strategy estimates a difference-in-differences model comparing families classified as affected by the TOSI reform with families classified as unaffected. I use spouses' common-share ownership and employment income to approximate the 10% ownership and 20-hour weekly participation exemptions, and restrict primary owners to those under age sixty-five. Treated spouses remain below the observable ownership and employment thresholds, while controls satisfy at least

[1] Evidence from the United Kingdom shows that the introduction of individual taxation led couples to shift investment income toward the lower-taxed spouse (Stephens & Ward-Batts, 2004). In Canada, Wolfson and Legree (2015) estimate that income splitting through private corporations reduced tax revenues by approximately $500 million in 2011.

[2] Examples include joint taxation of spouses in the United States and Germany, which generally limits the personal-income-tax benefit of reallocating income between spouses but does not extend to other family members; the clubbing and attribution rules in India and South Africa, which assign certain income transferred to family members back to the original taxpayer; the personal-services-income rules in Australia and New Zealand, which attribute qualifying income to the person whose services generated it; and Sweden's 3:12 rules, which restrict preferential taxation of dividends from closely held corporations. The United Kingdom's settlements rules attribute some transferred income to the transferor but permit dividends paid through genuine ownership interests in family businesses.

[3] Canadian corporations can pay different dividends to different classes of shares, allowing payments to be directed toward particular shareholders.

one exemption. The main sample is a balanced panel of couples who fully own their linked corporation, with the spouse holding shares in no other business. Spouse dividends initially rise by approximately 10.5% relative to controls in the 2017 announcement year, consistent with families bringing payments forward before the new rules took effect. I find that, relative to controls, treated spouses report 52.9% less dividend income after implementation on the intensive margin, measured among spouses reporting positive dividends. On the extensive margin, their probability of reporting any dividends falls by 16.8 percentage points. Both margins respond, consistent with evidence that dividend taxes affect payout amounts and the decision to pay dividends (Alstadsæter & Fjærli, 2009; Bilicka et al., 2025; Chetty & Saez, 2005). Primary owners report more dividends and total employment income, while wages paid to spouses by the family corporation fall. Much of the reduction in spouse income is therefore accompanied by higher income reported by the owner, although total couple dividends also decline. Family taxable income falls by only 1.2%, and family tax liability rises by 3.3%, with both log estimates significant at the 10% level and the dollar estimates imprecise. These much smaller family-level changes show why the decline in spouse dividends alone overstates the contraction of the family's personal tax base. The dividend response is larger among professional corporations, but is also present among non-professional corporations and families with shares in multiple businesses. A separate analysis of a broader sample that does not condition on exemption status shows little systematic movement in spouses' ownership or employment toward the observable exemption thresholds. Families primarily adjusted income payments, with little evidence of systematic changes in spouses' ownership or employment status.

The treatment–control comparison establishes the average response, but affected families differed in the tax advantage they previously received from income splitting. My second strategy restricts the sample to treated families and compares changes according to this pre-reform advantage. Using 2016 income and tax parameters, I define the shifting incentive as the log-difference between the spouse's and owner's net-of-dividend-tax rates. I calculate both rates after reallocating the spouse's dividends to the owner and hold this measure fixed. This captures the relative after-tax return to allocating dividends to the spouse rather than the owner, with a higher value indicating a stronger incentive to split income. The estimated elasticity is approximately negative 2.6, meaning that a 1% higher spouse-to-owner net-of-tax-rate ratio is associated with a roughly 2.6% larger post-reform decline in spouse dividends.[4] The probability of reporting any dividends also falls more among spouses with a larger pre-reform tax advantage from income splitting, so the adjustment includes whether families continue paying the spouse as well as the amount they pay. Owner dividends rise with the incentive, while couple dividends and family taxable income fall and log tax liability rises. The pattern complements evidence that business owners shift income more when the tax incentive is stronger (Alstadsæter & Jacob, 2016; Harju & Matikka, 2016a). The estimated spouse-dividend elasticities are similar across family-income quartiles, although the lowest-quartile estimate

[4]Small changes in shifting incentives can lead to large changes in spouse dividends because even a small tax-rate gap can make it worthwhile to pay substantial dividends before the spouses' marginal tax rates converge.

is less precise. Families at different income levels therefore show similar proportional responses to a comparable pre-reform incentive to split income. This finding offers a useful perspective on the larger taxable-income elasticities often estimated for high-income taxpayers (Gruber & Saez, 2002). One explanation for that pattern is that higher-income taxpayers have more opportunities to adjust how their income is reported (Saez et al., 2012; Slemrod & Kopczuk, 2002). Here, families throughout the income distribution have access to the same dividend-allocation channel, making their opportunities to respond more comparable. When this channel is available, I find similar responsiveness among moderate- and higher-income families. Although the elasticity concerns spouse dividends rather than total taxable income, the result supports the view that differences in access to tax-planning opportunities help explain differences in measured tax responsiveness across the income distribution. I also use this strategy to examine corporate outcomes and find no statistically detectable response in assets or retained earnings. Other settings exhibit profit retention following dividend or personal tax changes (Alstadsæter et al., 2014; Miller et al., 2024), while Yagan (2015) finds no investment or employee-compensation response to the US dividend tax cut. These measures do not establish where the income no longer reported by families ultimately goes.

The tax-rate advantage does not capture the amount of income allocated to the spouse. Two families with similar tax-rate advantages can therefore face different costs of maintaining their dividend allocations. My third strategy measures these financial stakes using the additional tax a family would owe if it maintained its observed 2016 income allocation under the expanded rules, divided by its total income in 2016. This calculation does not reassign dividends between spouses. The analysis retains treated and control families, with the penalty equal to zero for controls and before implementation. I use this predetermined relative penalty as an instrument for spouse dividend income and, separately, dividend receipt. Under the IV assumptions, the second stages link reform-induced reductions in these payments to changes in other family income. The receipt-IV estimates point to higher owner dividends, lower family taxable income, and higher tax liability. For an additional potential tax penalty equal to 1% of pre-reform family income, the estimates imply approximately 0.6% lower family taxable income and 1.5% higher family tax liability. Across income groups, the implied proportional tax-liability increase for a common relative penalty is larger in the second and third income quartiles than in the fourth. This highlights the importance of income splitting for the tax bills of moderate-income business-owning families, even though the largest dollar reductions in spouse dividends occur at higher incomes. The log-dividend IV also points to higher tax liability, although its average taxable-income estimate is imprecise. The tax-liability response includes higher tax on spouse dividends that continue to be paid as well as tax associated with changes in reported income. Comparisons with equal dollar penalties suggest that the stronger dividend response at higher incomes partly reflects the larger dollar amounts represented by a penalty equal to a given percentage of family income. Ranking treated families by their simulated dollar penalty shows that spouse-dividend declines are concentrated in the third and fourth penalty quartiles. The highest-penalty quartile also shows the clearest combination of

lower family taxable income and higher tax liability.

My research makes several contributions. To my knowledge, I provide the first causal evidence on how a rule targeting dividend-based income splitting in private businesses changes income reported by both the business owner and their spouse. Canadian studies document income splitting among business-owning couples (Schuetze, 2006; Wolfson & Legree, 2015), while Bauer et al. (2015) examine restrictions on dividends paid to minor children. Research on the taxation of couples shows changes in the allocation of assets and non-wage income (LaLumia, 2008; Stephens & Ward-Batts, 2004). Buettner et al. (2019) further show that couples' tax-planning choices depend on how those choices affect the distribution of net income between spouses. Creedy and Gemmell (2020) explain why joint decisions can alter individual taxable-income elasticities and revenue responses, and Alinaghi et al. (2023) find larger bunching-based elasticities among partnered individuals in New Zealand, particularly self-employed couples. I identify the response to a restriction on a specific dividend-allocation channel and follow both spouses and their corporation. This adds a family dimension to evidence on income shifting across corporate and personal tax bases, between income forms, through changes in business legal form, and across time (Gordon & Slemrod, 1998; Gorry et al., 2021; Harju & Matikka, 2016b; Le Maire & Schjerning, 2013; Pirttilä & Selin, 2011; Romanov, 2006; Sivadasan & Slemrod, 2008; Slemrod, 1995; Tazhitdinova, 2020). It also complements research on closing passive-loss shelters, offshore banking channels, and dividend-withholding loopholes (Casi et al., 2026; Johannesen & Zucman, 2014; Samwick, 1996). By measuring changes in the recipient, form, total amount, and taxation of family income, I show how restricting the targeted channel affects income that remains in the personal tax base. As Saez et al. (2012) emphasize, the revenue implications of a reported-income response depend on whether income is taxed elsewhere and at what rate. The family-level evidence here captures an important part of this adjustment by following income reported by both spouses and changes in their combined tax liability.

Second, I contribute to the literature on responses to dividend taxation by distinguishing the amount of income allocated through a tax-planning channel from responsiveness to a comparable incentive. Dividend-tax studies document changes in payouts and the composition of owner income (Alstadsæter & Fjærli, 2009; Bilicka et al., 2025; Chetty & Saez, 2005; Harju & Matikka, 2016a). The similar treatment-intensity elasticities across family-income quartiles provide a useful comparison with the larger taxable-income elasticities often estimated for higher-income taxpayers (Gruber & Saez, 2002). Taxable-income elasticities reflect the tax environment and the opportunities it provides for adjustment as well as preferences (Kopczuk, 2005; Saez et al., 2012; Slemrod & Kopczuk, 2002). Consistent with this view, Alstadsæter and Jacob (2017) show that participation in tax avoidance depends on incentives, access, and awareness, while Adam et al. (2021) document bunching among UK owner-managers that is generally absent among employees. Participation in tax planning and responsiveness once a channel is available are distinct. Larger potential savings can encourage greater use of a channel without implying a larger proportional response among families already able to use it. My sample compares families with access to the same dividend-allocation

channel and finds similar proportional responses to its tax advantage across income groups. The estimated elasticity concerns spouse dividends rather than total taxable income, but the result offers evidence that differences in opportunities to alter reported income can help explain differences in measured tax responsiveness. The relative- and dollar-penalty comparisons further show why an income gradient in responses to financial exposure does not by itself establish greater sensitivity to an equal dollar tax cost.

Third, I contribute to research on how tax planning and its consequences vary across the income distribution. Prior work documents income shifting and retention among high-income business owners (Miller et al., 2024; Pirttilä & Selin, 2011; Ring et al., 2026), while Koivisto (2025) finds strong dividend-tax responses among lower-income private-firm owners. I extend this work by linking dividend responses to changes in family taxable income and tax liability, and by comparing families according to both income and financial exposure. These comparisons distinguish families' economic resources from their use of the tax-planning channel and show that income splitting can have relatively large tax consequences for moderate-income families. Following income and tax payments together helps assess the distributional consequences of restricting tax planning without equating changes in reported income with changes in economic activity or welfare (Chetty, 2009).

The remainder of the paper is organized as follows. Section 2 describes the taxation of private-business income in Canada and the 2018 expansion of TOSI. Section 3 introduces the linked administrative data and explains the construction of the analysis sample. Section 4 presents the empirical strategies, and Section 5 reports the income responses. Section 6 examines differences by business type and multiple-business ownership. Section 7 concludes.

2 Institutional Setting

2.1 Taxation of Private-Business Income in Canada

In Canada, both the federal and provincial governments levy corporate and personal income taxes. For all but two provinces, provincial personal and corporate income taxes are administered by the federal government through the Canada Revenue Agency (CRA).[5] Although provinces independently determine their own tax rates and certain credits, taxable personal and corporate income is generally determined using a common federal tax base. Consequently, although statutory tax rates vary across provinces, the fundamental structure of Canada's corporate and personal income tax systems remains broadly consistent across the country.

Active business income earned by corporations is subject to corporate income tax before profits are distributed to shareholders. Qualifying Canadian-controlled private corporations (CCPCs) face a reduced corporate tax rate on active business income up to a prescribed business limit through

[5]Quebec administers both its personal and corporate income tax systems, while Alberta administers its own corporate income tax. Despite these administrative differences, taxpayers in all provinces remain subject to the federal Income Tax Act.

the Small Business Deduction, while active business income above that limit is taxed at higher corporate rate.[6] Personal income taxes are levied on individuals rather than households and are subject to a progressive rate schedule. Employment income is fully included in taxable income and taxed at the individual's applicable marginal tax rate. Capital gains receive preferential treatment through a 50% inclusion rate, meaning that only one-half of a realized capital gain is included in taxable income. For the period I study, Dividend income receives separate tax treatment relative to wage/salary income because corporate profits have already been subject to corporate income tax before being distributed to shareholders. Unlike dividends, salaries and bonuses are deductible when calculating corporate taxable income and are therefore taxed only once in the hands of the recipient. Dividends, in contrast, are paid from after-tax corporate profits and face a gross-up and tax credit in the hands of the shareholder to neutralize the corporate tax credit already paid

This gross up and tax credit system for dividend income is designed to promote horizontal equity between individuals who earn business income through incorporated businesses, those who operate unincorporated businesses, and those who derive their income through employment.[7] As a result, owner-managers generally derive only modest tax advantages from changing the form in which business income is received. Nevertheless, incorporation continues to provide considerable flexibility over the timing, form, and recipient of business income. For example, owner-managers may defer personal taxation by retaining earnings within the corporation rather than immediately distributing profits. In addition to determining the timing and form of business income, owner-managers may also determine who ultimately receives that income. By structuring ownership of the corporation among family members, dividends may be distributed across those shareholders rather than solely to the primary business owner. This practice is commonly referred to as income splitting.

Canada's progressive personal income tax system and its use of the individual, rather than the household, as the tax unit make income splitting a potentially lucrative form of tax planning. Rather than receiving all dividend income themselves, owner-managers may instead distribute dividends to lower-income family members who hold ownership interests in the corporation. Because dividends are taxed in the hands of the recipient, shifting dividend income from an individual facing a higher marginal tax rate to a family member facing a lower marginal tax rate can reduce the total amount of personal income tax paid on a given amount of corporate income. Income splitting through private corporations had long been recognized as a concern within the Canadian tax system. In response to the use of corporate and trust structures to transfer investment income to minor children, the federal government introduced the TOSI Rules—commonly known as the "kiddie tax"—in the 1999 federal budget, with the rules applying beginning in the 2000 taxation year. The tax subjected certain dividend and other property income received by individuals aged 17 or younger to the top marginal

[6] A Canadian-controlled private corporation (CCPC) is a private corporation resident in Canada that is not controlled, directly or indirectly, by one or more non-residents or public corporations, as defined under the Income Tax Act. Unless otherwise noted, all references to corporations throughout the paper refer to CCPCs.

[7] In Canada, income earned by unincorporated businesses is taxed as if it were labour income.

personal income tax rate rather than the ordinary progressive rate schedule. The stated objective was to discourage arrangements that shifted income from higher-income parents to lower-income children and thereby undermined the progressivity of the personal income tax system (Department of Finance Canada, 1999).

The original TOSI rules restricted income splitting with minor children, but dividends paid to adult family members generally remained subject to the recipient's ordinary progressive tax schedule. Business owners could therefore continue to reduce their family's tax liability by paying dividends to lower-income adult relatives. In 2018, the federal government expanded the TOSI rules to restrict this remaining channel, subject to several exemptions.

2.2 The 2018 Tax on Split Income Reform

On July 18, 2017, the federal government announced a package of proposed tax reforms targeting tax planning through private corporations, identifying income sprinkling (referred to throughout this paper as income splitting) as one of three principal areas of concern (Department of Finance Canada, 2017c).[8] Following revisions announced in October 2017 (Department of Finance Canada, 2017a), the revised Tax on Split Income (TOSI) rules took effect on January 1, 2018. Under the expanded rules, dividend income received by an adult family member from a related private corporation would be subject to the highest marginal tax rate unless the recipient satisfied one of several statutory exemption criteria. For affected individuals, the reform therefore represented a substantial increase in the marginal tax rate applied to dividend income.

The expanded TOSI rules provide several statutory exemptions under which dividend income would continue to be taxed under the ordinary personal income tax system. The exemption criteria differed depending on the recipient's age. For individuals aged 25 and older, dividend income remained subject to the ordinary dividend tax rules if any of the following conditions were satisfied. (i) the recipient had been actively engaged in the business for an average of at least 20 hours per week during the portion of the year in which the business operated in the current taxation year or in any five previous taxation years; (ii) the recipient qualified for the excluded shares exemption; (iii) the owner of the business paying out the dividend was aged 65 or older; or (iv) the dividend represented a reasonable return on the recipient's labour or capital contributions to the business. To qualify for the excluded shares exemption, the recipient must generally directly own shares representing at least 10% of both the voting rights and value of the corporation. Canadian corporations generally issue common and preferred shares. Common shares typically carry voting rights, while preferred shares

[8]The other two areas of concern were passive investment income earned within private corporations and the conversion of income into capital gains.

are typically non-voting.[9] The corporation must also satisfy several additional statutory conditions, including that it is not a professional corporation and that less than 90% of its business income is derived from the provision of services. With the exception of the reasonable return provision, the exemption criteria are based on objective statutory rules.

Individuals between the ages of 18 and 24 faced considerably more restrictive exemption criteria. In particular, the excluded shares exemption was unavailable, and the reasonable return provision was limited to prescribed factors such as contributed capital and labour. Consequently, fewer statutory exemptions were available to individuals aged 18 to 24 than to those aged 25 and older. The exemption criteria were intended to distinguish family members making genuine labour or capital contributions from those receiving dividends primarily for tax-planning purposes. The exemption criteria generate several observable margins that determine exposure to the reform. Some, such as age, are predetermined and cannot be manipulated. Others reflect pre-existing ownership structures or labour market participation within the corporation that could be altered. I exploit these statutory distinctions to identify the behavioural response to the expansion of TOSI.

3 Data

I utilize the Canadian Employer-Employee Dynamics Database (CEEDD) (Statistics Canada, 2022), which contains administrative tax records for the universe of Canadian individual tax filers and Canadian-controlled private corporations (CCPCs) owned by either individuals or other corporations.[10] The primary analysis focuses on the period from 2012 to 2020. A key feature of the CEEDD is a set of identifiers that enables linkages between the individual and corporate tax files, allowing individuals to be connected to the corporations in which they hold ownership interests, corporations to be connected through corporate ownership structures, and firms to be linked to their employees. These linkages allow me to establish longitudinal relationships between corporations, their individual owners, the family members of those owners, and their employees, while simultaneously observing outcomes at both the individual and corporate levels.

3.1 Information on Individuals and Firms

The individual-level data are drawn from the T1 Family File (T1FF), the T1 Personal Master File (T1PMF), and T4 Statements of Remuneration Paid. These records provide information on individuals' demographic characteristics and income sources, including age, gender, province of

[9]The rights attached to common and preferred shares are determined by the corporation's articles and need not follow these conventional characteristics. Corporations may issue multiple classes of common or preferred shares with different voting, dividend, and liquidation rights. For example, preferred shares may carry voting rights, while a class of common shares may be non-voting. Accordingly, eligibility for the excluded shares exemption depends on the voting rights and fair market value associated with the shares held by an individual rather than whether those shares are formally classified as common or preferred.

[10]Corporations owned by trusts are excluded from the CEEDD.

residence, employment income, dividend income, taxable income, and total income. They also contain a unique individual identifier and information on family relationships, including a family identifier, family type, an indicator for whether an individual is identified as a child within the family, and, when relevant, an identifier for the individual's spouse. The T4 records additionally identify the employer paying an individual's employment income. Together, these records allow me to follow individuals and their income over time and link them to family members and corporations.

I measure employment income using T4 earnings from all employers and, separately, earnings paid by the family corporation. Total T4 income captures changes beyond payments from the family business, including earnings associated with increased labour supply to outside employers. T4 income from the family corporation isolates changes in employment income paid through that business.

I measure dividend income using the cash amount reported on each individual's personal tax return, before the gross-up used to calculate taxable income. These records include dividends from all sources, whereas the corporate records report total dividends paid by the business without identifying how much each shareholder receives. The personal records therefore allow me to distinguish changes in spouse dividends from changes in owner dividends, including cases in which payments to one fall while payments to the other rise. Although these measures also include dividends from outside the family corporation, the treatment–control comparison accounts for changes in those dividends that are common to both groups. The identifying assumption requires that, absent the reform, these other dividend sources would not generate different trends between treated and control families.[11]

I use log income outcomes to express responses in proportional terms given the right-skewed income distribution and the concentration of income splitting among higher-income families (Bauer et al., 2015; Wolfson & Legree, 2015). The log-dividend specifications capture the intensive-margin response among individuals reporting positive dividends. To examine the extensive margin, I construct an indicator, $ClaimDiv$, equal to one if an individual reports any dividend income and zero otherwise. I construct this indicator separately for spouses and primary owners. All log-income specifications exclude observations with zero or negative values of the corresponding outcome.

At the couple level, I sum income across the primary owner and spouse to construct measures of dividend income, employment income, taxable income, and total income. I measure family tax liability by summing each spouse's total tax calculated in the personal tax records. This observed outcome is distinct from the simulated tax rates and liabilities used to construct the tax incentive and TOSI penalty. Together, these outcomes allow me to examine whether reductions in spousal dividends coincide with increases in other income received by the couple. In robustness analyses, I examine total dividends paid by the family corporation.

[11]Personal income is reported by calendar year, while corporate dividends, assets, and retained earnings are reported by corporate fiscal year, which may end in a different month. Payments received within the same calendar year can therefore fall in different corporate reporting periods, making the timing of annual personal and corporate outcomes less directly comparable.

The corporate-level data are drawn primarily from the T2 Schedule 50 Shareholder Information (T2S50) and the National Accounts Longitudinal Microdata File (NALMF). The T2S50 provides information on the common and preferred shares held by each shareholder. Shareholders are identified as either individuals or corporations, with identifiers that allow their ownership interests to be linked to the corresponding individual or corporate records. The NALMF provides information on corporate financial outcomes and characteristics, including revenue, profits, assets, retained earnings, and four-digit North American Industry Classification System (NAICS) industry codes. Using the corporate financial records, I construct measures of asset growth and retained earnings. I measure asset growth as the year-over-year percentage change in the corporation's total assets. Because retained earnings are reported at the beginning and end of each fiscal year, I measure average retained earnings during the year as the mean of beginning- and end-of-year retained earnings. I also calculate the year-over-year percentage change in retained earnings.

I use the Canadian Tax and Credit Simulator (CTaCS) to calculate marginal tax rates using each individual's province of residence, taxable income less cash dividends, cash dividend income, and their spouse's taxable income.[12] I calculate the dividend marginal tax rate as the change in simulated family tax liability resulting from a $100 increase in an individual's cash dividend income, divided by 100.[13] In addition to calculating rates at observed income, I construct a counterfactual dividend marginal tax rate for each spouse by setting their cash dividend income to zero while holding all other simulation inputs unchanged.[14] For the treatment-intensity measure, I also calculate the primary owner's dividend marginal tax rate after assigning the spouse's cash dividends to the owner. Equation (3) compares the spouse's and owner's net-of-tax rates under these counterfactual allocations. A larger spouse-to-owner ratio indicates a stronger incentive to allocate dividends to the spouse.

3.2 Linking Owners, Families, and Firms

I combine family, ownership, and employer identifiers to link each couple to its family corporation. Family identifiers connect individuals to their spouse, while the T2S50 shareholder records identify the corporations in which each spouse holds common or preferred shares. Employer identifiers on T4 slips then allow me to determine whether either spouse works for the family corporation

[12]Eligible and non-eligible dividends receive different gross-ups and dividend tax credits, with non-eligible dividends generally facing higher personal tax rates. This distinction reflects the corporate tax paid on the underlying income. Eligible dividends generally correspond to income taxed at the general corporate rate, while non-eligible dividends generally correspond to income taxed at a lower rate through provisions such as the small business deduction. This deduction generally applies to qualifying active business income up to $500,000. The majority of firms in my sample have profits below $500,000, so I simulate tax liabilities assuming all dividends are non-eligible. To the extent that individuals receive eligible dividends, this assumption may overstate their simulated personal tax liabilities.

[13]Appendix A.1 describes the CTaCS grid and marginal-rate calculation.

[14]Because individual dividend income includes dividends from all sources, this counterfactual also removes any dividends received from outside the family corporation. It therefore represents the spouse's tax rate in the absence of all dividend income, rather than only excluding dividends from the family corporation.

and to measure the employment income it pays them. Together, these linkages allow me to follow income reported by both spouses alongside the ownership, employment, and financial outcomes of their business.[15]

3.3 Measuring Exposure to the TOSI Reform

The exemptions introduced with the expansion of the TOSI rules depend in part on the spouse's ownership interest in and employment by the family corporation. I use these two dimensions to classify spouses according to their likely exposure to the TOSI reform. The ownership-based exemption generally requires the spouse to own at least 10% of the corporation's votes and value. The employment exemption generally requires the spouse to work at least 20 hours per week in the business. Because the administrative data report employment income rather than hours worked, I approximate the employment condition using an annual earnings threshold equal to 20 hours per week for 52 weeks at the applicable provincial minimum wage.

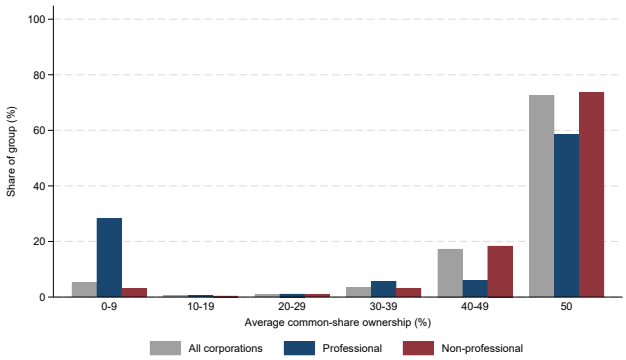
Within the age-restricted sample described below, I classify a spouse as *likely exposed* if they held less than 10% of the corporation's common shares in every observed year from 2012 through 2020 and never earned employment income from the corporation at or above the earnings threshold in any year from 2013 through 2020. I classify a spouse as *likely exempt* if they held at least 10% of the common shares in every observed year or earned employment income from the corporation at or above the threshold in every year from 2013 through 2020. All spouses in the analysis sample hold common or preferred shares and are therefore eligible to receive dividends directly from the corporation. Spouses whose ownership and employment histories satisfy neither definition are excluded from the treatment–control comparison. I refer to likely exposed spouses as the treated group and likely exempt spouses as the control group in the empirical analysis.

Figure 1 summarizes the pre-reform ownership and employment characteristics underlying this classification. Panel A shows that common-share ownership is concentrated at 50%, but low ownership is substantially more common among spouses associated with professional corporations. Approximately 28% of spouses associated with professional corporations hold less than 10% of the common shares, compared with approximately 3% of spouses associated with non-professional corporations. Panel B shows that approximately 49% of spouses are not employed by the family corporation, 8% are employed but earn less than the threshold, and 43% earn more than the threshold. The ownership and employment conditions therefore identify distinct sources of potential exposure to the reform.

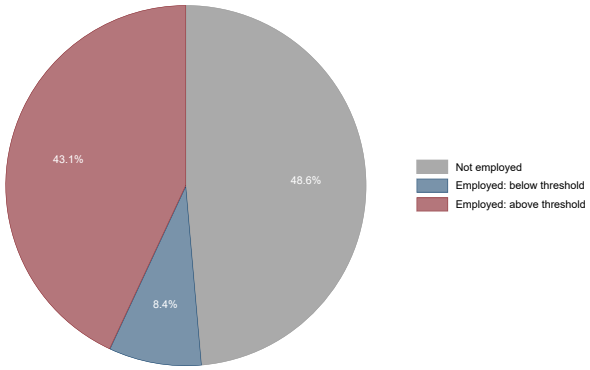
The main classification requires spouses in the treated group to remain below both observable exemption margins throughout the sample period. This restriction excludes spouses who appear exposed before the reform but subsequently move above either the ownership or employment threshold. Appendix Table A3 compares this definition with an alternative that identifies the treated group

[15]The family records also identify children living at the same address as their parents, but generally do not identify children living elsewhere or relationships between siblings. I therefore focus on income splitting between spouses.

Figure 1: Pre-Reform Ownership and Employment Characteristics



(a) Common-share ownership



(b) Employment relative to the earnings threshold

Notes: Panel A reports the distribution of the spouse's average common-share ownership through 2016 for the full sample and separately for professional and non-professional corporations. Panel B classifies spouses using their average pre-reform employment income from the family corporation. The earnings threshold equals the amount earned by working 20 hours per week for 52 weeks at the applicable provincial minimum wage. "Not employed" denotes spouses with no employment income from the corporation. The figure is descriptive; the treatment classification uses the annual ownership and employment histories described in the text. Counts are subject to the rounding requirements applied to the administrative data.

using only pre-reform characteristics. Requiring treated spouses to remain below both thresholds reduces the potential treatment–control sample from approximately 64,460 to 63,790 families. The number of potentially treated families falls from approximately 3,700 to 3,030, while the control group is unchanged.

Appendix Figure A1 examines whether this reduction reflects systematic adjustment toward the observable exemption margins. The figure uses the broader sample of spouses eligible to receive dividends and defines exposure using only the age-based rules. It therefore does not condition on spouses' ownership or employment choices. The figure shows little evidence of systematic post-reform movement into the ownership or employment categories associated with an exemption.

These classifications measure likely exposure rather than establish each spouse's statutory tax treatment with certainty. Common-share ownership does not reveal whether a spouse owns at least 10% of both the votes and value of the corporation, and the availability of the ownership-based exemption also depends on characteristics of the business. Employment income similarly does not reveal hours worked. A spouse earning more than the provincial minimum wage could exceed the earnings threshold while working fewer than 20 hours per week. The classification also does not measure every potential exemption, including whether dividends represent a reasonable return. As a result, some affected spouses may be assigned to the control group and some exempt spouses may be assigned to the treated group. Such misclassification would generally make the two groups more similar and attenuate the estimated differences between them.

I assess the sensitivity of the results to this classification using alternative definitions based on common-share ownership alone and on common-share ownership among spouses with no recorded employment in the family corporation. Both alternatives yield declines in spouse dividend income

and receipt alongside increases in primary-owner dividends, consistent with the main specification. Appendix A.5 describes these definitions and reports the estimates.

3.4 Sample Construction

My primary analysis examines income splitting between spouses. I restrict the sample to married or common-law individuals whose spouse can be identified in the administrative records. Within each couple, I designate the individual who owns the larger share of the corporation's common shares in 2016 as the primary owner and refer to the other individual as the spouse. If both individuals own equal shares, I assign these roles using a fixed ordering of individual identifiers. I hold the designations fixed throughout the analysis so that subsequent changes in ownership, employment, or income do not alter each individual's role within the couple.

Appendix Table A1 reports the cumulative sample restrictions. The linked data initially contain 1,114,185 married or common-law couples associated with at least one business. After requiring the variables needed to calculate tax rates to be observed and valid, the sample contains 1,078,135 couples. Among these couples, 397,585, or 36.9%, have observed share ownership for both spouses. I impose this restriction because an individual must own shares to receive dividends directly from the family corporation.[16]

I next check the completeness of the ownership linkages by comparing the corporations I identify through the shareholder records with an administrative count of the corporations associated with each individual. I retain individuals for whom these counts agree. Appendix A.2 describes the ownership-linkage checks.

For the primary analysis, I further restrict the sample to spouses who hold shares in only one business. This reduces the sample from 350,510 to 264,060 couples. I require the corporation associated with the spouse to be owned directly by individuals and to have no ownership links to other corporations. The primary owner may hold shares in additional businesses; the restriction applies only to the spouse and the corporation in which the spouse holds shares. These restrictions identify a single family corporation from which the spouse may receive dividends and reduce ambiguity about the business from which changes in the spouse's income originate. I report estimates that include spouses who hold shares in multiple businesses as a robustness exercise.

I also require members of the family to collectively own 100% of the corporation's common shares. Because common shares typically carry voting rights, this restriction focuses the analysis on corporations whose voting control is likely to rest entirely with the family. To keep age-based eligibility fixed during the post-reform period, I exclude spouses aged 23–24 in 2018 and families whose primary owner was aged 63–64 in 2018. Spouses aged 23–24 would cross the age-25 threshold during the post-reform period, while the spouse of a primary owner who reaches age 65 may become

[16] A family trust may hold shares in a private corporation and allocate dividend income to beneficiaries who do not directly own the shares. The CEEDD does not contain the information required to link family trusts to their beneficiaries, so I cannot identify income splitting conducted through these arrangements.

eligible for the age-based spousal exemption.

Finally, I construct a balanced panel of families observed throughout the analysis period. This restriction reduces the sample from 220,100 to 94,295 couples and prevents changes in sample composition from driving changes in average outcomes over time. I report estimates using an unbalanced panel as a robustness exercise in Appendix A.6. After retaining families classified as either likely exposed to or likely exempt from the TOSI reform, the final sample contains 63,790 couples. The final sample therefore contains approximately 16.0% of the 397,585 couples in which both spouses directly own shares. Outcome-specific estimation samples may be smaller when an outcome is missing or nonpositive and therefore excluded from a log specification.

Appendix Table A2 compares the final analysis sample with the broader population of business-owning couples in 2016. The restrictions select families in which the spouse has a stronger observable connection to the business. In the analysis sample, 52.5% of spouses receive dividends and 39.7% are employed by the family corporation, compared with 34.8% and 19.1%, respectively, in the broader population. The analysis sample also contains larger shares of likely service businesses and professional corporations. The estimates should therefore be interpreted as applying to business-owning couples whose ownership and income allocation can be observed clearly, rather than to all families associated with private corporations.

Table 1 reports 2016 summary statistics separately for families classified as likely exposed to and likely exempt from the TOSI reform. Section 3.3 defines these classifications in detail. The exposed group contains 3,030 couples, while the exempt group contains 60,755. Differences in common-share ownership, preferred-share ownership, and employment by the family corporation largely reflect the criteria used to construct the groups and should not be interpreted as balance tests.

The groups also differ substantially in business composition. Professional corporations account for 47.3% of the exposed group but only 7.2% of the exempt group.[17] Similarly, 70.1% of exposed families are associated with likely service businesses, compared with 32.3% of exempt families. Exposed families also report higher incomes. Average family taxable income is $218,068 in the exposed group and $137,804 in the exempt group. The concentration of professional and service corporations among exposed families is consistent with at least part of this income difference, although the summary statistics do not establish how much of the gap is attributable to business type. I therefore report estimates separately for professional and non-professional corporations as a robustness exercise.

[17]Provincial rules may also restrict voting shares to licensed practitioners. For example, Ontario permits family members to hold non-voting shares in physician and dentist corporations, while reserving voting shares for members of those professions. In the full analysis sample, exposed spouses hold 0.08% of common shares on average, compared with 47.9% among exempt spouses (Table 1).

Table 1: Summary Statistics by Exposure to the TOSI Reform

	Likely exposed (N = 3,030)		Likely exempt (N = 60,755)	
Panel A: Demographics				
Spouse age (years)	47.1	(8.6)	49.2	(8.3)
Primary-owner age (years)	47.6	(8.1)	49.3	(7.7)
Panel B: Personal and family income				
Spouse receives dividends (%)	65.5	(47.5)	51.8	(50.0)
Spouse dividend income	30,339	(44,585)	15,661	(30,690)
Primary-owner dividend income	44,255	(58,430)	17,262	(38,815)
Spouse T4 income	55,060	(90,389)	43,304	(57,765)
Primary-owner T4 income	66,930	(103,477)	48,496	(84,448)
Family taxable income	218,068	(155,565)	137,804	(135,963)
Family tax liability	45,508	(62,963)	25,900	(52,532)
Panel C: Characteristics relevant to treatment classification				
Spouse common-share ownership (%)	0.08	(0.75)	47.90	(6.54)
Spouse preferred-share ownership (%)	67.18	(34.86)	6.85	(21.91)
Spouse employed by the family corporation (%)	2.87	(16.70)	41.53	(49.28)
Likely service business (%)	70.05	(45.81)	32.28	(46.76)
Professional corporation (%)	47.26	(49.93)	7.18	(25.82)
Panel D: Business characteristics and outcomes				
Business age (years)	7.6	(5.4)	11.6	(7.6)
Businesses in which the primary owner holds shares	1.24	(0.64)	1.11	(0.46)
Labour units	2.10	(8.52)	4.33	(9.23)
Gross profits	264,035	(371,359)	211,272	(407,863)
Total assets	487,274	(771,819)	556,787	(1,197,583)
Retained earnings	341,787	(577,038)	237,937	(607,660)

Notes: The table reports 2016 means, with standard deviations in parentheses, separately for families classified as likely exposed to and likely exempt from the TOSI reform. Monetary values are in Canadian dollars and are rounded to the nearest dollar. Sample sizes may differ for payroll and financial-statement variables because of missing data. Counts are rounded in accordance with the disclosure requirements applied to the administrative data; separately rounded group counts may therefore differ slightly from the rounded total sample count.

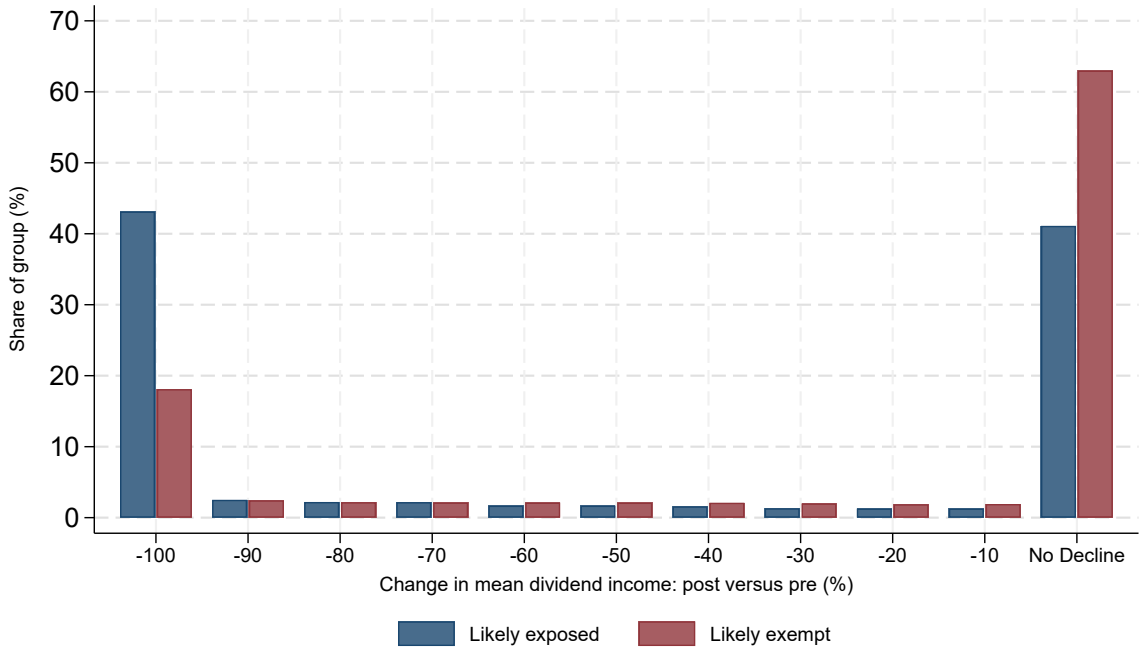
4 Empirical Strategy

I use three complementary empirical strategies to estimate responses to the TOSI reform. The regressions cover 2013–2020. First, I compare changes in personal and family income outcomes between families classified as treated and control, as defined in Section 3.3. This comparison estimates the average response among treated families. Second, I restrict the sample to treated families and examine how responses vary with the pre-reform tax advantage of allocating dividends to the spouse rather than the primary owner. This approach distinguishes exposure to the reform from the strength of the incentive it removed and allows me to estimate the elasticity of spouse dividends with respect to that incentive. Third, I measure financial exposure using the additional tax each family would owe if it maintained its pre-reform income allocation, divided by total family income in 2016. This measure incorporates both the amount of income previously allocated to the spouse and the resulting tax cost relative to the family's income. I use this relative penalty as an instrument for spouse dividends to examine how reform-induced changes in those payments are accompanied by changes in owner income, family taxable income, and tax liability.

4.1 Treatment-Control Specification

Figure 2 provides descriptive evidence on how spouse dividend income changed across the treatment and control groups defined in Section 3.3. For each spouse, I calculate the percentage change in average dividend income between the pre-reform period, 2012–2016, and the post-reform period, 2018–2020. The figure plots the distribution of these changes separately for treated and control families. Treated spouses are substantially more concentrated in the category representing a complete elimination of spouse dividends. Control spouses are more likely to experience no decline. The remaining reductions are distributed across similar intermediate categories in both groups. This descriptive pattern is consistent with larger reductions in spouse dividends among families classified as treated. Appendix Figure B1 complements this comparison by showing annual trends in spouse dividend income, dividend receipt, and family taxable income for treated and control families.

Figure 2: Changes in Spouse Dividend Income by Exposure Classification



Notes: The figure shows the distribution of the percentage change in each spouse's average dividend income between the pre-reform period, 2012–2016, and the post-reform period, 2018–2020. Reductions are grouped into 10-percentage-point bins. The category labelled "No decline" includes spouses whose average dividend income did not decline. The sample is restricted to spouses with positive average dividend income during the pre-reform period. Bars report the share of treated or control spouses in each category. Group shares are calculated using vetted, rounded cell counts.

I begin by estimating an event-study specification to examine differences in outcome trends before the TOSI reform and the timing of subsequent responses. The identifying assumption is that, in the absence of the reform, outcomes for treated and control families would have evolved in

parallel.[18] I estimate:

$$Y_{it} = \alpha + \sum_{\substack{y=2013 \\ y \neq 2016}}^{2020} \beta_y \left(\mathbb{1}[t = y] \times Treated_i \right) + \delta' X_{it} + \gamma_i + \lambda_{p(i,t),t} + \mu_{j(i,t),t} + \eta_{a^s(i,t)} + \kappa_{a^o(i,t)} + \varepsilon_{it}. \tag{1}$$

Here, i indexes couples and t indexes years. The dependent variable Y_{it} is an outcome for the spouse, primary owner, or couple, as defined in Section 3.1. $Treated_i$ equals one for families classified as treated and zero for families classified as control. The term γ_i denotes couple fixed effects. The terms $\lambda_{p(i,t),t}$ and $\mu_{j(i,t),t}$ denote province-by-year and four-digit NAICS-industry-by-year fixed effects, respectively.[19] The terms $\eta_{a^s(i,t)}$ and $\kappa_{a^o(i,t)}$ denote spouse-age and primary-owner-age fixed effects. The vector X_{it} contains the number of jobs, the numbers of main and subsidiary businesses owned by the couple, the total preferred-share ownership recorded for the family corporation, the number of individuals with recorded preferred-share holdings, and an indicator for whether the family has children. The omitted reference year is 2016, the final year before the reform was announced. Each β_y measures the difference between the change in outcomes for treated and control families from 2016 to year y. The pre-reform coefficients provide evidence on whether the two groups followed similar trends before the reform. I cluster standard errors at the family level.

I then estimate a more parsimonious difference-in-differences specification:

$$Y_{it} = \alpha + \beta_1 \left(Treated_i \times Announce_t \right) + \beta_2 \left(Treated_i \times Post_t \right) + \delta' X_{it} + \gamma_i + \lambda_{p(i,t),t} + \mu_{j(i,t),t} + \eta_{a^s(i,t)} + \kappa_{a^o(i,t)} + u_{it}. \tag{2}$$

The indicator $Announce_t = \mathbb{1}[t = 2017]$ identifies the announcement year, while $Post_t = \mathbb{1}[t \geq 2018]$ identifies the implementation period from 2018 through 2020. The coefficient β_1 measures the relative change in outcomes for treated families during the announcement year, and β_2 measures their average relative change after implementation. The controls and fixed effects are the same as in Equation (1). The identifying assumption is that, in the absence of the TOSI reform, the difference in outcomes between treated and control families would have remained stable after 2016.

[18] In 2018–2019, the federal government reduced the small-business corporate tax rate and made associated changes to dividend gross-ups and tax credits (Department of Finance Canada, 2017b). These changes applied to qualifying businesses and shareholders in both groups, rather than following the spouse-specific TOSI exemption criteria. Although they could affect the magnitude of the estimates, the decline in treated spouses' dividends alongside higher owner dividends is consistent with income reallocation. The spouse-dividend response also begins in 2018, before the passive-income measures took effect for taxation years beginning after 2018 (Department of Finance Canada, 2018).

[19] Province-by-year fixed effects absorb annual shocks shared by families in the same province, including common effects of provincial tax reforms. They do not absorb tax changes that affect income brackets differently. Industry-by-year fixed effects absorb annual shocks common to businesses in the same industry, such as changes in commodity prices or differences across industries in their exposure to the COVID-19 pandemic.

4.2 Tax Incentives and Treatment Intensity

The treatment and control groups are defined using the ownership and employment criteria associated with the TOSI exemptions. This classification captures whether a family was likely exposed to the reform, but it does not measure the strength of its pre-reform incentive to allocate dividends to the spouse. Treated families may have faced similar dividend marginal tax rates for both spouses or a substantial difference between them. I measure this variation using the marginal dividend tax rates faced by both members of the couple.

Starting from each family's observed 2016 income allocation, I construct a counterfactual in which the spouse's cash dividend income is set to zero and the same amount is added to the primary owner's dividend income. This holds the couple's total dividend income fixed while changing only which spouse reports it. Under this counterfactual allocation, I calculate the spouse's dividend marginal tax rate with no dividend income and the primary owner's dividend marginal tax rate after receiving the spouse's dividend income. I define the pre-reform tax incentive as

$$Incentive_i = \log\left(1 - \tau_{i,2016}^{Spouse,NoDiv}\right) - \log\left(1 - \tau_{i,2016}^{Owner,Reallocated}\right), \tag{3}$$

where $\tau_{i,2016}^{Spouse,NoDiv}$ is the spouse's dividend marginal tax rate after setting their dividend income to zero and $\tau_{i,2016}^{Owner,Reallocated}$ is the primary owner's dividend marginal tax rate after adding the spouse's dividend income to the owner's income. The measure is the log of the spouse-to-owner ratio of net-of-tax marginal dividend rates. A larger value of $Incentive_i$ indicates a stronger pre-reform incentive to allocate dividends to the spouse. Because the measure is constructed entirely from 2016 income and tax parameters, I hold it fixed throughout the analysis so that subsequent income responses do not alter the measured incentive.

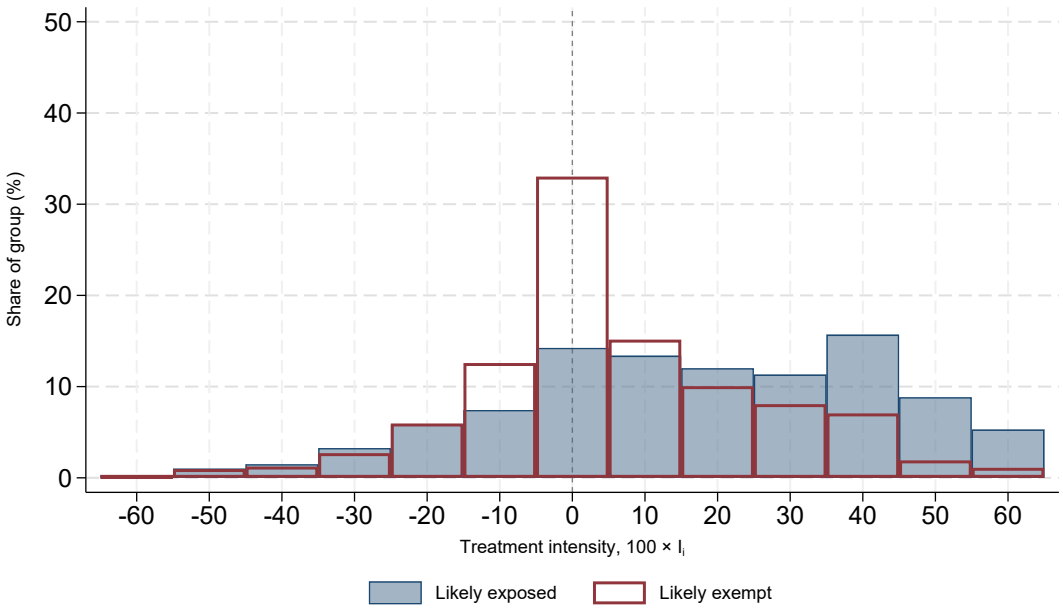
Figure 3 shows the distribution of the pre-reform tax incentive. Among treated families, the distribution is broad and shifted toward positive values. Many faced a substantial tax advantage from allocating dividends to the spouse, while others faced little or no advantage. The distribution for control families is more concentrated near zero. The variation within the treated group allows me to examine whether families with stronger initial incentives changed their income allocation more after the TOSI reform.

Restricting the sample to treated spouses, I first estimate the following event-study specification:

$$\begin{aligned} Y_{it} = \alpha + \sum_{\substack{y=2013 \\ y\neq 2016}}^{2020} \theta_y \left(\mathbb{1}[t=y] \times Incentive_i\right) + \delta' X_{it} \\ + \gamma_i + \lambda_{p(i,t),t} + \mu_{j(i,t),t} + \eta_{a^s(i,t)} + \kappa_{a^o(i,t)} + \varepsilon_{it}. \end{aligned} \tag{4}$$

The omitted reference year is 2016, and the controls and fixed effects are defined as in Equation (1). Each coefficient θ_y measures how the change in an outcome between 2016 and year y varies with the family's pre-reform incentive. When Y_{it} is expressed in logs, θ_y measures how the

Figure 3: Distribution of the Pre-Reform Incentive to Split Dividend Income



Notes: The figure shows the distribution of $100 \times Incentive_i$, where $Incentive_i$ is the log ratio of the spouse's counterfactual net-of-tax rate to the primary owner's counterfactual net-of-tax rate. The counterfactual assigns all of the spouse's 2016 dividend income to the primary owner before calculating their marginal dividend tax rates. A positive value indicates that allocating dividends to the spouse generated a tax advantage before the TOSI reform. The filled blue bars represent families classified as likely exposed, while the red outlines represent families classified as likely exempt. Tax rates are calculated using 2016 family characteristics and the Canadian Tax and Credit Simulator.

proportional change in the outcome varies with the spouse-to-owner net-of-tax-rate ratio.

I then estimate the following specification:

$$\begin{aligned} Y_{it} = \alpha + \theta_1 \left(Incentive_i \times Announce_t\right) + \theta_2 \left(Incentive_i \times Post_t\right) + \delta' X_{it} \\ + \gamma_i + \lambda_{p(i,t),t} + \mu_{j(i,t),t} + \eta_{a^s(i,t)} + \kappa_{a^o(i,t)} + u_{it}. \end{aligned} \tag{5}$$

The coefficient θ_1 measures how the change in an outcome during the 2017 announcement year varies with the pre-reform incentive, while θ_2 measures the corresponding relationship during 2018–2020. For log spouse dividend income, a negative value of θ_2 indicates that families with a stronger pre-reform incentive to allocate dividends to the spouse reduce those dividends by more after implementation. For example, a coefficient of negative one implies that a 1% higher spouse-to-owner net-of-tax-rate ratio would imply a roughly 1% larger decline in spouse dividend income.

A causal interpretation requires that, in the absence of the TOSI reform, changes in outcomes among treated families would not have varied systematically with their 2016 tax incentive, conditional on the included controls and fixed effects. The pre-reform event-study coefficients provide evidence on the plausibility of this assumption. This approach estimates how responses vary with the incentive to split income rather than the average effect of the reform among treated families. I cluster standard errors at the family level.

4.3 Financial Exposure to the TOSI Reform

The pre-reform tax incentive measures the relative tax advantage of allocating dividends to the spouse rather than the primary owner. It does not capture how much income the family allocated through this channel. Two families may face the same relative net-of-tax advantage but owe different amounts of additional tax under TOSI because they reported different spouse dividends before the reform. The importance of a given tax cost can also depend on the family's income. I therefore measure financial exposure using the additional tax on the spouse's 2016 dividends as a share of the family's total income in 2016.

Holding each family's observed 2016 incomes fixed, I calculate its total personal tax liability in each year under two tax systems. The first applies the expanded TOSI rules to the spouse's observed dividend income, while the second taxes that income under the ordinary personal income-tax system. Unlike the counterfactual used to construct the pre-reform tax incentive, no income is reassigned between spouses in either calculation. I define the TOSI penalty as

$$Penalty_{it} = Treated_i \left[Tax_t^{TOSI} \left(FamInc_{i,2016} \right) - Tax_t^{NoTOSI} \left(FamInc_{i,2016} \right) \right], \tag{6}$$

where $FamInc_{i,2016}$ denotes the observed 2016 income components of the spouse and primary owner. The same income amounts are used in both calculations; only the tax treatment applied to the spouse's dividend income differs. Both liabilities are calculated using year-t federal and provincial tax parameters, allowing tax rules to vary over time while holding the family's income fixed.

The analysis retains both treated and control families, but the penalty applies only to treated families because control families remain subject to the ordinary personal income-tax system. The measure is zero for every family before the expanded TOSI rules take effect and for control families in every year. Beginning in 2018, it varies across treated families according to the additional tax applied to the spouse's 2016 dividend income. I divide this dollar penalty by the family's total income in 2016 to obtain relative financial exposure,

$$RelativePenalty_{it} = \frac{Penalty_{it}}{FamilyTotalIncome_{i,2016}}. \tag{7}$$

Both the income allocation used to calculate the penalty and the income denominator are fixed at their 2016 values. Subsequent income responses therefore do not change the measured exposure. A ratio of 0.01 represents additional tax equal to 1% of the family's baseline total income. Dividing the dollar penalty by family income changes the variation used for identification, while multiplying the resulting ratio by 100 would only change its units.[20]

I use the relative TOSI penalty as an instrument for dividend income reported by the spouse. The first-stage specification is

[20] I also use the dollar penalty, measured in thousands of dollars, as an alternative instrument. Appendix D compares the dividend and family-outcome estimates by family-income quartile.

$$\log\left(Div_{it}^{Spouse}\right) = \alpha + \pi RelativePenalty_{it} + \delta' X_{it} + \gamma_i + \lambda_{p(i,t),t} + \mu_{j(i,t),t} + \eta_{a^s(i,t)} + \kappa_{a^o(i,t)} + \nu_{it}, \tag{8}$$

The controls and fixed effects are defined as in Equation (1). In this log-dividend specification, the instrument enters as a decimal ratio. The first-stage coefficients and standard errors in the tables and figures are the estimated decimal-unit values divided by 100, so they show the change in log spouse dividends for a one-percentage-point increase in the penalty-to-income ratio. I also estimate a specification that replaces log spouse dividends with an indicator for receiving any spouse dividends. This version includes families with zero dividends and used the ratio multiplied by 100 in estimation. Its first-stage coefficients give the probability change for a one-percentage-point increase in relative exposure; multiplying them by 100 expresses that change in percentage points. Changing the instrument's units leaves the second-stage estimates and first-stage F-statistics unchanged.

To estimate how family taxable income changes when TOSI induces a reduction in spouse dividends, I estimate

$$\log\left(FamilyTaxableIncome_{it}\right) = \alpha + \beta \log\left(\widehat{Div_{it}^{Spouse}}\right) + \delta' X_{it} + \gamma_i + \lambda_{p(i,t),t} + \mu_{j(i,t),t} + \eta_{a^s(i,t)} + \kappa_{a^o(i,t)} + \epsilon_{it}. \tag{9}$$

In Equation (9), spouse dividend income is instrumented using the relative TOSI penalty. The coefficient β relates the response of family taxable income to the instrument-induced change in spouse dividends. Its interpretation requires that, conditional on the included controls and fixed effects, the instrument affects the outcome through the response captured by the instrumented spouse-dividend measure. The coefficient should not be interpreted as mechanically changing spouse dividends while holding every other income source fixed. I estimate corresponding specifications for owner dividends, employment income, and family tax liability, and report both the log-dividend and dividend-receipt IVs. TOSI can change tax liability directly on dividends that continue to be paid, so its second-stage coefficient does not isolate the causal effect of changing spouse dividends alone. Multiplying each second-stage coefficient by its matching first-stage coefficient recovers the reduced-form response of the outcome to relative exposure in the same estimation sample. I use these products when comparing responses to a common change in exposure. I cluster standard errors at the family level.

5 Income Responses

I first examine how the TOSI reform changed the allocation of dividend and employment income between spouses and primary owners. I then examine the implications for family taxable income and tax liability. Throughout, I report average responses and differences across quartiles of family income measured in 2016 to assess how the form and magnitude of adjustment vary across the

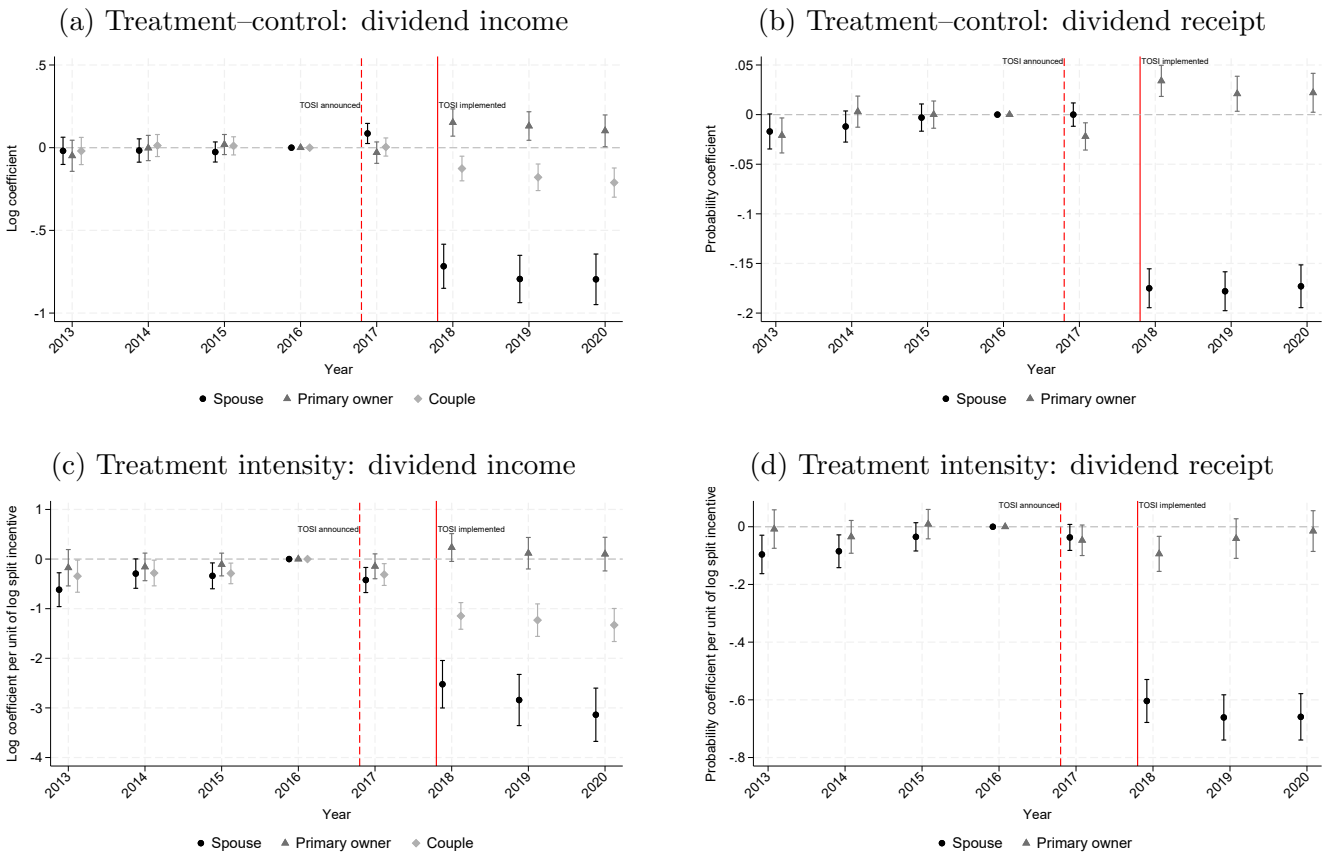
income distribution. The main tables report log-income estimates and dividend receipt; dollar and PPML estimates appear in the appendix.

5.1 Income Allocation Between Spouses and Primary Owners

5.1.1 Dividend Income

Spouse dividends rise relative to the control group in 2017, consistent with families bringing payments forward before the TOSI reform took effect. They fall sharply in 2018 and remain lower through 2020 (Figure 4). Primary-owner dividends rise after implementation, but couple dividends fall. The decline in the probability that treated spouses report any dividends shows that families respond by stopping some payments altogether, as well as by reducing the amounts paid to continuing recipients.

Figure 4: Event-Study Estimates of Dividend Responses



Notes: Panels A and B report treatment–control event-study estimates; Panels C and D report treatment-intensity estimates among treated families. Panels A and C show log dividend income for the spouse, primary owner, and couple. Panels B and D show indicators for dividend receipt by the spouse and primary owner. Treatment-intensity coefficients relate the outcome in each year to the fixed 2016 tax incentive. The omitted reference year is 2016. Regressions include the controls and fixed effects in Equations (1) and (4), respectively. Bars report 95% confidence intervals based on standard errors clustered at the family level. The dashed vertical line marks the 2017 announcement, and the solid vertical line marks the 2018 implementation of the TOSI reform.

After implementation, treated spouses report 52.9% less dividend income[21] and are 16.8 percentage points less likely to report any dividends than controls (Table 2). Primary owners report 14.3% more dividends, and their probability of receipt rises by 2.9 percentage points. In dollars, the $12,391 decline in spouse dividends is partly offset by a $7,124 increase in owner dividends, leaving a $5,267 decline in couple dividends. The couple log estimate similarly implies a 15.9% decline. The PPML estimates, which include zero dividend outcomes, support the same pattern. Families thus redirect substantial dividend income toward owners, but they do not replace all of the dividends no longer reported by spouses. Because personal tax records include dividends from sources other than the family corporation, these estimates do not trace each forgone spouse dollar to a particular destination.

Table 2: Dividend Responses to the TOSI Reform

	Log dividend income			Dividend receipt	
	Spouse	Primary owner	Couple	Spouse	Primary owner
Panel A: Treatment–control, 2018–2020					
Treated × Post	-0.753***	0.134***	-0.173***	-0.168***	0.029***
	(0.066)	(0.037)	(0.035)	(0.009)	(0.007)
Observations	239,795	255,870	295,125	478,735	478,735
Panel B: Treatment intensity, 2018–2020					
2016 incentive × Post	-2.552***	0.256**	-1.018***	-0.591***	-0.043
	(0.238)	(0.130)	(0.134)	(0.038)	(0.029)
Observations	11,665	15,900	17,830	21,835	21,835

Panel C: Relative financial exposure

	Log spouse dividends IV		Dividend receipt IV	
Dependent variable	Log spouse dividends	Log owner dividends	Spouse dividend receipt	Log owner dividends
First stage	-0.071***	—	-0.022***	—
	(0.007)		(0.001)	
Second stage	—	-0.315***	—	-0.780***
		(0.061)		(0.162)
First-stage F-statistic	125.4		397.7	
Observations	200,680		255,855	

Notes: Panels A and B report implementation-period TC and TI estimates. Panel B interacts the fixed 2016 log spouse-to-owner net-of-tax-rate ratio with the post-reform indicator among treated families. Log-income coefficients in Panel B are elasticities with respect to that ratio. Panel C uses relative financial exposure to instrument log spouse dividends or dividend receipt. Both second stages use log owner dividends. First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome has its own estimation sample. Family-clustered standard errors appear in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Dollar and PPML TC and TI estimates appear in Appendix Table B1; dollar-penalty IV comparisons by family-income quartile appear in Appendix D.

[21]The log-dividend regression excludes observations with zero dividend income; the dividend-receipt and dollar specifications include them.

The treatment–control estimates show that the dividend response varies substantially across the 2016 family-income distribution (Figure 5). The spouse-dividend decline grows from about $2,000 for the lowest income quartile Q1 to $20,100 for Q4, while the log estimate is imprecise in Q1 and becomes increasingly negative from Q2 through Q4. Only Q4 owners show a clear increase in dividend income, and even their increase is smaller than the decline for spouses. Higher-income families therefore account for much larger adjustments in dividend payments. These estimates alone cannot tell whether they are more responsive to a comparable tax incentive or had more dividend income available to reallocate. Appendix Section B.3 reports the underlying log and dollar treatment–control and treatment-intensity estimates, together with both financial-exposure IV specifications for all four quartiles.

The treatment-intensity estimates address this distinction within the treated group. A 0.1 larger pre-reform spouse–owner tax incentive is associated with a roughly 22.5% larger post-reform decline in spouse dividend income and a 5.9-percentage-point larger decline in the probability of receipt. Owner dividends rise by about 2.6% for the same difference in the incentive, but couple dividends decline by about 9.7%. The direction of these changes is consistent with families scaling back income splitting when its tax advantage is removed, while only partly replacing spouse dividends with owner dividends. Before the reform, the spouse-dividend coefficients move toward zero; they turn sharply negative in 2018. The post-reform decline is therefore not a continuation of the earlier trend. The spouse PPML estimate is negative but imprecise; the evidence for an incentive gradient is clearer for log dividend income and receipt.

Across Q2–Q4, the estimated elasticity of spouse dividends with respect to the pre-reform incentive has no clear income gradient. The coefficients range from -2.81 to -2.15, while Q1 is much less precise. In dollars, the same 0.1 incentive difference is associated with progressively larger declines at higher family incomes. The large dollar losses at the top are therefore consistent with a similar proportional response applied to larger initial dividend amounts, rather than greater sensitivity to a comparable incentive. The decline in spouse dividend receipt across Q2–Q4 reinforces that this pattern involves whether families continue paying the spouse as well as how much they pay.

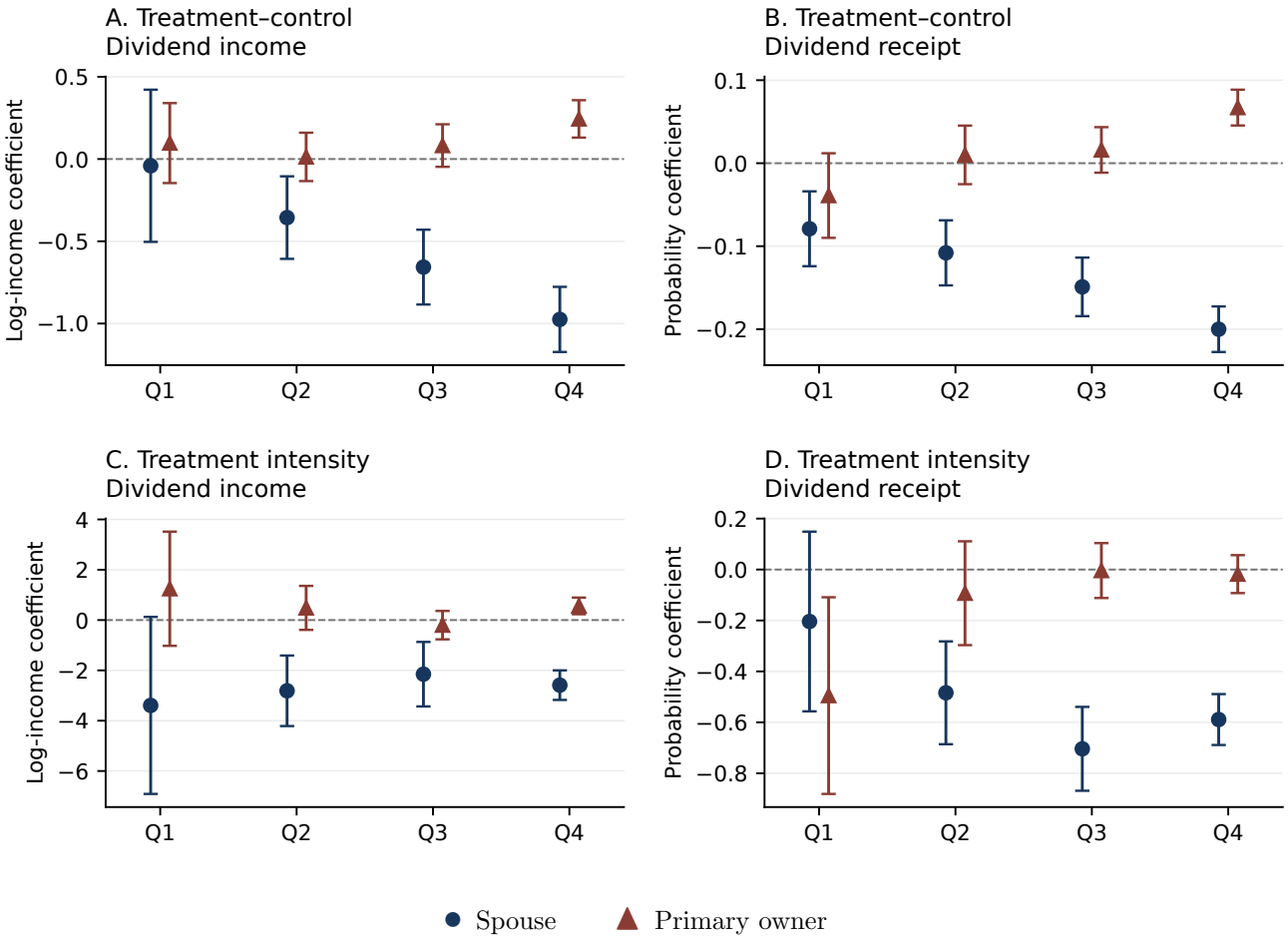
The financial-exposure analysis brings the amount initially allocated to the spouse into the comparison and measures the additional tax relative to baseline family income. In the owner-dividend estimation sample, a penalty equal to 1% of 2016 family income predicts a 0.071-log-point reduction in spouse dividends, or approximately 7%. The second-stage coefficient is -0.315, so a 10% reduction in spouse dividends induced by greater relative exposure is associated with roughly 3% more owner dividends. Instrumenting spouse dividend receipt gives a second-stage coefficient of -0.780, also pointing to an increase in owner dividends as spouse receipt falls. The response therefore appears on both the amount and receipt margins. Under the IV assumptions, the owner increase accompanies the reform-induced reduction in spouse dividends, although the records do not trace a particular payment directly from one spouse to the other.

Figure 6 shows that spouse dividends respond more strongly at higher family incomes to the

same potential tax penalty measured as a share of 2016 family income. A one-percentage-point increase in that ratio predicts declines of 0.031, 0.071, and 0.155 log points in spouse dividends in Q2, Q3, and Q4, respectively. These estimates use positive dividend observations; the separate first stage for dividend receipt also shows a larger decline from Q2 through Q4. When exposure is instead measured in thousands of dollars, the corresponding log-dividend coefficients are -0.033, -0.047, and -0.051 in Q2–Q4 (Appendix Table D2). The dollar responses are much closer, particularly in Q3 and Q4.[22] The similar treatment-intensity elasticities across Q2–Q4 are consistent with this interpretation, but that specification measures a tax-rate advantage rather than the total financial stakes. Under either IV version, an increase in owner dividends is clearest in Q4. Q1 is omitted from the main figure because its first stages are weak and its IV confidence intervals are wide.

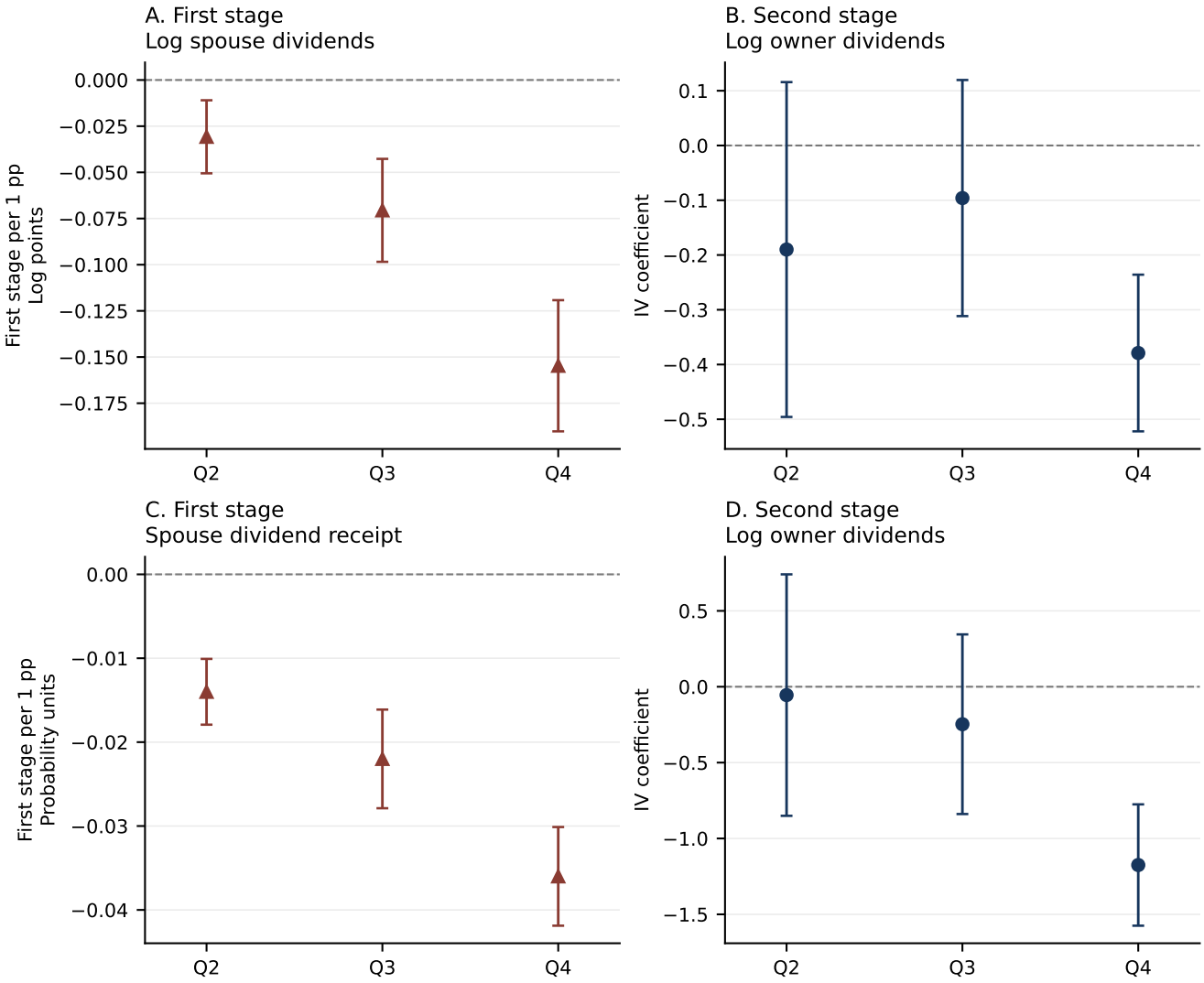
[22] Appendix D.1 uses income benchmarks to illustrate how a common penalty relative to income represents larger dollar amounts at higher incomes, contributing to differences in the estimated dividend response.

Figure 5: Dividend Responses by 2016 Family-Income Quartile



Notes: Points are estimates and bars are 95% confidence intervals based on standard errors clustered at the family level. Panels A and B show implementation-period treatment–control estimates for log dividend income and dividend receipt, respectively, for both spouses and primary owners. Panels C and D show the corresponding treatment-intensity estimates among treated families. Receipt coefficients are in probability units. Treatment-intensity coefficients retain their original scale. Log-income coefficients in Panel C are elasticities with respect to the spouse-to-owner net-of-tax-rate ratio. Blue circles denote spouses and red triangles denote primary owners. The coefficient scales differ across strategies.

Figure 6: Owner Dividend Responses by Family-Income Quartile Using Relative Financial Exposure



Notes: Panels A and B instrument log spouse dividends; Panels C and D instrument spouse dividend receipt. First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each second-stage outcome has its own sample and first stage. Red triangles denote first stages. Blue circles denote second-stage income outcomes. Q1 is omitted because of weak first stages and wide IV confidence intervals. Appendix Table B3 reports estimates for all four quartiles. Bars show 95% confidence intervals based on family-clustered standard errors. The two IV versions have different samples and second-stage units. Vertical scales differ across panels.

5.1.2 Labour Income

Employment income also adjusts, but the estimates do not show a simple replacement of spouse dividends with wages paid by the family corporation. In the treatment–control comparison, primary owners' total T4 income rises by 16.5%, while T4 income paid to spouses by the family corporation falls by 12.5% (Panel A of Appendix Table B4). There is no clear increase in owners' business-paid T4 income in the log specification, so the rise in their total earnings cannot be attributed solely to the family corporation. The decline in spouses' business-paid wages is notable because the TOSI reform did not directly change their taxation. One interpretation is that owners also reduce their use of wage payments as an income-splitting channel, perhaps because continuing these payments or restructuring the business immediately after the reform could draw scrutiny from the Canada Revenue Agency. This interpretation is consistent with the limited movement toward the ownership and employment thresholds associated with exemption (Appendix Figure A1). Higher owner employment income therefore accompanies the dividend reallocation, while the average response provides no evidence of a broad substitution toward spouse wages.

The treatment-intensity estimates show that wage adjustment varies with the pre-reform splitting incentive. Among treated families, stronger incentives are associated with higher total T4 income and higher wages paid to spouses by the family corporation, while neither owners' total T4 income nor their wages from the family corporation change significantly in the log specifications (Panel B of Appendix Table B4). The positive spouse business-wage response suggests that some families adjust compensation through this channel, even though these payments fall in the treatment–control comparison. The average decline therefore coexists with greater use of spouse wages among families with stronger initial incentives to split income.

The financial-exposure estimates provide clearer evidence of employment-income adjustment on the dividend-receipt margin (Panel C2 of Appendix Table B4). The log-dividend IV yields no statistically significant T4 response, while the dividend-receipt IV links reductions in spouse dividend receipt to higher total T4 income for both spouses and owners. This pattern is consistent with families replacing some of the income lost when spouse dividends cease with employment income received by either spouse. The source of these additional earnings is less clear, since total T4 income includes earnings from other employers and the estimates for wages paid by the family corporation are not statistically significant.

5.1.3 Business Activity

I examine corporate outcomes using the treatment-intensity specification, which compares firms within the treated group and therefore reduces reliance on comparisons between businesses with different ownership structures and industry composition.

Because couple dividends fall, families might leave some of the income no longer paid to spouses inside their corporations. The treatment-intensity estimates provide no clear evidence of this re-

sponse. Firms with stronger pre-reform incentives to split income do not show statistically detectable increases in retained earnings or assets, whether measured in logs or as annual growth (Appendix Table B10). Asset and retained-earnings growth also show no detectable response in Q3 or Q4, where stronger splitting incentives are associated with lower family taxable income (Appendix Table B11). The annual-growth estimates are imprecise, so they cannot rule out corporate retention. They do not, however, show a balance-sheet increase corresponding to the reduction in income reported by families. The relative-penalty IV estimates provide some evidence of asset adjustment. The log-dividend IV estimates are imprecise, while the receipt-based IV points to increases in assets and annual asset growth as spouse dividend receipt falls; the asset-level estimate is significant at the 10% level and the growth estimate at the 5% level. Neither IV specification shows a clear change in retained earnings (Appendix Table B12).

Professional corporations account for a large share of the treated sample and may have less need to acquire productive assets. This could help explain the lack of an asset response, but income left inside even a low-investment firm could still raise retained earnings. I also find no clear asset-growth response among non-professional corporations, as discussed in Section 6.1. The concentration of professional firms therefore does not by itself explain the absence of a detectable balance-sheet response. These outcomes do not establish where the income missing from family tax returns ultimately goes. One possibility is that firms spread the adjustment across several uses of funds, including other payments and business expenses, so that no single balance-sheet measure changes enough to be detected.

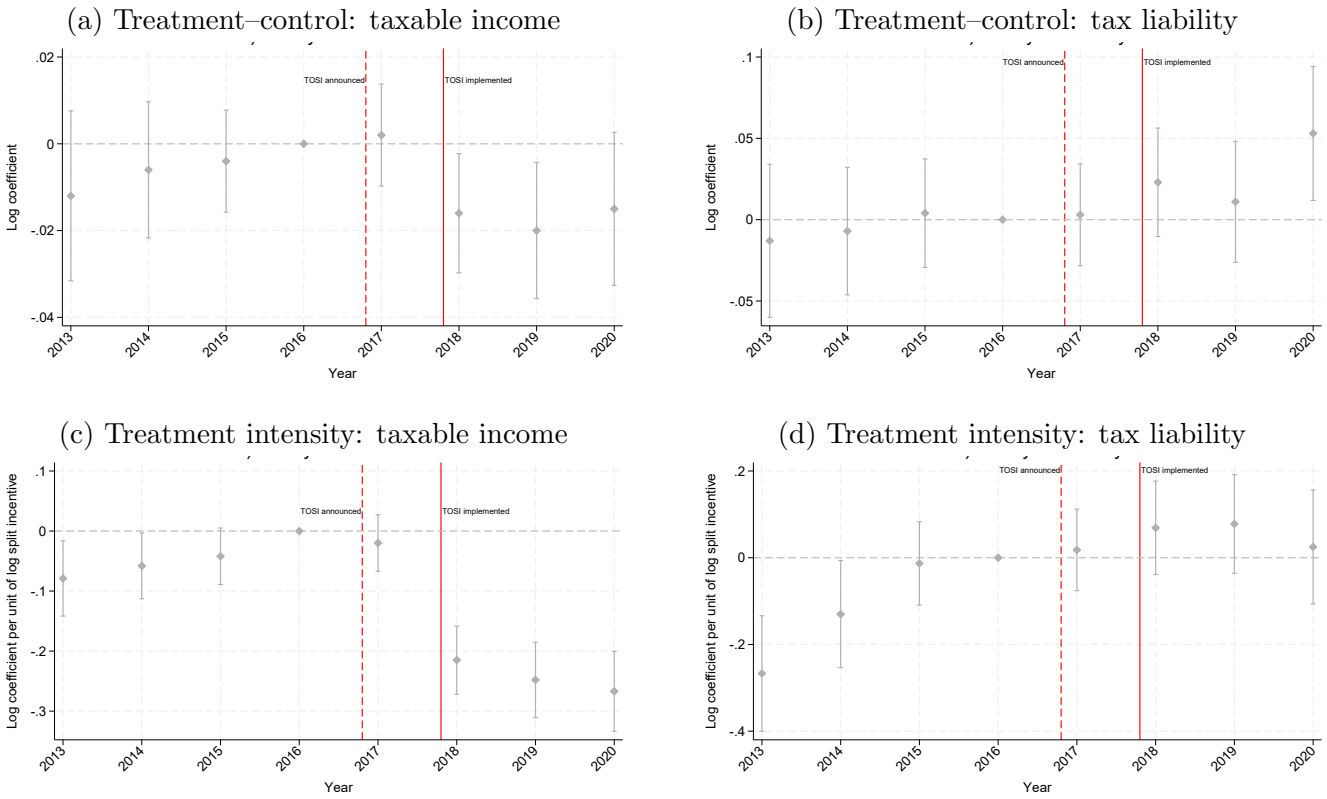
5.2 Family Taxable Income and Tax Liability

The fall in couple dividends raises two questions. How much income remains on the family's personal tax returns, and how much tax does the family pay after changing its income allocation? Family taxable income addresses the first question, and combined tax liability addresses the second. A family can report less taxable income yet owe more tax if income moves from a lower-taxed spouse to a higher-taxed owner. The main table reports log estimates; dollar estimates appear in the appendix.

The treatment–control event studies show little change in family taxable income in the 2017 announcement year and a modest decline after implementation (Figure 7). Tax liability rises, although the annual increase is statistically clear only in 2020. Over 2018–2020, the average estimates imply a 1.2% decline in family taxable income and a 3.3% increase in tax liability, both significant at the 10% level (Table 3). The corresponding dollar estimates are imprecise. Taxable income rises by $928 with a standard error of $1,875, and tax liability rises by $225 with a standard error of $755. The large decline in spouse dividends therefore produces much smaller average changes in the family's tax base and tax bill. This is consistent with much of the income no longer reported by spouses appearing elsewhere on family tax returns. The estimates do not provide an exact ac-

counting of those dollars because taxable income also includes other sources and incorporates the dividend gross-up, as described in Section 3.

Figure 7: Event-Study Estimates of Family Taxable Income and Tax Liability



Notes: Panels A and B report treatment–control estimates. Panels C and D report treatment-intensity estimates among treated families. All outcomes are in logs. The omitted reference year is 2016. Regressions include the controls and fixed effects in Equations (1) and (4), respectively. The dashed vertical line marks the 2017 announcement and the solid line marks the 2018 implementation of the TOSI reform. Bars show 95% confidence intervals based on standard errors clustered at the family level. The vertical scales differ across panels. Treatment-intensity coefficients retain their original scale.

Within the treated group, families with a larger pre-reform advantage from allocating dividends to the spouse show a larger decline in family taxable income. The treatment-intensity coefficient is -0.201 in logs, and the dollar estimate of -$40,197 is also statistically significant. The log tax-liability coefficient is positive at 0.150, although its dollar counterpart is imprecise. These results connect the family tax response to the strength of the tax-planning advantage that the reform removed. They are consistent with families moving some income to the higher-taxed owner while reducing the amount entering the personal tax base. The taxable-income event study supports this interpretation because its pre-reform coefficients move toward zero before turning sharply negative in 2018. The tax-liability path is less distinct, since it was already rising with the incentive before the reform.

Relative financial exposure combines the tax-rate change with the share of baseline family income previously routed through the spouse. In the log-dividend IV, the second-stage coefficient for family taxable income is 0.019 and is not statistically distinguishable from zero, while the tax-liability

Table 3: Family Taxable Income and Tax Liability Responses

	Log family taxable income	Log family tax liability
Panel A: Treatment–control (TC), implementation period		
Treated × Post	-0.012*	0.032*
	(0.007)	(0.017)
Observations	478,260	456,090
Panel B: Treatment intensity (TI), implementation period		
2016 incentive × Post	-0.201***	0.150***
	(0.025)	(0.055)
Observations	21,835	21,455

Panel C: Relative financial exposure

C1. Log spouse dividends instrumented

	Family taxable income regression		Family tax liability regression	
Dependent variable	Log spouse dividends	Log family taxable income	Log spouse dividends	Log family tax liability
First stage	-0.080***	—	-0.084***	—
	(0.007)		(0.007)	
Second stage	—	0.019	—	-0.156***
		(0.014)		(0.039)
First-stage F-statistic	125.4		125.4	
Observations	239,730		232,285	

C2. Spouse dividend receipt instrumented

	Family taxable income regression		Family tax liability regression	
Dependent variable	Spouse dividend receipt	Log family taxable income	Spouse dividend receipt	Log family tax liability
First stage	-0.027***	—	-0.028***	—
	(0.001)		(0.001)	
Second stage	—	0.222***	—	-0.528***
		(0.034)		(0.088)
First-stage F-statistic	397.7		397.7	
Observations	478,150		456,030	

Notes: Panels A and B report implementation-period TC and TI estimates. First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome has its own estimation sample. Family-clustered standard errors appear in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The two IV versions use different samples and second-stage units. Tax-liability estimates can also reflect direct effects of TOSI on taxes. Dollar TC and TI estimates appear in Appendix Table B13; dollar-penalty IV comparisons by family-income quartile appear in Appendix D.

coefficient is −0.156 and is statistically significant. Multiplying by the matching first stages implies approximately 0.15% lower taxable income and 1.3% higher tax liability for a penalty equal to 1% of baseline family income, although the taxable-income change remains imprecise. The receipt-based IV gives second-stage coefficients of 0.222 for family taxable income and −0.528 for tax liability. Using the matching first stages, the point estimates imply about 0.6% lower family taxable income and 1.5% higher tax liability for a penalty equal to 1% of baseline income. These results link greater financial exposure to a smaller personal tax base and higher taxes, while the increases in owner income show that much of the adjustment is absorbed through other reported income. The tax-liability estimate captures the overall response to exposure, including direct tax on spouse dividends still paid after the reform, and cannot be attributed entirely to dividend reallocation.

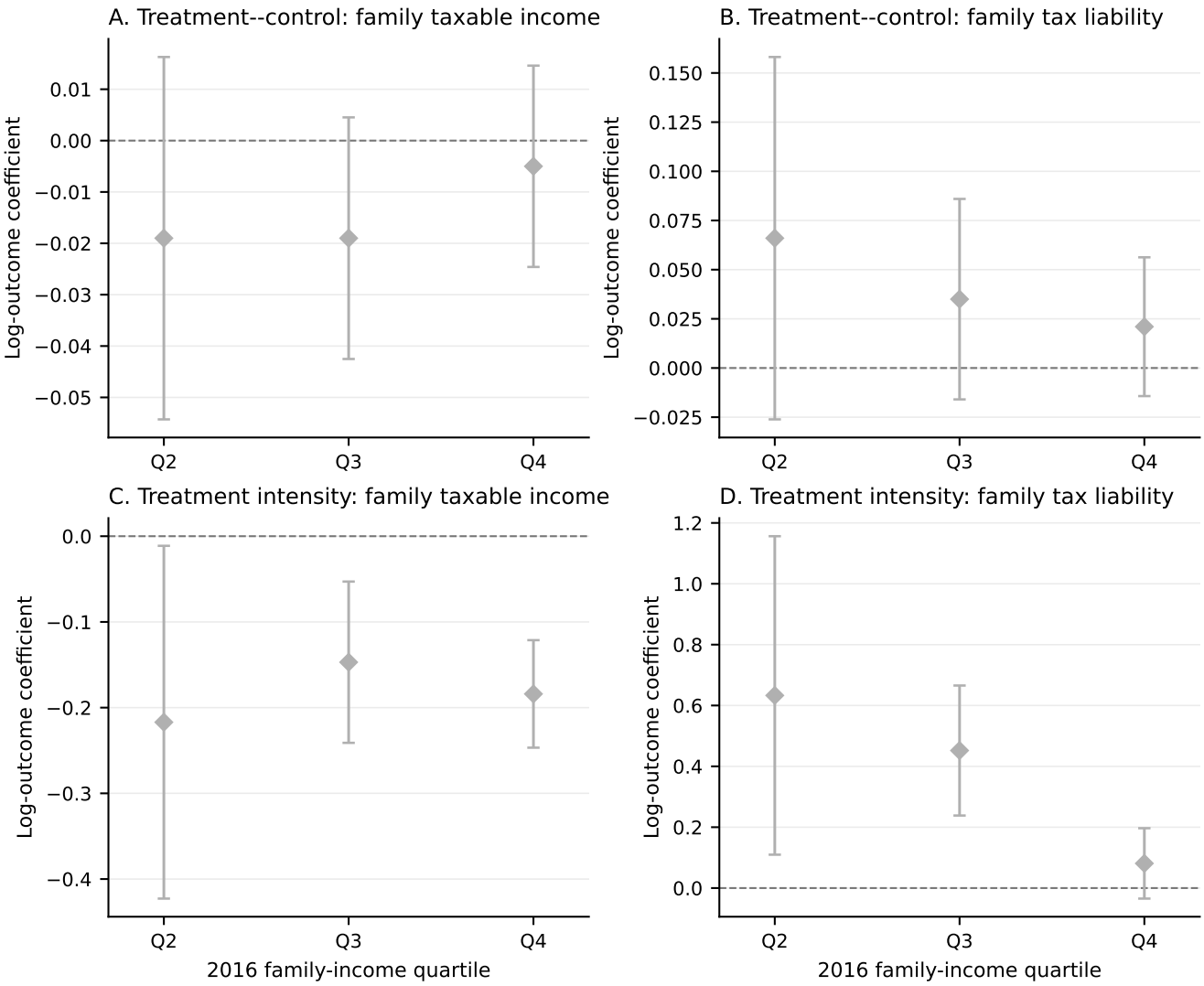
The average estimates conceal different patterns across 2016 family-income quartiles (Figure 8). The treatment–control comparison shows no statistically clear change in either log outcome in any quartile.[23] No quartile has a clear dollar increase in tax liability (Appendix Table B14). The larger spouse-dividend reductions among high-income families therefore do not translate into a correspondingly larger treatment–control tax increase.

Among treated families, stronger initial splitting incentives are associated with lower taxable income for Q2–Q4, but higher tax liability is clearest for Q2 and Q3. For a 0.1 difference in the incentive, the Q3 estimates imply about 4.6% higher tax liability and an increase of $901. For Q4, the same incentive difference implies about 1.8% lower taxable income and a decline of $5,797, while neither tax-liability estimate is statistically clear (Appendix Table B14). The Q1 estimates are imprecise. One possible explanation is that moving income to a higher-taxed owner raises taxes in the middle quartiles, whereas a larger reduction in income entering the personal tax base limits that increase for Q4. The treatment–control and treatment-intensity Q4 results answer different questions, since the former compares treated families with controls and the latter compares treated families with different initial incentives. Neither design establishes where income missing from personal returns ultimately goes, and the estimates do not establish statistically significant differences between quartiles.

The family-income results show why the larger Q4 response in spouse dividends does not imply that Q4 families experience the largest proportional tax increase. In the dividend-receipt IV, an equal induced decline in the probability of reporting spouse dividends is associated with larger proportional changes in family outcomes at more moderate incomes (Figure 9). The second-stage coefficients for log family taxable income are 0.458, 0.234, and 0.150 from Q2 to Q4; those for log tax liability are -1.058, -0.561, and -0.226. These second-stage patterns are nearly unchanged when the dollar penalty replaces the income-normalized penalty as the instrument (Appendix Table D5). With the relative penalty, the first-stage decline in dividend receipt grows toward Q4. Combining each first stage with its matching second stage implies broadly similar proportional declines in

[23]The dollar estimates for family taxable income are negative and significant at the 10% level for Q2 and Q3, despite imprecise log estimates. This may reflect large dollar reductions that are less apparent in proportional terms.

Figure 8: Family Taxable Income and Tax Liability Responses by 2016 Family-Income Quartile

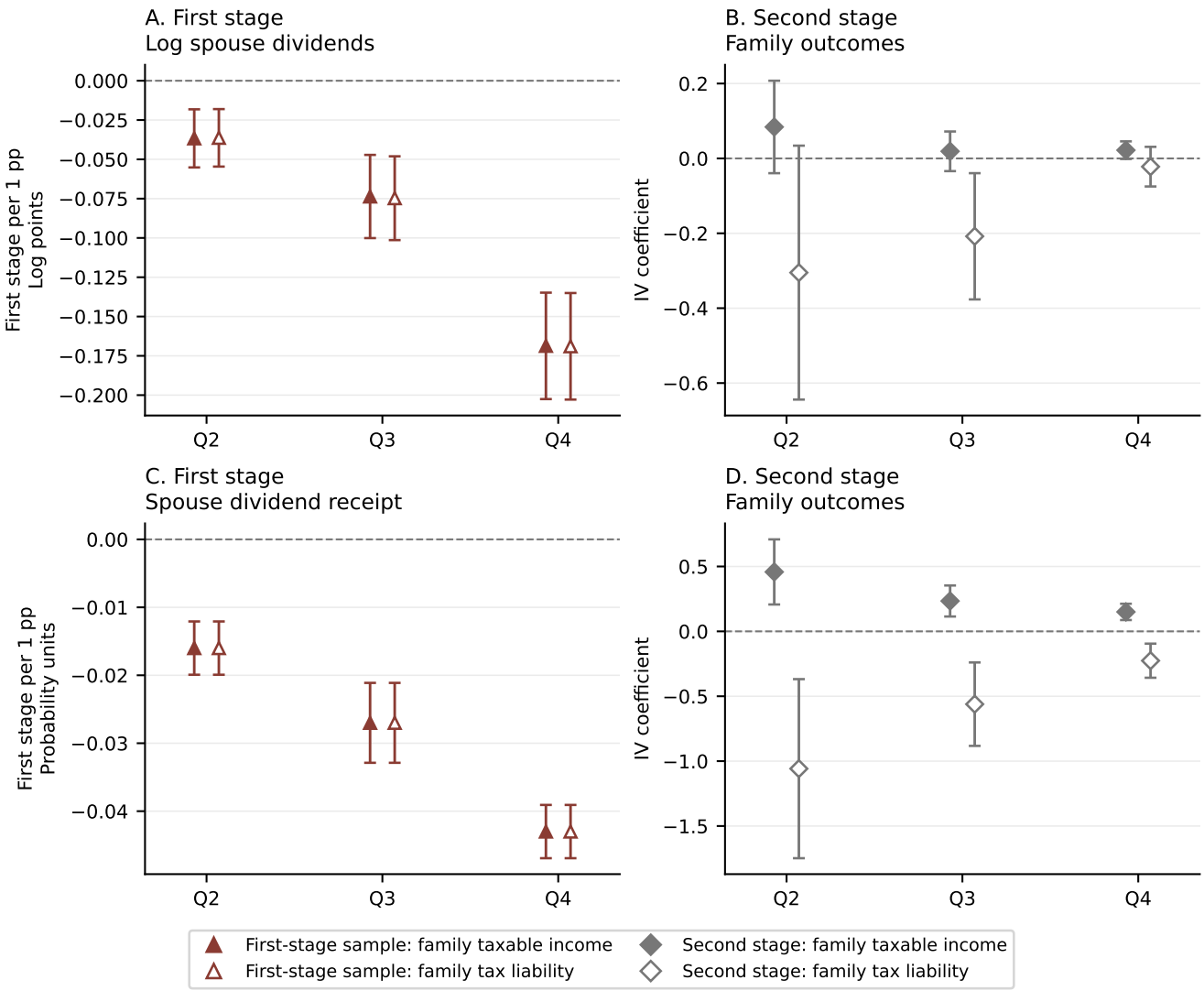


Notes: Panels A and B report treatment–control estimates for 2018–2020. Panels C and D report treatment-intensity estimates for the same period among treated families. All outcomes are in logs. Grey diamonds denote estimates, and bars show 95% confidence intervals based on standard errors clustered at the family level. Q1 is omitted from this figure. Appendix Table B14 reports estimates for all four quartiles. The vertical scales differ across panels. Treatment-intensity coefficients retain their original scale.

family taxable income across Q2–Q4 for a common penalty relative to baseline income, but larger proportional tax-liability increases for Q2 and Q3 than for Q4.[24] The income normalization therefore matters for the size of the spouse-dividend first stage without overturning the second-stage pattern in family income and taxes. The log-dividend IV provides less uniform evidence for family taxable income, whose decline is clearest for Q4; its tax-liability increase is clearest for Q2 and Q3. Together with the dividend results and the employment-income evidence in Appendix B.5, these estimates suggest that the form of adjustment differs across income groups, but they cannot trace each forgone spouse payment to its destination. TOSI can also raise tax liability directly on dividends still paid

[24]Part of this difference may reflect smaller initial tax bills relative to income. Appendix D.1 illustrates how a given income-relative penalty can represent a larger percentage of a family's initial tax bill.

Figure 9: Family Taxable Income and Tax Liability Responses by Family-Income Quartile Using Relative Financial Exposure



Notes: Panels A and B instrument log spouse dividends; Panels C and D instrument spouse dividend receipt. First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each second-stage outcome has its own sample and first stage. Red triangles denote first stages. Grey diamonds denote second-stage family outcomes. Filled and hollow markers distinguish the two outcome samples as indicated in the legend. Q1 is omitted because of weak first stages and wide IV confidence intervals. Appendix Table B15 reports estimates for all four quartiles. Bars show 95% confidence intervals based on family-clustered standard errors. The two IV versions have different samples and second-stage units. Tax liability can also be affected directly by TOSI. Vertical scales differ across panels.

to spouses, so the IV tax-liability coefficients should be read as part of the overall response to exposure rather than as the effect of stopping spouse dividends alone. Q1 remains imprecise.

Family-income rank therefore does not by itself identify the families with the greatest financial exposure to the reform.

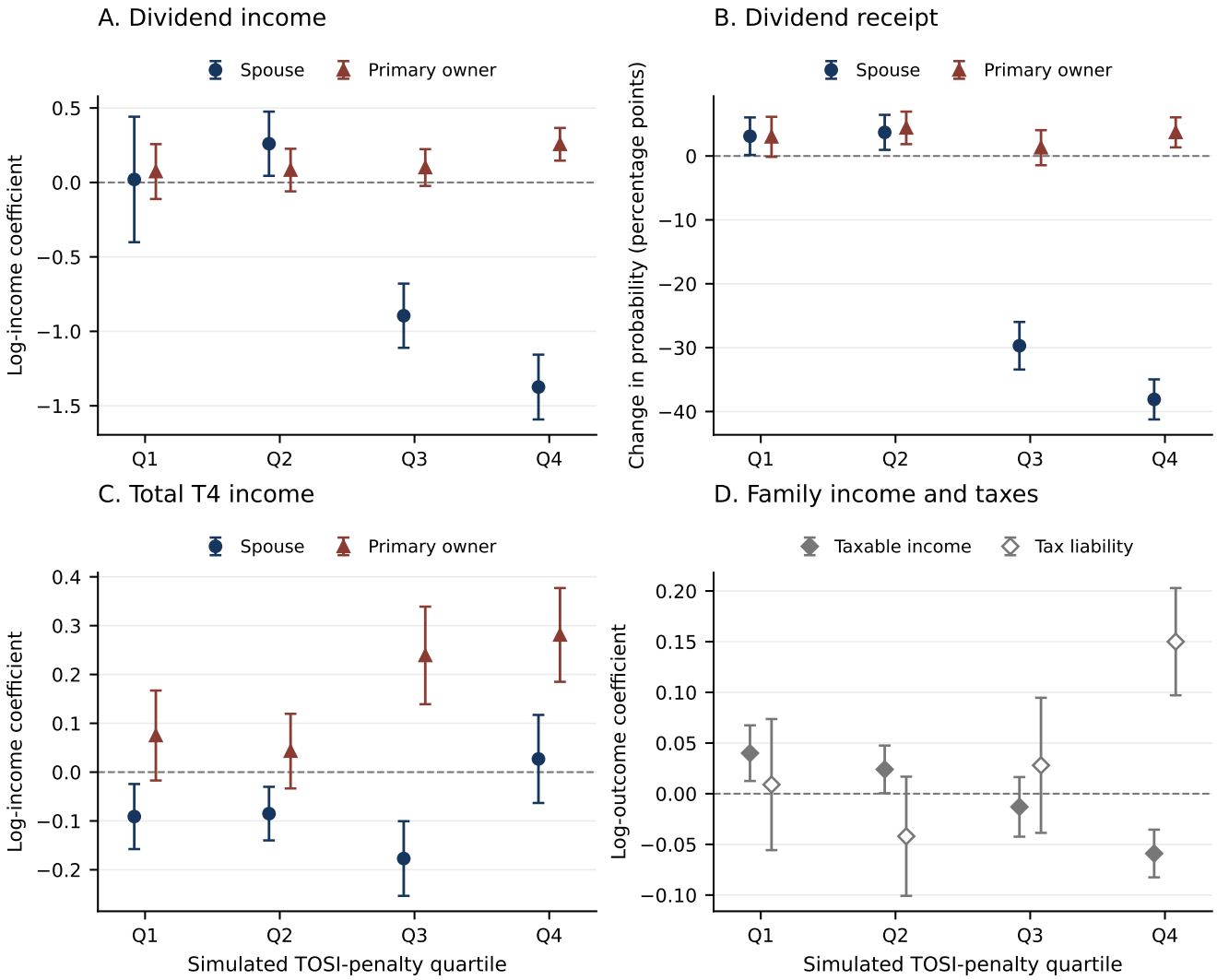
5.3 Heterogeneity by Tax Penalty

Two families with similar total income can face different tax penalties if they allocated different amounts of dividends to the spouse before the reform. I rank treated families by their average simulated dollar penalty for 2018–2020, calculated with 2016 income held fixed, and compare each quartile separately with the same control group. The ranking reflects both the change in tax treatment and the amount the family previously allocated to the spouse. Unlike the main IV instrument, this ranking uses the dollar penalty rather than the penalty as a share of family income. The resulting dividend responses differ sharply across quartiles (Figure 10). Spouse dividend income does not fall in Q1 and rises in Q2, but falls by 0.895 log points in Q3 and 1.374 log points in Q4, or roughly 59% and 75%, respectively. The probability that the spouse reports dividends falls by 29.7 percentage points in Q3 and 38.1 percentage points in Q4. The estimates therefore do not show a uniform reduction in spouse dividends among families classified as treated. The largest reductions occur where maintaining the pre-reform payment pattern would have been most costly.

The income and tax results show why ranking families by the amount at stake adds to the family-income analysis. Primary-owner dividends increase clearly only in Q4, while owners' total T4 income rises in Q3 and Q4. In Q4, family taxable income falls by 0.059 log points, yet tax liability rises by 0.150 log points, or about 16%. The dollar specification estimates a tax increase of approximately $4,840. Thus, income reported by the owner increases while the family's personal tax base contracts and its tax bill rises. This can occur if some income moves to a higher-taxed owner or if dividends still paid to the spouse face TOSI, even as less total income enters the personal tax base. The Q4 dollar estimate for taxable income is negative but imprecise, so the evidence for a decline in the tax base is stronger in proportional terms. Taxable income rises in Q1 and Q2 without a clear tax increase. In Q3, neither log family outcome changes clearly, although the dollar estimate shows a decline in tax liability of about $3,800. The clearest tax increase is therefore among families with the largest simulated penalties, a pattern that the ranking by family income alone did not reveal.

Employment-income results provide additional context. In the two highest penalty quartiles, owners' total T4 income rises and business-paid spouse T4 income falls, but there is no clear increase in wages paid to owners by the family corporation (Appendix Table B17). The event studies in Appendix Figures B3–B5 make the timing of the main results clearer. Spouse dividends in Q3 and Q4 had been moving toward the control group before the reform but fall sharply starting in 2018. Q4 family taxable income similarly reverses an earlier relative increase. These prior paths do not

Figure 10: Treatment–Control Responses by Simulated TOSI-Penalty Quartile



Notes: Treated families are ranked into quartiles by their average simulated TOSI penalty for 2018–2020, calculated using fixed 2016 income. Each treated quartile is compared separately with the same control group. Points report treatment–control implementation-period estimates for 2018–2020. Panels A, C, and D use log outcomes. Panel B reports changes in the probability of receiving dividends, expressed in percentage points. Blue circles denote spouses and red triangles denote primary owners. In Panel D, filled grey diamonds denote family taxable income and hollow grey diamonds denote family tax liability. T4 income includes employment income from all employers. Bars show 95% confidence intervals based on family-clustered standard errors. Outcome samples and vertical scales differ across panels. Dollar estimates appear in Appendix Table B18; detailed log and receipt estimates appear in Appendix Tables B16–B17.

explain the post-reform declines. The Q4 tax-liability path had already been rising, so the increase in taxes has a less distinct break at implementation. In Q1 and Q2, the positive spouse-dividend estimates are already visible in 2017 and change little in 2018; I therefore do not interpret them as an increase caused by implementation.

6 Business Type and Multiple-Business Ownership

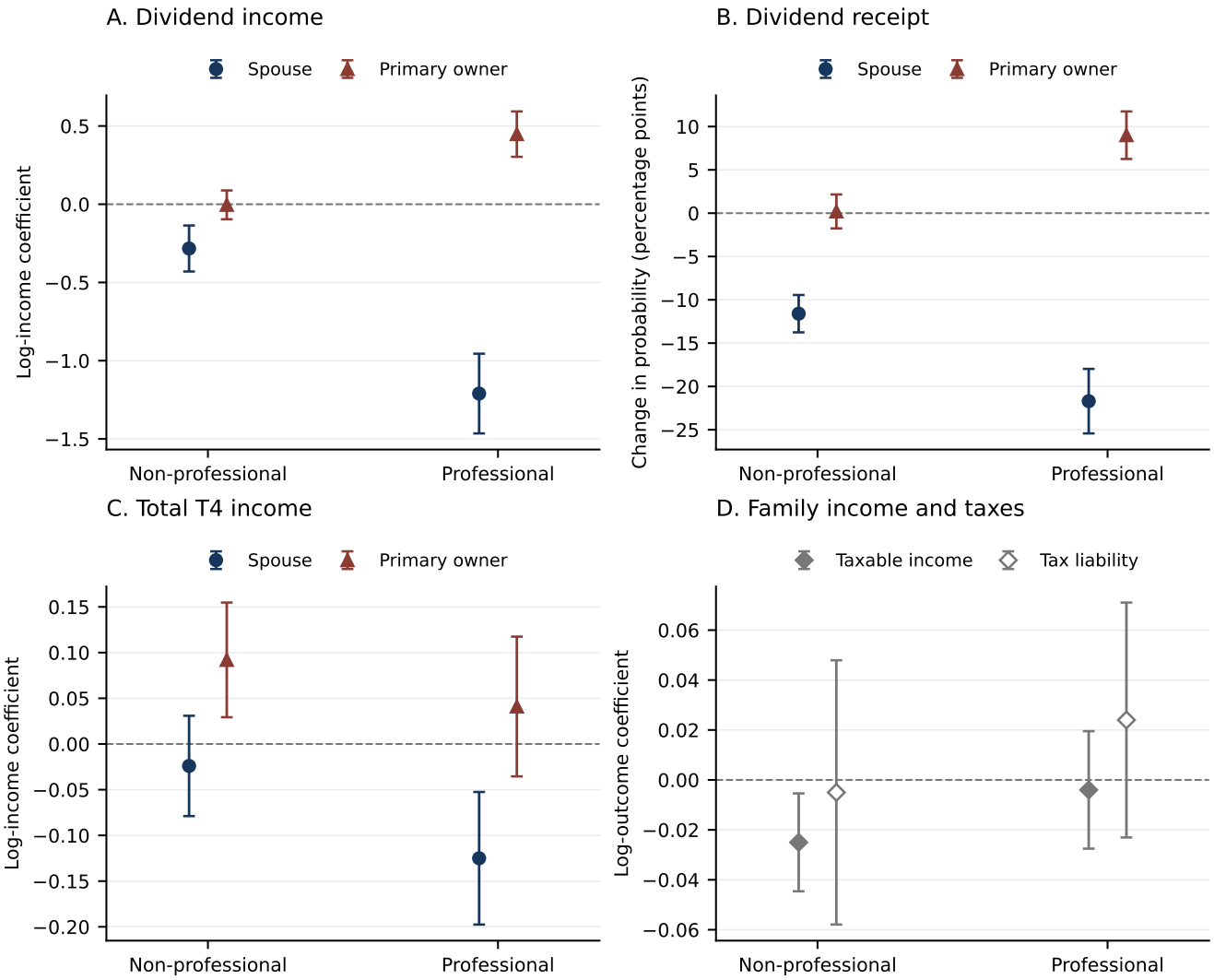
6.1 Professional and Non-Professional Corporations

Professional corporations account for a substantially larger share of treated families than of controls. Their ownership rules also differ, and the 10% excluded-shares exemption is unavailable to professional corporations. I therefore examine whether their concentration explains the pooled results. Figure 11 shows that the treatment–control estimate for spouse dividends is -1.210 log points among professionals, compared with -0.283 among non-professionals. The probability that a spouse receives dividends falls by 21.7 and 11.6 percentage points, respectively. Professionals contribute substantially to the large pooled decline, but the reform also reduces spouse dividends among non-professional families. The accompanying income changes differ. Owner dividends rise among professionals, whereas owner total T4 income rises among non-professionals. Spouse total T4 income falls among professionals. Thus, the response does not follow a single pattern of replacement across business types. The large professional dividend response occurs without a detectable decline in family taxable income, while non-professional families have a small decline of 0.025 log points. Neither group has a statistically detectable treatment–control change in family tax liability. These comparisons describe responses within the two business types; they do not establish that professional status caused the difference in their estimated responses. Appendix Table C1 reports the log, dollar, and dividend-receipt estimates.

The treatment-intensity estimates show that the pre-reform incentive to split income matters within both types of business. The spouse-dividend coefficients are -2.199 for non-professionals and -2.812 for professionals, both statistically significant. A stronger initial incentive is also associated with lower family taxable income in both groups. Owner dividends and family tax liability rise more clearly among professionals. The relative financial-exposure estimates in Figure 12 distinguish the amount of spouse dividends from the decision to report any. A penalty equal to 1% of baseline family income predicts a 0.143-log-point decline in spouse dividends among professionals and a 0.038-log-point decline among non-professionals in the family-income samples. Professionals also show a larger first-stage decline in dividend receipt. In the receipt-IV specification, greater relative exposure is associated with lower family taxable income and higher tax liability in both groups. In the log-dividend IV, taxable income falls most clearly among professionals, while tax liability rises in both groups. The response to relative exposure therefore extends to non-professional firms even though professional firms account for much of the average dividend decline. Appendix Tables C2, C3, and C4 report the underlying estimates.

Separating the two business types also helps assess whether the concentration of professionals explains the limited corporate response. Families with professional corporations may have less reason to acquire productive assets, but the treatment-intensity estimates show no clear asset-growth response among non-professional firms either. Retained-earnings growth has a positive coefficient among professionals that is significant at the 10% level, while the non-professional estimate is

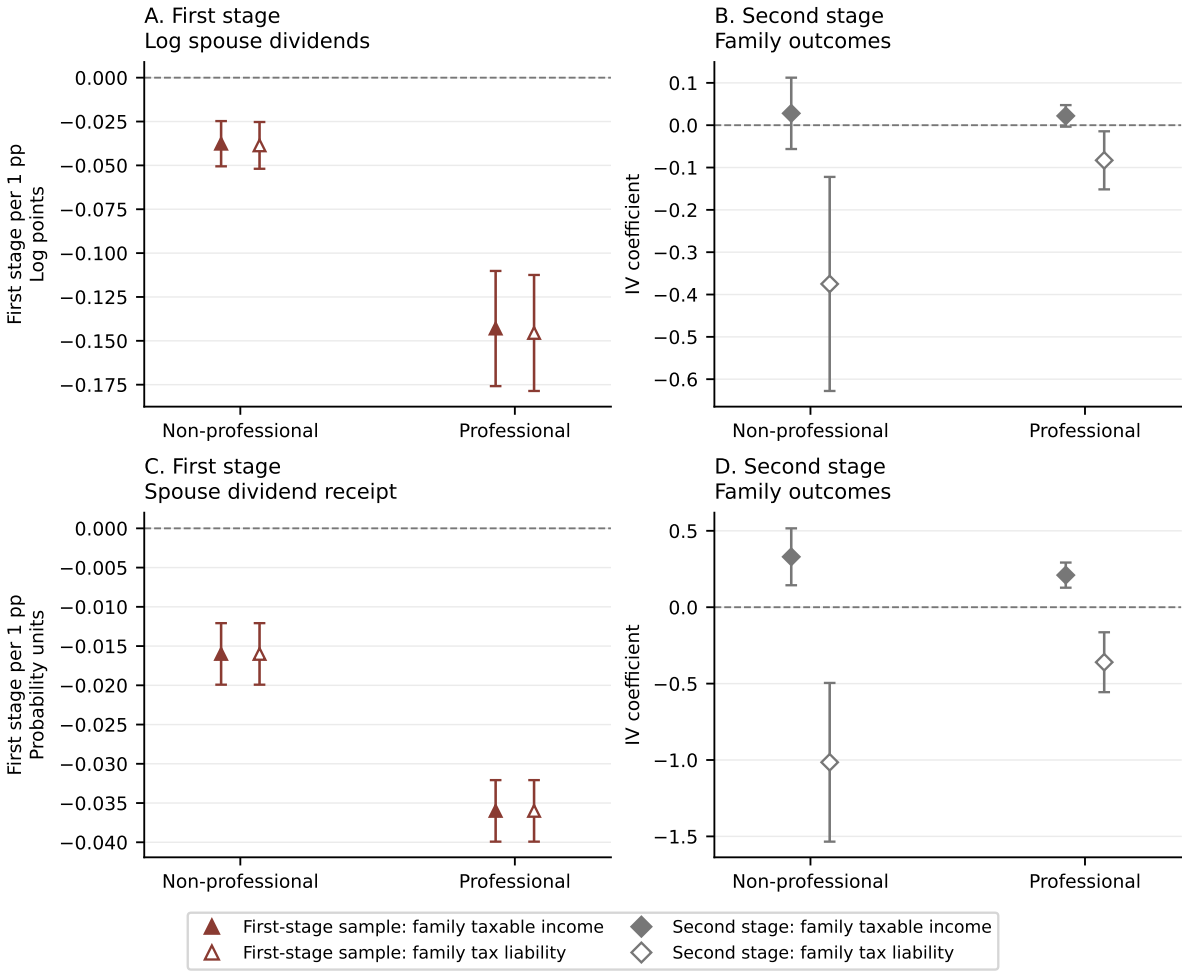
Figure 11: Treatment–Control Responses by Corporation Type



Notes: Coefficients compare implementation-period changes between treated and control families separately within non-professional and professional corporations. Panels A, C, and D use log outcomes; Panel B shows changes in dividend receipt in percentage points. Blue circles denote spouses and red triangles denote primary owners. In Panel D, filled grey diamonds denote family taxable income and hollow grey diamonds denote family tax liability. T4 income includes all employers. Bars show 95% confidence intervals based on family-clustered standard errors. Outcome samples and vertical scales differ. Appendix Table C1 reports numerical estimates in logs and dollars and for dividend receipt.

imprecise. The relative-penalty receipt-IV estimates show no clear change in the levels of assets or retained earnings for either group. Professional composition alone therefore does not explain the absence of a consistent balance-sheet response. These measures still cannot show where all income that leaves family tax returns ultimately goes. Appendix Table C5 reports the TI growth estimates and relative-penalty receipt-IV estimates for corporate assets and retained earnings.

Figure 12: Relative Financial-Exposure IV Estimates by Corporation Type



Notes: Panels A and B instrument log spouse dividends; Panels C and D instrument spouse dividend receipt. First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Red triangles denote first stages and grey diamonds denote second-stage family outcomes. Filled markers identify the taxable-income regression and hollow markers identify the tax-liability regression, including their separate first stages. Bars show 95% confidence intervals based on family-clustered standard errors. Tax liability can also be affected directly by TOSI. Appendix Tables C3 and C4 report the estimates.

6.2 Multiple-Business Ownership

The main sample restricts spouses to shares in one business so that their dividend income can be related more clearly to the family corporation. To see whether the response extends beyond this sample, I also study spouses who hold shares in multiple businesses. In the treatment–control comparison, their spouse dividends fall by 0.383 log points and their probability of receiving dividends falls by 7.6 percentage points, compared with 0.753 log points and 16.8 percentage points in the main sample. The fall in spouse-reported dividends therefore extends to families with more complex ownership structures, although the estimated decline is smaller. Owner dividend receipt rises in both samples, but owner dividend income rises significantly only in the main sample. Among multiple-business families, owner total T4 income increases, and T4 income paid to the spouse by the linked corporation falls. There is no statistically detectable change in family taxable income or tax liability for this group. Because personal tax returns combine dividends from all the corporations a person owns, I cannot establish which firm reduced the spouse's payment or where that income went. The differences between the two samples are descriptive and need not be caused by ownership of another business. Appendix Table C6 reports these estimates.

The treatment-intensity and relative financial-exposure results show why the main sample provides a clearer measure of tax responsiveness. For families with multiple businesses, the treatment-intensity estimate for log spouse dividends is negative but imprecise, at -0.465, although the estimate for dividend receipt remains negative and statistically significant. A stronger splitting incentive is also associated with higher family tax liability in this group, while the family taxable-income estimate is imprecise. The relative penalty predicts total spouse dividends less strongly than in the main sample. In the family-taxable-income sample, a penalty equal to 1% of baseline income predicts a 0.030-log-point decline in spouse dividends, compared with 0.080 in the main sample, and the reported first-stage F-statistic falls from 125.4 to 7.54. The receipt first stage is also smaller, although its reported F-statistic remains 28.37. The receipt IV points to higher family tax liability but no clear change in owner dividends or family taxable income. A penalty calculated from the family's 2016 income allocation may be less closely related to later total dividends when spouses receive payments from several firms. The dividend response extends beyond single-business families, while the single-business restriction makes it easier to connect the measured incentive, reported income, and linked corporation. Appendix Tables C7, C8, and C9 report the full results.

7 Conclusion

This paper examines how business-owning families responded to Canada's 2018 expansion of the Tax on Split Income rules, which restricted the tax advantage of paying dividends to family members. The reform sharply reduced dividends reported by affected spouses. In the treatment–control comparison, dividends and T4 income reported by owners rise at the same time, offsetting

much of the lost spouse dividend income. The decline in family taxable income is consequently far smaller than the decline in spouse dividends. The log estimates show a modest decline in family taxable income and a modest increase in tax liability, while the dollar tax-liability estimate is imprecise. Among treated families, stronger pre-reform incentives to split income are associated with larger adjustments in reported income. The financial-exposure estimates further connect reductions in spouse dividends to changes elsewhere on family tax returns. The main response is therefore a reallocation of reported income within the family, alongside a smaller change in the family tax base. This matters because the drop in spouse dividends, viewed alone, would greatly overstate the reduction in income entering the family's personal tax base. The offsetting rise in owner income shows that a rule restricting income splitting can reduce payments through the targeted channel without a comparable fall in reported family income.

The distributional results show why approaches based on family income and approaches based on financial exposure answer different questions. Higher-income families have larger dollar reductions in spouse dividends, but the estimated elasticity of spouse dividends with respect to the pre-reform splitting incentive is similar across income quartiles. The taxable-income literature often finds larger elasticities among high-income taxpayers (Gruber & Saez, 2002). Responsiveness also depends on opportunities to adjust reported income (Kopczuk, 2005). My estimate concerns spouse dividends rather than total taxable income, but it isolates a channel available to business-owning families at each income level. Their similar elasticities suggest that the larger taxable-income elasticities often found at the top may partly reflect the greater net gains from tax-planning at higher incomes rather than. a greater underlying behavioural responsiveness a given channel. Relative financial exposure measures a different dimension of the reform. For the same simulated tax penalty as a share of baseline income, spouse dividends and dividend receipt fall more at higher incomes. When I compare equal dollar penalties, the proportional decline in spouse dividends is much more similar across quartiles, especially in Q3 and Q4. Part of the relative-penalty gradient may therefore reflect the larger dollar stakes represented by a fixed share-of-income penalty at higher incomes, although these estimates do not quantify that contribution. A corresponding rise in owner dividends is clearest in the highest-income quartile. Yet the receipt-IV estimates imply broadly similar proportional declines in family taxable income across Q2–Q4 and larger proportional tax-liability increases in Q2 and Q3 than in Q4. The form of adjustment therefore changes across income groups even when exposure is measured relative to family income. Ranking families by the dollar penalty reveals that spouse dividends fall most sharply in the third and fourth penalty quartiles, and the fourth has the clearest decline in family taxable income and increase in tax liability. Families with the highest incomes do not always face the largest penalties, because the penalty depends on both the tax-rate change and the amount previously allocated to the spouse. The reform affects income splitting across the distribution, while income rank and exposure help explain different parts of the adjustment.

Business structure affects the magnitude of the response but does not confine it to one type

of family. Professional corporations account for much of the pooled decline in spouse dividends, yet dividends also fall among non-professional families and among spouses with shares in multiple businesses. I find little consistent evidence of changes in corporate assets or retained earnings in either professional or non-professional firms. The available corporate measures therefore do not show where income that leaves family tax returns ultimately goes or establish the effects on real activity or welfare. Changes in taxable income alone do not establish the welfare cost of tax avoidance (Chetty, 2009). Evaluating this type of rule requires following income reported by both family members, the family's tax liability, and the business, rather than relying only on the decline in payments to the targeted spouse.

This paper focuses on a restricted sample that allows me to trace income flows between spouses, business owners, and a single family corporation. Future work could relax these restrictions to study a broader range of business structures and assess how income splitting affects the progressivity of Canada's personal income-tax system. The limited evidence of changes in business outcomes also raises the question of whether firms outside my sample respond differently to restrictions on income splitting. Finally, because income splitting gave some business owners an additional way to reduce their tax liability, it would be valuable to test whether owners who used this channel respond differently to broader tax changes than owners who did not, or could not, split income within their families.

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A Data and Treatment Classification

A.1 Tax Simulation Grid

I construct a lookup grid for marginal tax rates using the Canadian Tax and Credit Simulator (CTaCS). The grid covers 2010–2021 and 13 provincial and territorial tax jurisdictions, varying own income, non-eligible dividend income, and spouse income for a married tax filer. The own-income and dividend grids contain \$1, values from \$1,000 to \$50,000 in \$1,000 steps, values from \$55,000 to \$150,000 in \$5,000 steps, values from \$160,000 to \$200,000 in \$10,000 steps, and values from \$225,000 to \$350,000 in \$25,000 steps. The spouse-income grid contains \$1, \$10,000 to \$50,000 in \$10,000 steps, and \$75,000, \$100,000, \$150,000, \$200,000, \$300,000, and \$500,000. Spouse dividend income is set to zero in the grid inputs. CTaCS first calculates total family tax at each grid point. I then rerun it after separately adding \$100 to own earnings and to non-eligible dividends. Each change in total family tax divided by \$100 gives the corresponding earnings or dividend marginal tax rate. I match the resulting rates to the income allocations used in the analysis.

A.2 Sample Construction and Representativeness

I construct the broader linked sample using the individual and corporate shareholder identifiers in the T2S50. When a shareholder is itself a corporation, I trace ownership through the corporate identifiers to identify the ultimate individual owners. I retain corporations owned directly by individuals or indirectly through a single intermediate corporation. This allows conventional holding-company–operating-company structures but excludes ownership chains requiring more than one intermediate corporation to identify the individual owner.

I assess the completeness of these linkages using an administrative measure of the total number of corporations associated with each individual, either directly or indirectly. I compare this count with the number of corporations I independently identify through the ownership records and exclude individuals for whom the two counts differ. This check reduces the risk of treating an individual as associated with only one corporation when additional ownership interests are missing from the constructed linkages.

The main analysis imposes further restrictions on this broader sample. The spouse must hold shares in only one business, and that corporation must be owned directly by individuals and have no ownership links to other corporations. The primary owner may hold shares in additional businesses. Table A1 reports the cumulative sample restrictions, and Table A2 compares the final analysis sample with the broader population of business-owning couples.

Table A1: Construction of the Analysis Sample

	Cumulative sample restriction	Couples
0	Linked married or common-law couples with observed business ownership	1,114,185
1	Require treatment status to remain constant across observed years	1,112,625
2	Require the tax-rate variables used in the analysis to be observed and valid	1,078,135
3	Require observed share ownership	397,585
4	Exclude observations with incomplete corporate ownership links	369,765
5	Exclude families that cross a relevant age threshold	350,510
6	Require the spouse to hold shares in only one business	264,060
7	Require the family to own all common shares of that business	220,100
8	Require a balanced analysis panel	94,295
9	Retain families classified as likely exposed or likely exempt	63,790

Notes: Restrictions are cumulative and are applied in the order shown. One spouse identifies one couple and family in the analysis. Counts are rounded in accordance with the disclosure requirements applied to the administrative data. Additional outcome-specific restrictions may cause estimation samples to differ across regressions.

Table A2: Analysis Sample Compared with the Broader Population

	Broader population (up to N = 1,078,135)	Analysis sample (up to N = 63,790)
Panel A: Demographics		
Spouse age (years)	50.2 (13.2)	49.1 (8.4)
Primary-owner age (years)	51.6 (13.1)	49.2 (7.7)
Panel B: Personal and family income		
Spouse receives dividends (%)	34.8 (47.6)	52.5 (49.9)
Spouse dividend income	9,287 (41,124)	16,359 (31,643)
Primary-owner dividend income	20,674 (101,922)	18,545 (40,376)
Spouse T4 income	40,522 (80,356)	43,862 (59,772)
Primary-owner T4 income	52,018 (138,820)	49,372 (85,537)
Family taxable income	152,921 (278,060)	141,619 (138,017)
Family tax liability	31,703 (100,266)	26,832 (53,238)
Panel C: Ownership, employment, and business type		
Spouse common-share ownership (%)	37.96 (17.84)	45.62 (12.01)
Spouse preferred-share ownership (%)	12.80 (28.20)	9.72 (26.07)
Spouse employed by the family corporation (%)	19.1 (39.3)	39.7 (48.9)
Likely service business (%)	10.5 (30.7)	34.1 (47.4)
Professional corporation (%)	3.1 (17.3)	9.1 (28.7)
Panel D: Business characteristics and outcomes		
Business age (years)	10.8 (8.1)	11.1 (6.6)
Labour units	5.66 (22.74)	4.25 (9.21)
Gross profits	201,763 (1,342,287)	213,789 (406,348)
Total assets	875,169 (7,402,883)	553,480 (1,180,901)
Retained earnings	395,333 (4,217,809)	242,901 (606,632)

Notes: The table reports 2016 means, with standard deviations in parentheses. The broader population contains business-owning spouses in the linked administrative records before the restrictions used to construct the analysis sample. The analysis sample contains families classified as either likely exposed to or likely exempt from the TOSI reform. Monetary values are in Canadian dollars and are rounded to the nearest dollar. Sample sizes vary across variables because ownership, payroll, and financial-statement information is not available for every observation. The maximum sample reported here is based on observations with nonmissing information in the summary-statistics file and therefore differs from the unrestricted linked-data count reported in Table A1. Counts are subject to the rounding requirements applied to the administrative data.

A.3 Stability of the Exposure Classification

Table A3: Effect of Requiring a Fixed Exposure Classification

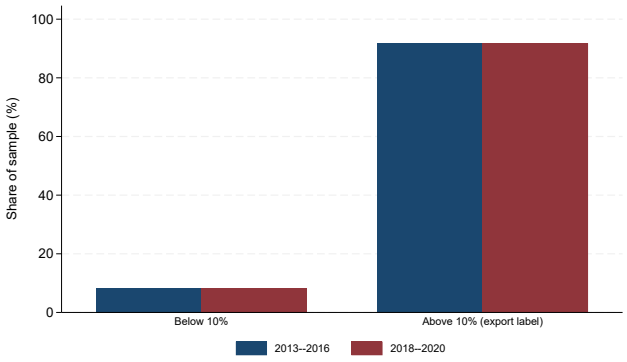
	Potential treated	Control	Combined
Before fixed-classification requirement	3,700	60,755	64,460
After fixed-classification requirement	3,030	60,755	63,790
Approximate number excluded	670	0	670
Share excluded	18.1%	0.0%	1.0%

Notes: The first row classifies spouses without requiring their ownership and employment classifications to remain fixed throughout the sample period. The second row requires these classifications to remain fixed across the pre- and post-reform periods. Counts are independently rounded in accordance with the disclosure requirements applied to the administrative data and may therefore not sum exactly.

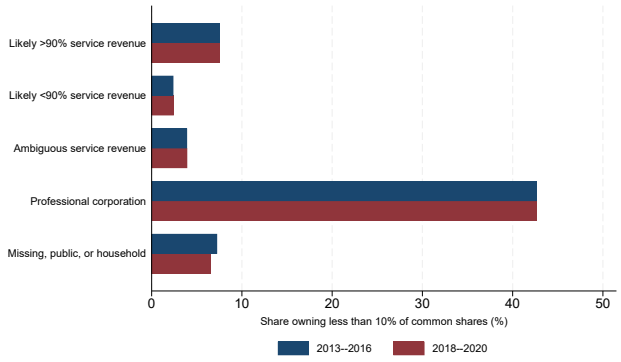
A.4 Changes Along the Observable TOSI Exemption Margins

Figure A1 examines whether spouses adjusted their common-share ownership or employment in response to the TOSI reform. The sample consists of spouses eligible to receive dividends, and exposure is defined only by the age-based rules. The sample therefore does not condition on spouses remaining on the same side of the ownership or employment criteria used in the main treatment classification.

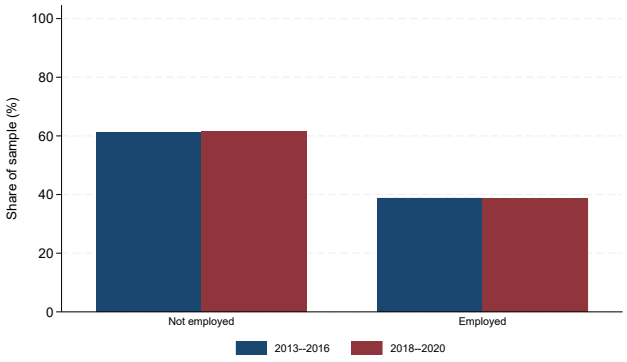
Figure A1: Changes Along the Observable TOSI Exemption Margins



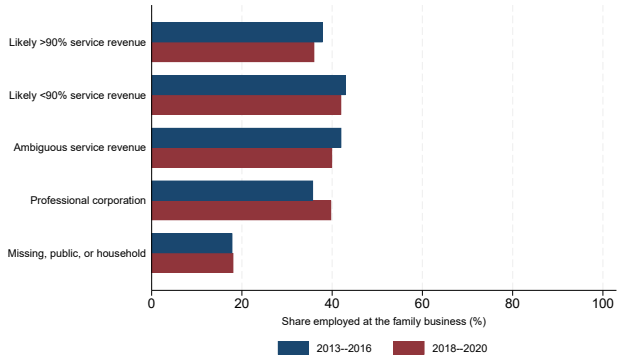
(a) Common-share ownership



(b) Common-share ownership by business category



(c) Employment in the family corporation



(d) Employment by business category

Notes: The sample consists of spouses eligible to receive dividends. Age-exposed and age-exempt spouses are classified using only the age-based rules and without conditioning on their ownership or employment choices. Panels A and B compare spouses' common-share ownership during 2013–2016 with ownership during 2018–2020. Panel A reports the shares below and at or above the 10% threshold. Panel B reports the share below the threshold by business category. Panels C and D compare employment in the family corporation during 2013–2016 with employment during 2018–2020, overall and by business category. All comparisons are descriptive.

A.5 Alternative Exposure Definitions

The two alternative definitions retain the primary owner's age restriction but change how spouses are classified. A3 compares spouses whose common-share ownership remains below 10% with spouses whose ownership remains at or above 10%, without using the employment criterion. A3B makes the same comparison among spouse-year observations with no recorded employment in the family corporation. Under both definitions, spouse dividend income and dividend receipt decline after implementation, while primary-owner dividend income rises. The estimated decline in log spouse dividends is similar to the main estimate under A3 and somewhat smaller under A3B. The positive estimate for family tax liability in the main specification is not statistically significant under either alternative. Because employment in A3B is observed each year and may itself respond to the reform, I treat that specification as a sensitivity check.

Table A4: Income Responses under Alternative Exposure Definition A3

	Dividend income			Dividend receipt	T4 income		T4 from family business	
	Log	Raw	PPML	Indicator	Log	Raw	Log	Raw
Panel A: Spouse								
Treated × 2017	0.091***	5,244***	0.118***	0.014***	-0.038***	-893	0.003	-1,640**
	(0.025)	(586)	(0.016)	(0.005)	(0.014)	(1,272)	(0.017)	(704)
Treated × Post	-0.735***	-14,201***	-0.899***	-0.159***	-0.026*	-3,865***	0.029	-4,100***
	(0.049)	(711)	(0.037)	(0.007)	(0.015)	(809)	(0.020)	(901)
Observations	248,530	492,680	365,535	492,680	389,025	492,680	222,280	222,475
Panel B: Primary owner								
Treated × 2017	-0.030	421	-0.018	-0.014**	0.030**	1,588*	-0.005	762
	(0.028)	(1,611)	(0.038)	(0.006)	(0.014)	(901)	(0.017)	(1,209)
Treated × Post	0.196***	9,521***	0.189***	0.040***	0.105***	8,887***	0.030	8,085***
	(0.033)	(990)	(0.024)	(0.006)	(0.016)	(1,316)	(0.021)	(1,714)
Observations	265,565	492,680	376,975	492,680	386,295	492,680	183,055	183,245

Panel C: Couple outcomes

	Taxable income		Tax liability	
	Log	Raw	Log	Raw
Treated × 2017	0.006	5,668**	-0.008	934
	(0.006)	(2,475)	(0.013)	(979)
Treated × Post	-0.011*	1,405	0.018	1,508*
	(0.006)	(1,985)	(0.014)	(900)
Observations	492,205	492,680	469,850	492,680

Notes: A3 compares spouses whose common-share ownership remains below 10% with spouses whose ownership remains at or above 10%, retaining the age restriction and without using employment to classify exposure. Post covers 2018–2020. Raw outcomes are in dollars; receipt is a zero-one indicator. PPML uses dividend income in levels and retains zero outcomes. Standard errors are in parentheses; observation counts are reported below each panel. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A5: Income Responses under Alternative Exposure Definition A3B

	Dividend income			Dividend receipt	T4 income	
	Log	Raw	PPML	Indicator	Log	Raw
Panel A: Spouse						
Treated × 2017	0.127***	3,401***	0.099***	0.004	-0.047**	-27
	(0.035)	(707)	(0.023)	(0.007)	(0.020)	(2,150)
Treated × Post	-0.624***	-10,723***	-1.002***	-0.128***	-0.053***	-3,706***
	(0.067)	(848)	(0.056)	(0.009)	(0.020)	(1,261)
Observations	112,655	226,090	163,585	226,090	156,615	226,090
Panel B: Primary owner						
Treated × 2017	-0.009	1,295	0.001	-0.006	0.024	1,953
	(0.036)	(2,690)	(0.055)	(0.007)	(0.021)	(1,190)
Treated × Post	0.113***	5,500***	0.102***	0.035***	0.108***	8,591***
	(0.042)	(1,126)	(0.029)	(0.008)	(0.023)	(1,483)
Observations	127,270	226,090	174,615	226,090	153,380	226,090

Panel C: Couple outcomes

	Taxable income		Tax liability	
	Log	Raw	Log	Raw
Treated × 2017	-0.004	3,596	-0.026	143
	(0.007)	(3,896)	(0.017)	(1,552)
Treated × Post	-0.013*	-1,646	-0.002	-464
	(0.007)	(1,939)	(0.018)	(812)
Observations	225,865	226,090	217,740	226,090

Notes: A3B applies the A3 ownership comparison only to spouse-year observations with no recorded employment at the family corporation. Family-business T4 outcomes are omitted. Post covers 2018–2020. Raw outcomes are in dollars; receipt is a zero-one indicator. PPML uses dividend income in levels and retains zero outcomes. Standard errors are in parentheses; observation counts are reported below each panel. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.6 Unbalanced Panel

The dividend-income and receipt estimates remain close to the balanced-panel results. The spouse family-business T4 estimate is smaller and no longer statistically clear in the unbalanced panel. Table A6 reports the comparison.

Table A6: Income Responses in the Balanced and Unbalanced Panels

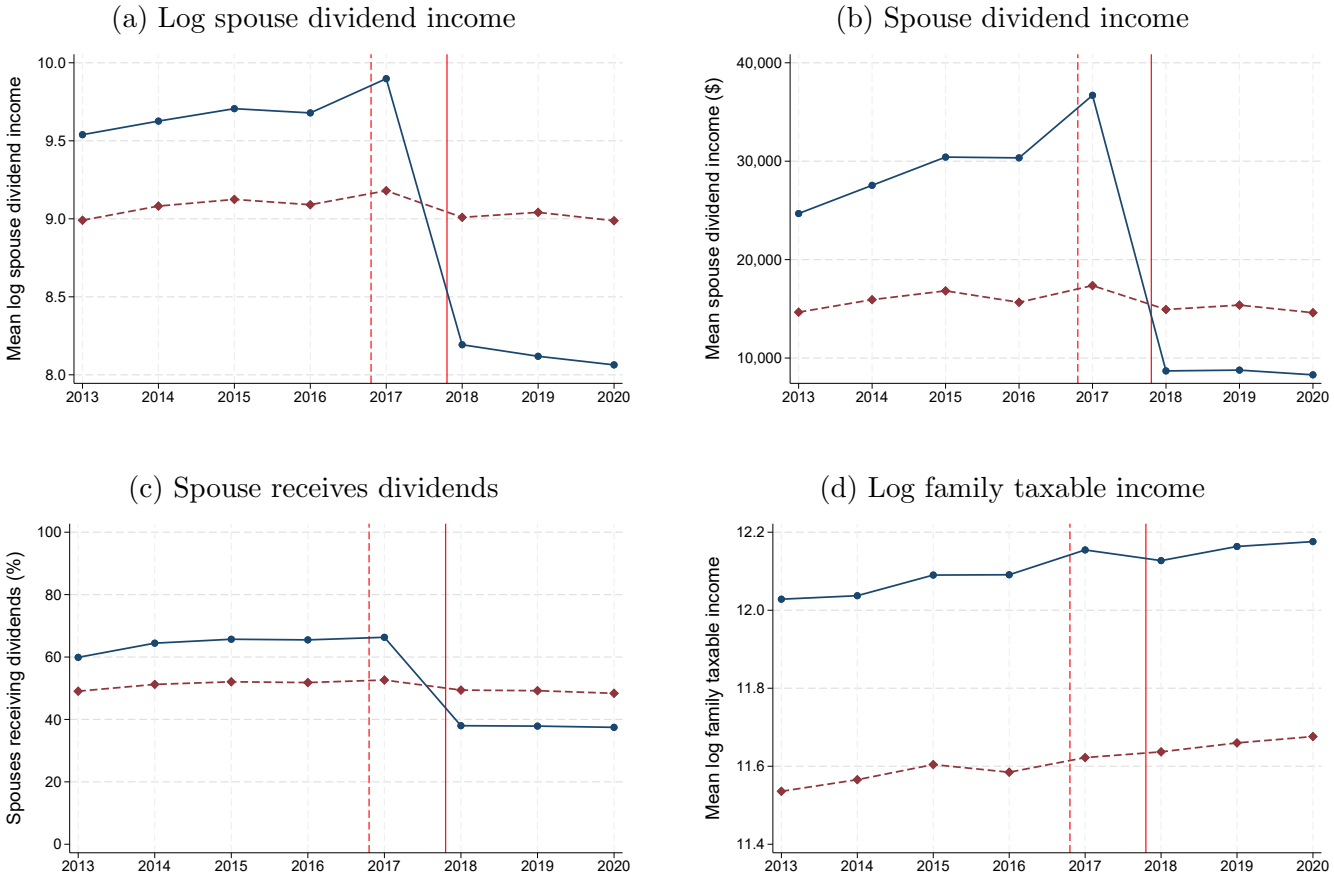
	Balanced panel	Unbalanced panel
A. Dividends		
Spouse dividend income	-0.753***	-0.739***
	(0.066)	(0.047)
Observations	239,795	414,920
Primary-owner dividend income	0.134***	0.124***
	(0.037)	(0.029)
Observations	255,870	451,240
Spouse receives dividends	-0.168***	-0.168***
	(0.009)	(0.006)
Observations	478,735	904,315
Primary owner receives dividends	0.029***	0.032***
	(0.007)	(0.006)
Observations	478,735	904,315
B. Employment income		
Spouse T4 income	-0.079***	-0.088***
	(0.020)	(0.014)
Observations	377,555	692,140
Primary-owner T4 income	0.153***	0.143***
	(0.024)	(0.017)
Observations	374,695	683,610
Spouse T4 from family business	-0.133***	-0.062
	(0.047)	(0.041)
Observations	215,465	356,555
Primary-owner T4 from family business	-0.040	-0.051
	(0.051)	(0.054)
Observations	177,420	288,940
C. Family outcomes		
Family taxable income	-0.012*	-0.006
	(0.007)	(0.005)
Observations	478,260	902,805
Family tax liability	0.032*	0.027**
	(0.017)	(0.012)
Observations	456,090	853,635

Notes: TC implementation-period estimates for 2018–2020. Income outcomes are in logs and dividend receipt is a zero-one indicator. Family-clustered standard errors are in parentheses. Observations are rounded as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Additional Results

B.1 Descriptive Income Trends

Figure B1: Spouse Dividends and Family Taxable Income Around the TOSI Reform



Notes: The figure reports annual means separately for families classified as likely exposed and likely exempt. The solid blue line with circles represents likely exposed families, while the dashed red line with diamonds represents likely exempt families. Panel A reports mean log spouse dividend income among spouses reporting positive dividends. Panel B reports mean spouse dividend income in dollars, including zero outcomes. Panel C reports the percentage of spouses receiving dividends. Panel D reports mean log family taxable income. The dashed vertical line marks the July 2017 announcement, and the solid vertical line marks the January 2018 implementation of the expanded TOSI rules.

B.2 Average Dividend Responses in Dollars and PPML

B.3 Dividend Responses by Family Income

Table B1: Dividend Responses in Dollars and PPML

	Dividend income ($)			PPML dividend income	
	Spouse	Primary owner	Couple	Spouse	Primary owner
Panel A: Treatment–control, 2018–2020					
Treated × Post	-12,391***	7,124***	-5,267***	-1.061***	0.125***
	(815)	(1,149)	(1,313)	(0.052)	(0.026)
Observations	478,735	478,735	478,735	353,100	364,340
Panel B: Treatment intensity, 2018–2020					
2016 incentive × Post	-59,999***	18,989***	-41,010***	-0.271	0.249***
	(3,036)	(4,310)	(4,766)	(0.268)	(0.089)
Observations	21,835	21,835	21,835	17,015	19,670

Notes: This table supplements Table 2. Dollar coefficients and standard errors are measured in dollars. PPML includes zero dividend outcomes; its coefficients are on the log-link scale. Panel A compares treated and control families. Panel B reports coefficients on the fixed 2016 tax incentive interacted with the post-reform indicator among treated families. Standard errors clustered at the family level appear in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B2: Dividend Responses by 2016 Family-Income Quartile

	Q1	Q2	Q3	Q4
Panel A. Treatment–control, log dividends				
Spouse dividend income	-0.041	-0.356***	-0.657***	-0.975***
	(0.236)	(0.128)	(0.116)	(0.101)
Observations	32,775	62,030	72,345	70,650
Primary-owner dividend income	0.097	0.013	0.082	0.244***
	(0.124)	(0.075)	(0.066)	(0.058)
Observations	35,120	66,110	76,855	75,765
Panel B. Treatment intensity, log dividends				
Spouse dividend income	-3.393*	-2.812***	-2.151***	-2.587***
	(1.795)	(0.715)	(0.654)	(0.300)
Observations	215	1,190	2,495	7,035
Primary-owner dividend income	1.245	0.484	-0.202	0.563***
	(1.158)	(0.446)	(0.288)	(0.169)
Observations	520	1,890	3,615	9,075
Panel C. Treatment–control, dividends in dollars				
Spouse dividends	-2,015***	-4,569***	-8,315***	-20,099***
	(715)	(810)	(910)	(1,561)
Primary-owner dividends	187	500	2,400	13,112***
	(1,185)	(1,363)	(1,666)	(2,175)
Panel D. Treatment intensity, dividends in dollars				
Spouse dividends	-7,421	-18,663***	-39,811***	-74,919***
	(4,713)	(4,746)	(4,078)	(4,603)
Primary-owner dividends	-6,630	9,433	15,327**	26,684***
	(10,598)	(6,554)	(6,560)	(6,310)

Notes: Quartiles use family income in 2016. All coefficients cover 2018–2020. Panels A and C report TC estimates; Panels B and D report TI estimates among treated families. Panel B retains the original coefficient scale, so its log-income coefficients are elasticities with respect to the spouse-to-owner net-of-tax-rate ratio. Panel D reports dollar changes for a 0.1 increase in that fixed incentive. Log and dollar specifications can use different samples. Standard errors clustered at the family level are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B3: Owner Dividend Responses by Family-Income Quartile Using Relative Financial Exposure

	Q1	Q2	Q3	Q4
Panel A. Log spouse dividends instrumented				
Owner dividends				
First stage	-0.010	-0.031***	-0.071***	-0.155***
	(0.010)	(0.010)	(0.014)	(0.018)
Second stage	-0.226	-0.190	-0.096	-0.379***
	(0.648)	(0.156)	(0.110)	(0.073)
Observations	26,040	52,435	60,555	59,820
First-stage F-statistic	3.06	15.15	30.19	95.44
Panel B. Spouse dividend receipt instrumented				
Owner dividends				
First stage	-0.001	-0.014***	-0.022***	-0.036***
	(0.003)	(0.002)	(0.003)	(0.003)
Second stage	2.542	-0.055	-0.247	-1.175***
	(9.143)	(0.406)	(0.302)	(0.204)
Observations	35,105	66,110	76,855	75,765
First-stage F-statistic	0.79	48.28	111.10	311.00

Notes: The second-stage outcome is log owner dividends. Panel A instruments log spouse dividends; Panel B instruments spouse dividend receipt. The two second-stage coefficients have different units. First stages report the effect of a one-percentage-point increase in the simulated TOSI penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome and IV specification retains its own estimation sample and first stage. Family-clustered standard errors are in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

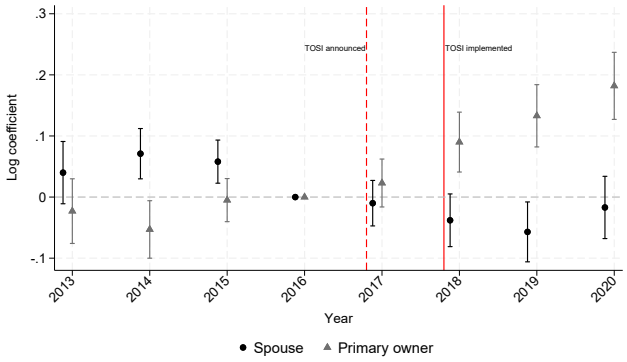
B.4 Labour Income Results

This appendix reports the employment-income evidence summarized in Section 5.1.2, distinguishing total T4 income from payments by the family corporation.

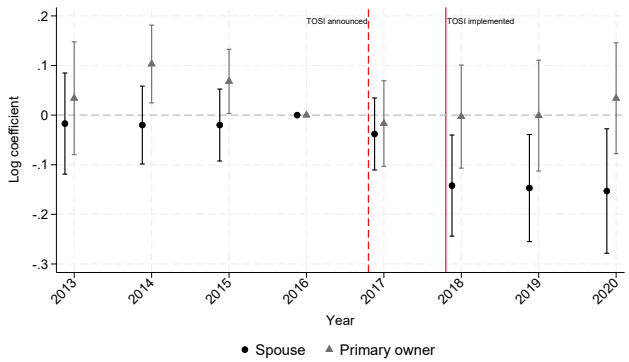
Employment income does not provide a simple replacement for the forgone spouse dividends. The treatment–control event studies show that T4 income paid to spouses by the family corporation falls after implementation, while primary owners' total T4 income rises (Figure B2). During 2018–2020, spouses' total T4 income falls by 7.6% ($4,419) and owners' rises by 16.5% ($10,867); spouses' T4 income from the family corporation falls by 12.5% ($11,503) (Table B4). Treated and control spouses followed different trends in total T4 income before the reform, so that decline is harder to attribute to TOSI than the decline in wages paid by the family business. The smaller estimated fall in total than in firm-paid spouse T4 is consistent with earnings from other employers providing an offset, but the outcomes use different samples and cannot measure it directly. The rise in owners' total T4 income also cannot be assigned confidently to the family business, whose log estimate is imprecise despite a positive dollar estimate. The clearest business-specific result is that families did not broadly replace spouse dividends with higher wages paid to spouses by the same firm. Changes in spouses' actual work could also affect their wages, however; I do not observe hours worked or families' expectations of an audit.

Figure B2: Event-Study Estimates of Labour Income Responses

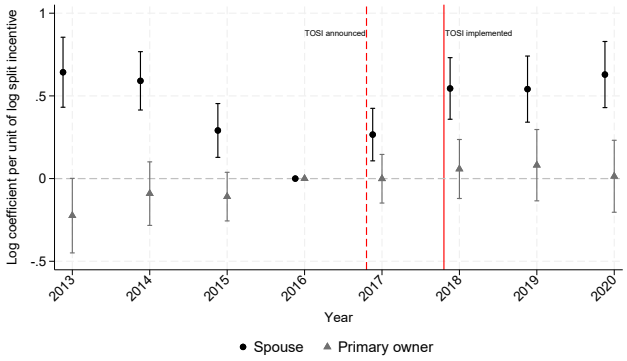
(a) Treatment–control, total T4 income



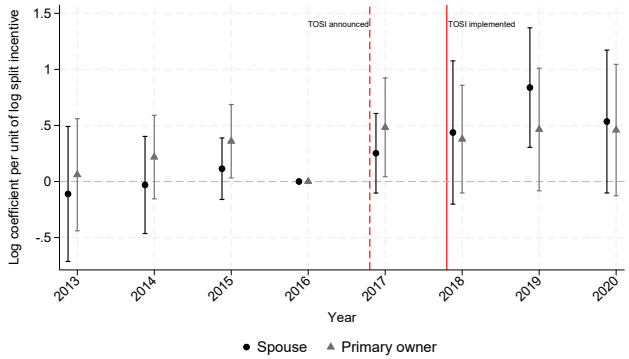
(b) Treatment–control, family-corporation T4 income



(c) Treatment intensity, total T4 income



(d) Treatment intensity, family-corporation T4 income



Notes: Panels A and B report treatment–control estimates, and Panels C and D report treatment-intensity estimates among treated families. Each panel plots spouse and primary-owner log T4 income. Total T4 income includes all employers. The omitted year is 2016. Regressions use the controls and fixed effects specified in the empirical strategy. Bars show 95% confidence intervals based on standard errors clustered at the family level. Dashed and solid vertical lines mark the 2017 announcement and 2018 implementation of the TOSI reform, respectively. Treatment-intensity coefficients retain their original scale.

The treatment-intensity estimates show a different wage pattern within treated families. A stronger pre-reform incentive to split dividends is associated with higher spouse total T4 income in logs but lower spouse total T4 income in dollars. T4 income paid to spouses by the family corporation also rises with the incentive, with a log coefficient of 0.663. For a 0.1 difference in the incentive, this implies about 6.9% higher business-paid spouse wages, although this estimate uses a smaller sample of 1,150 observations. Primary-owner total T4 income rises in dollars, while its log estimate is imprecise. The positive spouse business-wage response suggests that some families with stronger initial splitting incentives adjust compensation through this channel, even though business-paid spouse wages fall in the treatment–control comparison. These results describe different comparisons and do not imply a uniform replacement of spouse dividends with wages.

Table B4: Labour Income Responses to the TOSI Reform

	Log T4 income, all employers		Log T4 income, family corporation	
	Spouse	Primary owner	Spouse	Primary owner
Panel A: Treatment–control (TC), implementation period				
Treated × Post	-0.079***	0.153***	-0.133***	-0.040
	(0.020)	(0.024)	(0.047)	(0.051)
Observations	377,555	374,695	215,465	177,420
Panel B: Treatment intensity (TI), implementation period				
2016 incentive × Post	0.211***	0.149	0.663***	0.281
	(0.075)	(0.097)	(0.249)	(0.251)
Observations	15,565	14,910	1,150	1,005

Panel C: Relative financial exposure

C1. First-stage estimates

	Estimation sample			
	T4 income, all employers		T4 income, family corporation	
Dependent variable	Spouse	Primary owner	Spouse	Primary owner
Log spouse dividends	-0.141***	-0.120***	-0.077***	-0.076***
	(0.015)	(0.013)	(0.020)	(0.022)
First-stage F-statistic	125.4	125.4	125.4	125.4
Observations	174,435	176,015	104,605	86,790
Spouse dividend receipt	-0.043***	-0.038***	-0.026***	-0.024***
	(0.002)	(0.002)	(0.004)	(0.004)
First-stage F-statistic	397.7	397.7	397.7	397.7
Observations	377,515	374,655	215,390	177,280

C2. Second-stage estimates

	Dependent variable			
	T4 income, all employers		T4 income, family corporation	
Instrumented variable	Spouse	Primary owner	Spouse	Primary owner
Log spouse dividends	-0.045	-0.024	0.020	0.069
	(0.046)	(0.034)	(0.072)	(0.059)
Spouse dividend receipt	-0.252**	-0.676***	0.008	0.146
	(0.102)	(0.104)	(0.186)	(0.193)

Notes: Panels A and B report implementation-period TC and TI estimates. All T4 outcomes are in logs. Panel C1 reports first stages, and Panel C2 reports second stages. First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome has its own estimation sample. Family-clustered standard errors appear in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Dollar TC and TI estimates appear in Appendix Table B5.

Relative financial exposure produces different results for dividend amounts and for the decision to report any dividends. Instrumenting log spouse dividends yields no statistically clear average T4 response. In the receipt-based IV, a reduction in the probability of spouse dividend receipt is associated with higher total T4 income for both spouses and owners. The second-stage coefficients are -0.252 for spouses and -0.676 for owners. Neither person's T4 income from the family corporation changes clearly in the average receipt-IV estimates. The increase in total T4 therefore does not show that the family corporation replaced spouse dividends with wages. The treatment–control, treatment-intensity, and IV estimates use different samples and describe different comparisons.

Table B5: Labour Income Responses in Dollars

	2017	2018–2020
A. Treatment–control		
Spouse, all employers	278	-4,419***
	(2,071)	(1,136)
Primary owner, all employers	2,442**	10,867***
	(1,138)	(1,505)
Spouse, family corporation	-1,846*	-11,503***
	(1,063)	(1,296)
Primary owner, family corporation	-3,222	9,729*
	(3,447)	(5,174)
B. Treatment intensity		
Spouse, all employers	-18,644***	-16,726***
	(5,950)	(5,662)
Primary owner, all employers	3,947	29,196***
	(4,903)	(6,326)
Spouse, family corporation	-10,794*	7,473*
	(6,163)	(4,224)
Primary owner, family corporation	-4,976	-2,061
	(15,536)	(24,921)

Notes: Outcomes are in dollars. Post covers 2018–2020. Panel A reports TC estimates. Panel B reports original coefficients on the fixed 2016 log spouse-to-owner net-of-tax-rate ratio interacted with the period indicator among treated families. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.5 Labour Income Responses by Family Income

Tables B6 and B7 report the log and dollar treatment–control and treatment-intensity estimates. Tables B8 and B9 report the two relative financial-exposure IV specifications. The tables retain all four income quartiles and distinguish total T4 income from payments by the family corporation.

The treatment–control increase in primary owners' total T4 income appears across the income distribution, whereas a clear decline in spouses' total T4 income is concentrated in Q4 (Table B6). In that quartile, spouses report $9,535 less and owners $11,058 more in total T4 income. These estimates are consistent with a shift in reported employment income toward the owner among higher-income families, but they do not trace the same earnings between spouses. The treatment-intensity estimates again differ between log and dollar outcomes, so they do not establish a comparable wage-reallocation pattern by income rank.

The receipt-IV estimates show higher owner total T4 income in Q2–Q4, while higher spouse T4 income is clearest in Q2 and Q3 (Table B9). In Q2 and Q3, owner T4 rises more clearly than owner dividends in the receipt-IV results, suggesting a different mix of reported income than in Q4. The spouse T4 increase contrasts with the treatment–control estimate of a decline, which is clearest in

Table B6: Treatment–Control Labour Income Responses by 2016 Family-Income Quartile

	Q1	Q2	Q3	Q4
A. Log income				
Spouse, all employers	0.150	0.042	-0.043	-0.146***
	(0.130)	(0.050)	(0.035)	(0.027)
Observations	75,695	104,500	106,340	88,640
Primary owner, all employers	0.177*	0.183***	0.150***	0.093***
	(0.091)	(0.063)	(0.045)	(0.032)
Observations	75,415	103,280	104,845	88,780
Spouse, family corporation	0.178	0.033	0.022	-0.304***
	(0.127)	(0.102)	(0.089)	(0.070)
Observations	52,680	62,700	55,105	42,995
Primary owner, family corporation	0.035	-0.166	-0.004	-0.037
	(0.099)	(0.175)	(0.114)	(0.062)
Observations	45,775	52,145	43,520	34,175
B. Income in dollars				
Spouse, all employers	2,027**	578	-1,721*	-9,535***
	(1,026)	(873)	(985)	(2,389)
Observations	104,425	132,450	131,085	108,060
Primary owner, all employers	966	3,050*	4,480***	11,058***
	(2,163)	(1,742)	(1,720)	(2,927)
Observations	104,425	132,450	131,085	108,060
Spouse, family corporation	-555	-3,039***	-4,021**	-21,318***
	(1,351)	(1,017)	(1,804)	(2,497)
Observations	52,740	62,770	55,140	43,030
Primary owner, family corporation	-342	-3,852*	389	18,423**
	(2,445)	(2,162)	(4,736)	(8,579)
Observations	45,850	52,195	43,550	34,220

Notes: Implementation-period estimates for 2018–2020. Income quartiles are fixed using 2016 family income. Standard errors clustered at the family level are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. A dash indicates an unreported specification.

Q4. The comparisons use different samples and capture different margins of adjustment. Q1 IV estimates are imprecise; Tables B8 and B9 retain the numerical estimates for all four quartiles.

Table B7: Treatment-Intensity Labour Income Responses by 2016 Family-Income Quartile

	Q1	Q2	Q3	Q4
A. Log income				
Spouse, all employers	2.054	0.218	0.410**	0.202**
	(1.739)	(0.317)	(0.199)	(0.085)
Observations	485	1,865	3,725	8,640
Primary owner, all employers	-1.344	0.199	0.325	0.180
	(0.811)	(0.352)	(0.237)	(0.121)
Observations	545	1,670	3,235	8,605
Spouse, family corporation	—	—	—	1.016***
				(0.284)
Observations				625
Primary owner, family corporation	—	—	—	0.063
				(0.294)
Observations				550
B. Income in dollars				
Spouse, all employers	-9,830	-12,178**	-7,783*	-22,163**
	(6,357)	(5,810)	(4,324)	(8,987)
Observations	1,100	2,980	5,240	11,495
Primary owner, all employers	-28,252	16,522	28,706***	28,026***
	(18,156)	(11,506)	(8,106)	(9,712)
Observations	1,100	2,980	5,240	11,495
Spouse, family corporation	—	—	—	14,814*
				(8,254)
Observations				625
Primary owner, family corporation	—	—	—	-14,287
				(58,261)
Observations				550

Notes: Implementation-period estimates for 2018–2020. Income quartiles are fixed using 2016 family income. Treatment-intensity estimates retain the original incentive scale. Standard errors clustered at the family level are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. A dash indicates an unreported specification.

Table B8: Labour Income Responses by Family-Income Quartile Using Relative Financial Exposure with Log Spouse Dividends

	Q1	Q2	Q3	Q4
Spouse total T4				
First stage	-0.061*	-0.030*	-0.103***	-0.273***
	(0.035)	(0.017)	(0.022)	(0.032)
Second stage	-0.418	-0.344	-0.132	0.015
	(0.307)	(0.446)	(0.117)	(0.042)
Observations	20,570	43,000	53,625	55,415
First-stage F-statistic	3.06	15.15	30.19	95.44
Owner total T4				
First stage	-0.003	-0.034**	-0.107***	-0.218***
	(0.015)	(0.015)	(0.024)	(0.026)
Second stage	-7.907	-0.069	-0.031	0.008
	(36.650)	(0.208)	(0.063)	(0.033)
Observations	20,280	43,260	54,265	56,370
First-stage F-statistic	3.06	15.15	30.19	95.44
Spouse T4 from business				
First stage	-0.024**	-0.030	-0.071*	-0.199***
	(0.009)	(0.027)	(0.040)	(0.049)
Second stage	-0.399	-0.189	0.082	0.042
	(0.255)	(0.275)	(0.145)	(0.086)
Observations	14,360	27,880	31,210	29,520
First-stage F-statistic	3.06	15.15	30.19	95.44
Owner T4 from business				
First stage	-0.039	-0.030	-0.068*	-0.205***
	(0.028)	(0.028)	(0.040)	(0.054)
Second stage	-0.683	-0.133	0.145	0.056
	(0.715)	(0.250)	(0.137)	(0.060)
Observations	12,500	23,625	25,255	23,830
First-stage F-statistic	3.06	15.15	30.19	95.44

Notes: First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome has its own estimation sample. Family-clustered standard errors appear in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The endogenous variable is log spouse dividends.

Table B9: Labour Income Responses by Family-Income Quartile Using Relative Financial Exposure with Dividend Receipt

	Q1	Q2	Q3	Q4
Spouse total T4				
First stage	-0.012*	-0.022***	-0.037***	-0.060***
	(0.007)	(0.004)	(0.004)	(0.004)
Second stage	-2.327*	-0.836**	-0.504**	-0.024
	(1.401)	(0.421)	(0.228)	(0.116)
Observations	75,655	104,500	106,340	88,640
First-stage F-statistic	0.79	48.28	111.10	311.00
Owner total T4				
First stage	-0.007	-0.021***	-0.033***	-0.050***
	(0.009)	(0.004)	(0.004)	(0.003)
Second stage	-1.914	-0.737**	-0.754***	-0.531***
	(4.102)	(0.314)	(0.239)	(0.123)
Observations	75,375	103,275	104,845	88,780
First-stage F-statistic	0.79	48.28	111.10	311.00
Spouse T4 from business				
First stage	-0.020	-0.020***	-0.015**	-0.047***
	(0.016)	(0.007)	(0.007)	(0.008)
Second stage	-0.635	-0.402	-0.546	0.370
	(0.511)	(0.399)	(0.473)	(0.227)
Observations	52,645	62,675	55,085	43,015
First-stage F-statistic	0.79	48.28	111.10	311.00
Owner T4 from business				
First stage	-0.038	-0.019***	-0.013*	-0.044***
	(0.025)	(0.007)	(0.007)	(0.009)
Second stage	-0.576	0.507	-0.054	0.006
	(0.487)	(0.497)	(0.541)	(0.194)
Observations	45,730	52,115	43,455	34,195
First-stage F-statistic	0.79	48.28	111.10	311.00

Notes: First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome has its own estimation sample. Family-clustered standard errors appear in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The endogenous variable is an indicator for spouse dividend receipt.

B.6 Business Activity Results

Table B10: Family-Corporation Balance-Sheet Outcomes by Treatment Intensity

	Log average retained earnings	Log total assets	Annual change in assets	Annual change in retained earnings
$I_i \times$ 2017	-0.153***	-0.103*	1,588.465	-1.174
	(0.058)	(0.058)	(2,131.480)	(43.764)
$I_i \times$ Post	-0.080	-0.062	1,547.614	21.825
	(0.073)	(0.075)	(2,059.330)	(38.844)
Observations	18,100	21,685	19,520	19,180

Notes: Treatment-intensity estimates among families classified as treated. I_i is the fixed 2016 log ratio of the spouse's to the primary owner's net-of-tax marginal dividend rates. Post covers 2018–2020. Annual changes in assets and retained earnings are year-over-year percentage changes, multiplied by 100. Standard errors are clustered at the family level and shown in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B11: Corporate Balance-Sheet Growth by Family-Income Quartile

	Q1	Q2	Q3	Q4
Panel A: Year-over-year change in assets				
$I_i \times$ 2017	1,729.424	39.010	-1,173.945	1,577.855
	(1,765.129)	(303.609)	(2,101.771)	(1,914.665)
$I_i \times$ Post	414.336	1,051.605	-751.755	1,274.613
	(816.627)	(882.389)	(2,133.617)	(1,626.054)
Observations	975	2,635	4,675	10,350
Panel B: Year-over-year change in retained earnings				
$I_i \times$ 2017	-254.703	422.161	-187.054	-25.100
	(689.127)	(312.594)	(129.459)	(44.197)
$I_i \times$ Post	-1,054.419	-17.856	-39.096	53.572
	(680.453)	(96.643)	(90.538)	(50.714)
Observations	965	2,600	4,605	10,155

Notes: Each column reports a separate treatment-intensity difference-in-differences regression within the indicated quartile of 2016 family income. I_i is the fixed 2016 log ratio of the spouse's to the primary owner's net-of-tax marginal dividend rates. Post covers 2018–2020. Annual changes are year-over-year percentage changes, multiplied by 100. Standard errors are clustered at the family level and shown in parentheses. None of the reported interaction estimates is statistically significant at conventional levels. The sample size varies across outcomes and quartiles because of outcome availability.

Table B12: Corporate Responses Using Relative Financial Exposure

Outcome	First stage	Second stage	Observations	F-statistic
Panel A. Log spouse dividends instrumented				
Log retained earnings	-0.088***	-0.021	194,170	125.4
	(0.008)	(0.033)		
Log corporate assets	-0.080***	-0.023	238,620	125.4
	(0.007)	(0.032)		
Annual asset growth	-0.081***	-183.914	212,310	125.4
	(0.007)	(362.304)		
Annual retained-earnings growth	-0.081***	1.694	210,230	125.4
	(0.007)	(118.126)		
Panel B. Spouse dividend receipt instrumented				
Log retained earnings	-0.029***	-0.036	344,040	397.7
	(0.002)	(0.079)		
Log corporate assets	-0.027***	-0.150*	474,165	397.7
	(0.001)	(0.084)		
Annual asset growth	-0.028***	-4,285.989**	423,840	397.7
	(0.001)	(2,035.554)		
Annual retained-earnings growth	-0.028***	133.813	419,420	397.7
	(0.001)	(148.765)		

Notes: Corporate outcomes are in logs or annual percentage growth, as labelled. Panel A instruments log spouse dividends; Panel B instruments spouse dividend receipt. First stages report the effect of a one-percentage-point increase in the simulated TOSI penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome and IV specification retains its own estimation sample and first stage. Family-clustered standard errors are in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.7 Family Taxable Income and Tax Liability Results

Table B13: Family Taxable Income and Tax Liability Responses in Dollars

	Family taxable income	Family tax liability
Panel A: Treatment–control, 2018–2020		
Treated × Post	928	225
	(1,875)	(755)
Observations	478,735	478,735
Panel B: Treatment intensity, 2018–2020		
2016 incentive × Post	-40,197***	2,109
	(8,174)	(3,787)
Observations	21,835	21,835

Notes: This table supplements Table 3. Coefficients and standard errors are in dollars. Panel A compares treated and control families. Panel B reports coefficients on the fixed 2016 tax incentive interacted with the post-reform indicator among treated families. Standard errors clustered at the family level appear in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B14: Family Taxable Income and Tax Liability Responses by 2016 Family-Income Quartile

	Q1	Q2	Q3	Q4
Panel A. Treatment–control, log outcomes				
Family taxable income	-0.007	-0.019	-0.019	-0.005
	(0.040)	(0.018)	(0.012)	(0.010)
Family tax liability	0.205	0.066	0.035	0.021
	(0.152)	(0.047)	(0.026)	(0.018)
Panel B. Treatment–control, dollar outcomes				
Family taxable income	533	-4,211*	-4,656*	-4,552
	(2,851)	(2,544)	(2,380)	(3,909)
Family tax liability	162	-819	-374	-1,918
	(766)	(792)	(695)	(1,612)
Panel C. Treatment intensity, log outcomes				
Family taxable income	-0.390	-0.217**	-0.147***	-0.184***
	(0.326)	(0.105)	(0.048)	(0.032)
Family tax liability	0.716	0.633**	0.452***	0.081
	(1.145)	(0.267)	(0.109)	(0.059)
Panel D. Treatment intensity, dollar outcomes				
Family taxable income	-12,145	-8,706	-15,406*	-57,972***
	(31,412)	(13,667)	(8,440)	(12,983)
Family tax liability	-1,242	3,772	9,007***	-2,335
	(7,697)	(4,234)	(2,580)	(6,078)

Notes: All estimates cover 2018–2020. Panels A and B report TC estimates; Panels C and D report TI estimates among treated families. TI coefficients retain the original scale of the fixed 2016 log spouse-to-owner net-of-tax-rate ratio. Dollar outcomes are in dollars. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B15: Family Income and Tax Responses by Family-Income Quartile Using Relative Financial Exposure

	Q1	Q2	Q3	Q4
Panel A. Log spouse dividends instrumented				
Family taxable income				
First stage	-0.022*	-0.037***	-0.074***	-0.169***
	(0.012)	(0.009)	(0.014)	(0.017)
Second stage	0.106	0.084	0.019	0.022*
	(0.120)	(0.063)	(0.027)	(0.012)
Observations	32,735	62,025	72,335	70,640
First-stage F-statistic	3.06	15.15	30.19	95.44
Family tax liability				
First stage	-0.027*	-0.036***	-0.075***	-0.169***
	(0.015)	(0.009)	(0.014)	(0.017)
Second stage	-0.585	-0.305*	-0.208**	-0.022
	(0.524)	(0.173)	(0.086)	(0.027)
Observations	27,470	60,635	71,780	70,425
First-stage F-statistic	3.06	15.15	30.19	95.44
Panel B. Spouse dividend receipt instrumented				
Family taxable income				
First stage	-0.007*	-0.016***	-0.027***	-0.043***
	(0.003)	(0.002)	(0.003)	(0.002)
Second stage	1.147*	0.458***	0.234***	0.150***
	(0.615)	(0.128)	(0.061)	(0.032)
Observations	103,935	132,410	131,055	108,035
First-stage F-statistic	0.79	48.28	111.10	311.00
Family tax liability				
First stage	-0.004	-0.016***	-0.027***	-0.043***
	(0.004)	(0.002)	(0.003)	(0.002)
Second stage	-4.896	-1.058***	-0.561***	-0.226***
	(6.914)	(0.352)	(0.164)	(0.067)
Observations	85,960	129,740	130,060	107,640
First-stage F-statistic	0.79	48.28	111.10	311.00

Notes: Second-stage outcomes are log family taxable income and log family tax liability. Panel A instruments log spouse dividends; Panel B instruments spouse dividend receipt. The two second-stage coefficients have different units. TOSI can also affect tax liability directly. First stages report the effect of a one-percentage-point increase in the simulated TOSI penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome and IV specification retains its own estimation sample and first stage. Family-clustered standard errors are in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.8 Responses by Simulated TOSI-Penalty Quartile

Table B16: Dividend and Family Income Responses by Simulated TOSI-Penalty Quartile

	Q1	Q2	Q3	Q4
Panel A. Dividend income and receipt				
Spouse dividend income	0.020	0.260**	-0.895***	-1.374***
	(0.215)	(0.110)	(0.110)	(0.111)
Observations	228,505	229,735	231,555	233,145
Primary-owner dividend income	0.073	0.083	0.100	0.256***
	(0.094)	(0.073)	(0.063)	(0.056)
Observations	242,160	242,975	243,855	245,500
Spouse receives dividends	0.031**	0.037***	-0.297***	-0.381***
	(0.015)	(0.014)	(0.019)	(0.016)
Observations	460,780	462,080	461,720	463,535
Primary owner receives dividends	0.030*	0.044***	0.013	0.037***
	(0.016)	(0.013)	(0.014)	(0.012)
Observations	460,780	462,080	461,720	463,535
Panel B. Family taxable income and tax liability				
Family taxable income	0.040***	0.024**	-0.013	-0.059***
	(0.014)	(0.012)	(0.015)	(0.012)
Observations	460,310	461,610	461,250	463,065
Family tax liability	0.009	-0.042	0.028	0.150***
	(0.033)	(0.030)	(0.034)	(0.027)
Observations	438,395	439,725	439,375	441,175

Notes: Treated families are ranked by their average simulated 2018–2020 dollar penalty calculated using fixed 2016 income. Each quartile is compared separately with the same control group. Coefficients cover 2018–2020. Income and tax outcomes are in logs; dividend receipt is a zero-one indicator. Observation counts are rounded as released. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.9 Event Studies by Simulated TOSI-Penalty Quartile

Table B17: Treatment–Control T4 Income Responses by Simulated TOSI-Penalty Quartile

	Average simulated TOSI-penalty quartile			
	Q1	Q2	Q3	Q4
Spouse T4 income	-0.091***	-0.085***	-0.177***	0.027
	(0.034)	(0.028)	(0.039)	(0.046)
	365,115	366,220	365,570	365,260
Primary-owner T4 income	0.075	0.043	0.239***	0.281***
	(0.047)	(0.039)	(0.051)	(0.049)
	362,405	363,385	363,040	363,920
Spouse T4 from family business	-0.283	0.010	-0.159*	-0.135**
	(0.199)	(0.112)	(0.097)	(0.061)
	214,005	214,145	214,365	214,875
Primary-owner T4 from family business	-0.305	0.086	-0.208	0.026
	(0.203)	(0.077)	(0.153)	(0.058)
	176,050	176,120	176,420	176,910

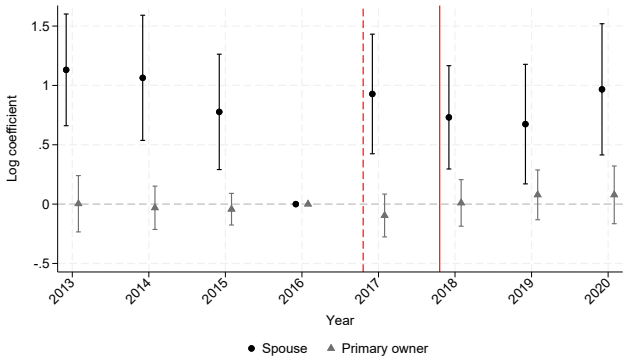
Notes: Treated families are ranked by their average simulated 2018–2020 TOSI penalty, calculated using fixed 2016 income. Each quartile is compared separately with the same control group. Coefficients are treatment–control estimates for 2018–2020. All T4 income outcomes are in logs. The third line for each outcome gives the rounded number of observations. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B18: Dollar Responses by Simulated TOSI-Penalty Quartile

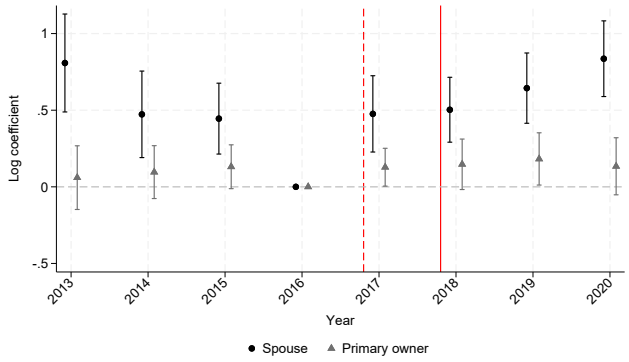
	Q1	Q2	Q3	Q4
Spouse dividends	2.93	2.83	-6.16	-40.76
	(0.68)	(0.69)	(1.07)	(1.79)
Primary-owner dividends	2.60	1.92	2.52	19.82
	(1.96)	(1.42)	(2.20)	(2.51)
Primary-owner total T4 income	4.96	0.14	10.23	28.60
	(2.37)	(2.09)	(3.53)	(3.19)
Family taxable income	12.80	5.31	-2.30	-3.02
	(3.06)	(2.79)	(4.20)	(3.38)
Family tax liability	0.72	-1.07	-3.80	4.84
	(0.97)	(1.24)	(1.76)	(1.32)

Notes: Entries are treatment–control implementation-period estimates for 2018–2020, in thousands of dollars; family-clustered standard errors are in parentheses. Treated families are ranked by their average simulated penalty for 2018–2020 using fixed 2016 income, and each quartile is compared separately with the same control group. Outcome samples can differ. Log specifications appear in Appendix Table B16.

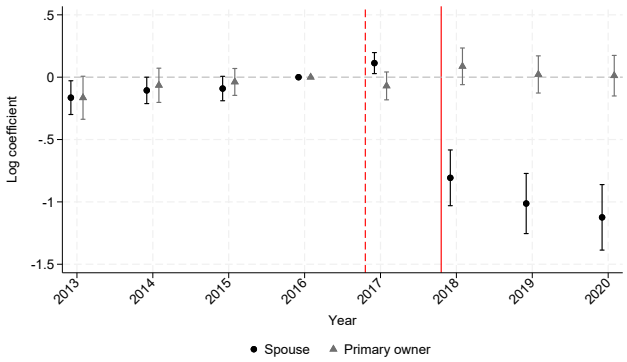
Figure B3: Event-Study Estimates of Dividend Income by Simulated TOSI-Penalty Quartile



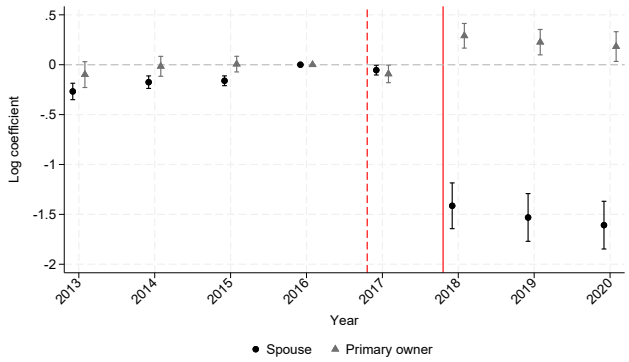
(a) Penalty quartile 1



(b) Penalty quartile 2



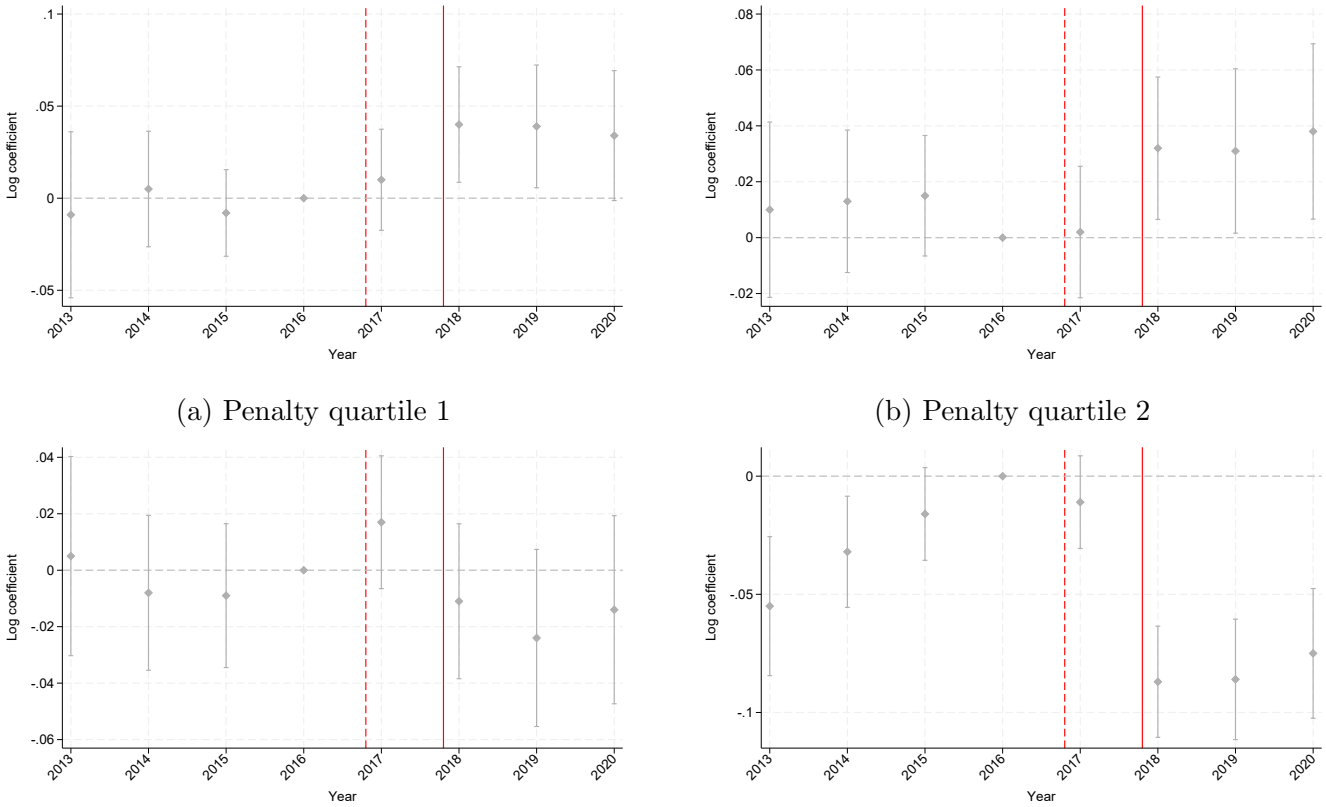
(c) Penalty quartile 3



(d) Penalty quartile 4

Notes: Each panel compares one quartile of treated families, ranked by their average simulated 2018–2020 TOSI penalty using fixed 2016 income, with the same control group. Estimates are relative to 2016. Dots show treatment–control event-study coefficients and bars show 95% confidence intervals based on family-clustered standard errors. The dashed line separates the pre-reform years from the 2017 announcement year; the solid line separates the announcement year from implementation in 2018.

Figure B4: Event-Study Estimates of Family Taxable Income by Simulated TOSI-Penalty Quartile



(a) Penalty quartile 1

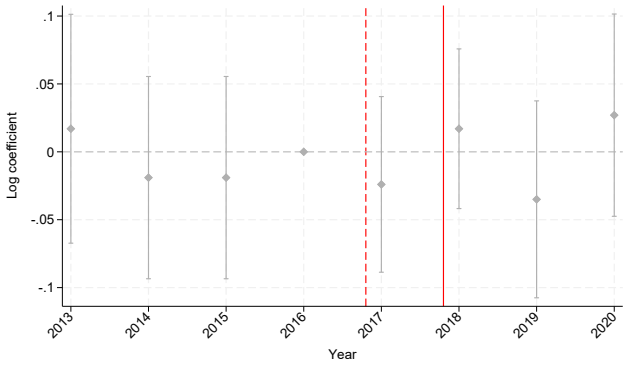
(b) Penalty quartile 2

(c) Penalty quartile 3

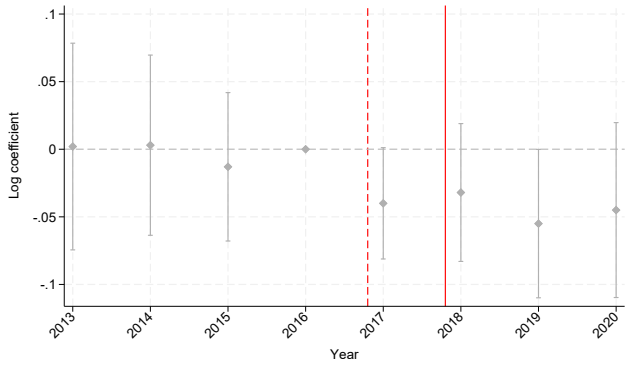
(d) Penalty quartile 4

Notes: Each panel compares one quartile of treated families, ranked by their average simulated 2018–2020 TOSI penalty using fixed 2016 income, with the same control group. Estimates are relative to 2016. Dots show treatment–control event-study coefficients and bars show 95% confidence intervals based on family-clustered standard errors. The dashed line separates the pre-reform years from the 2017 announcement year; the solid line separates the announcement year from implementation in 2018.

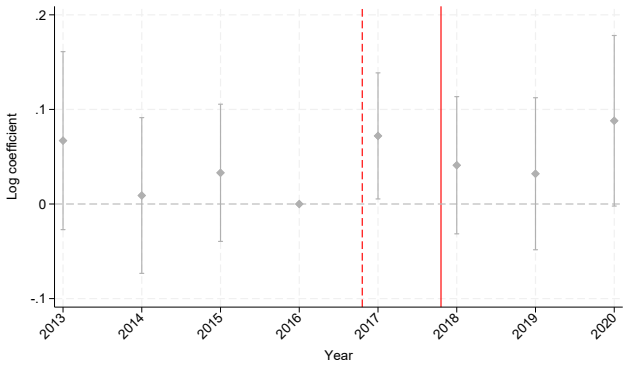
Figure B5: Event-Study Estimates of Family Tax Liability by Simulated TOSI-Penalty Quartile



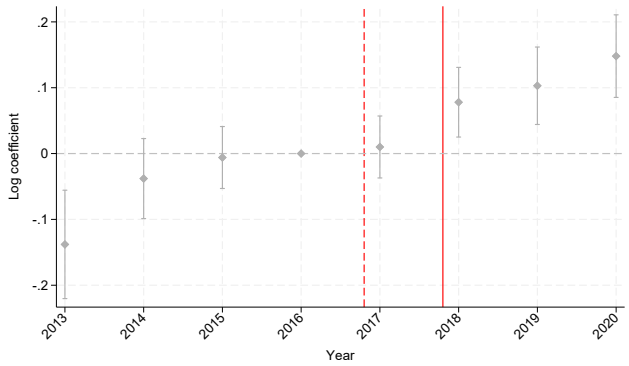
(a) Penalty quartile 1



(b) Penalty quartile 2



(c) Penalty quartile 3



(d) Penalty quartile 4

Notes: Each panel compares one quartile of treated families, ranked by their average simulated 2018–2020 TOSI penalty using fixed 2016 income, with the same control group. Estimates are relative to 2016. Dots show treatment–control event-study coefficients and bars show 95% confidence intervals based on family-clustered standard errors. The dashed line separates the pre-reform years from the 2017 announcement year; the solid line separates the announcement year from implementation in 2018.

C Business Type and Ownership Comparisons

C.1 Professional and Non-Professional Corporations

Table C1: TC Income Responses by Corporation Type

	Non-professional		Professional	
	Log income	$1,000	Log income	$1,000
Spouse dividends	-0.283***	-4.996***	-1.210***	-24.875***
	(0.075)	(0.609)	(0.130)	(2.202)
Observations	199,880	418,760	25,465	34,945
Owner dividends	-0.004	1.861*	0.448***	20.384***
	(0.047)	(0.951)	(0.074)	(2.914)
Observations	212,680	418,760	27,275	34,945
Spouse total T4	-0.024	-2.615*	-0.125***	-7.047***
	(0.028)	(1.500)	(0.037)	(2.191)
Observations	329,330	418,760	28,355	34,945
Owner total T4	0.092***	1.972	0.041	5.220
	(0.032)	(1.771)	(0.039)	(3.202)
Observations	326,510	418,760	28,550	34,945
Spouse T4 from family business	0.027	-2.970**	-0.243***	-20.343***
	(0.077)	(1.204)	(0.065)	(2.946)
Observations	189,460	189,640	16,065	16,065
Owner T4 from family business	-0.144	-3.030	-0.032	13.978*
	(0.094)	(1.986)	(0.055)	(7.990)
Observations	156,335	156,510	13,050	13,060
Family taxable income	-0.025**	-4.772**	-0.004	-2.170
	(0.010)	(2.173)	(0.012)	(4.164)
Observations	418,295	418,760	34,940	34,945
Family tax liability	-0.005	-1.011	0.024	-0.185
	(0.027)	(0.932)	(0.024)	(1.704)
Observations	396,825	418,760	34,830	34,945
Dividend receipt				
Spouse receives dividends	-0.116***		-0.217***	
	(0.011)		(0.019)	
Observations	418,760		34,945	
Owner receives dividends	0.002		0.090***	
	(0.010)		(0.014)	
Observations	418,760		34,945	

Notes: TC coefficients compare treated and control families during 2018–2020 within each corporation type. Dollar outcomes are in thousands; receipt outcomes are zero-one indicators. Each outcome has its own sample. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C2: TI Income Responses by Corporation Type

	Non-professional		Professional	
	Log income	$1,000	Log income	$1,000
Spouse dividends	-2.199***	-21.906***	-2.812***	-88.588***
	(0.312)	(3.099)	(0.399)	(5.661)
Observations	4,275	9,760	5,560	8,700
Owner dividends	0.008	9.207**	0.500**	27.810***
	(0.192)	(4.069)	(0.206)	(8.553)
Observations	6,145	9,760	7,125	8,700
Spouse total T4	0.119	-6.439	0.287***	-27.401**
	(0.130)	(5.409)	(0.106)	(10.759)
Observations	6,790	9,760	6,260	8,700
Owner total T4	-0.209	10.368	0.292**	36.680***
	(0.167)	(9.489)	(0.148)	(9.693)
Observations	6,095	9,760	6,410	8,700
Spouse T4 from family business	1.124***	3.630	0.571	27.054*
	(0.293)	(3.060)	(0.372)	(14.030)
Observations	340	340	650	650
Owner T4 from family business	1.704***	44.828***	-0.429	-63.865
	(0.481)	(8.958)	(0.303)	(56.119)
Observations	260	260	605	605
Family taxable income	-0.173***	-14.122	-0.218***	-69.372***
	(0.041)	(10.886)	(0.037)	(13.978)
Observations	9,755	9,760	8,695	8,700
Family tax liability	0.150	4.089	0.179**	-3.751
	(0.096)	(5.250)	(0.071)	(6.167)
Observations	9,445	9,760	8,660	8,700
Dividend receipt				
Spouse receives dividends	-0.329***		-0.766***	
	(0.054)		(0.065)	
Observations	9,760		8,700	
Owner receives dividends	-0.080*		0.016	
	(0.043)		(0.044)	
Observations	9,760		8,700	

Notes: TI coefficients interact the fixed 2016 log net-of-tax-rate ratio with Post among treated families within each corporation type. Dollar outcomes are in thousands; receipt outcomes are zero-one indicators. Each outcome has its own sample. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C3: Owner Dividend and Family Income Responses by Corporation Type with Log Spouse Dividends

	Non-professional	Professional
Owner dividends		
First stage	-0.035***	-0.125***
	(0.007)	(0.018)
Second stage	-0.168	-0.463***
	(0.118)	(0.095)
Observations	167,565	21,210
First-stage F-statistic	32.24	74.39
Family taxable income		
First stage	-0.038***	-0.143***
	(0.007)	(0.017)
Second stage	0.028	0.022*
	(0.043)	(0.013)
Observations	199,820	25,465
First-stage F-statistic	32.24	74.39
Family tax liability		
First stage	-0.039***	-0.146***
	(0.007)	(0.017)
Second stage	-0.375***	-0.083**
	(0.129)	(0.035)
Observations	192,700	25,410
First-stage F-statistic	32.24	74.39

Notes: Second-stage outcomes are log owner dividends, log family taxable income, and log family tax liability. First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome has its own estimation sample. Family-clustered standard errors appear in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The endogenous variable is log spouse dividends. TOSI can also affect tax liability directly.

Table C4: Owner Dividend and Family Income Responses by Corporation Type with Dividend Receipt

	Non-professional	Professional
Owner dividends		
First stage	-0.012***	-0.031***
	(0.002)	(0.003)
Second stage	-0.327	-1.297***
	(0.366)	(0.248)
Observations	212,665	27,275
First-stage F-statistic	82.08	222.50
Family taxable income		
First stage	-0.016***	-0.036***
	(0.002)	(0.002)
Second stage	0.330***	0.210***
	(0.095)	(0.042)
Observations	418,185	34,940
First-stage F-statistic	82.08	222.50
Family tax liability		
First stage	-0.016***	-0.036***
	(0.002)	(0.002)
Second stage	-1.015***	-0.360***
	(0.265)	(0.100)
Observations	396,765	34,830
First-stage F-statistic	82.08	222.50

Notes: Second-stage outcomes are log owner dividends, log family taxable income, and log family tax liability. First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome has its own estimation sample. Family-clustered standard errors appear in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The endogenous variable is an indicator for spouse dividend receipt. TOSI can also affect tax liability directly.

Table C5: Corporate Responses by Corporation Type

	Non-professional	Professional
Panel A: Treatment intensity		
Annual asset growth	3,241.369	-89.867
	(5,409.044)	(101.448)
Observations	8,635	7,700
Annual retained-earnings growth	-35.641	61.453*
	(75.066)	(34.364)
Observations	8,550	7,600
Panel B: Relative-penalty dividend-receipt IV		
Log retained earnings		
First stage	-0.016***	-0.036***
	(0.002)	(0.002)
Second stage	-0.000	0.016
	(0.249)	(0.083)
Observations	291,920	32,270
Log total assets		
First stage	-0.016***	-0.035***
	(0.002)	(0.002)
Second stage	-0.182	-0.016
	(0.245)	(0.098)
Observations	414,480	34,890

Notes: Panel A reports TI coefficients on the fixed 2016 log net-of-tax-rate ratio interacted with Post; its outcomes are annual percentage growth. Panel B instruments spouse dividend receipt with relative financial exposure and reports outcome-specific first and second stages. Its first stages are in probability units per one-percentage-point increase in the penalty relative to baseline income; second-stage outcomes are in logs. Family-clustered standard errors are in parentheses. Observations are rounded as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.2 Multiple-Business Ownership

Table C6: Treatment–Control Income Responses by Spouse Business Ownership

	Main sample		Multiple businesses	
	Log income	$1,000	Log income	$1,000
Spouse dividend income	-0.753***	-12.391***	-0.383***	-6.274**
	(0.066)	(0.815)	(0.114)	(2.929)
Observations	239,795	478,735	87,780	140,655
Primary-owner dividend income	0.134***	7.124***	-0.120	4.397
	(0.037)	(1.149)	(0.095)	(3.214)
Observations	255,870	478,735	93,425	140,655
Spouse T4 income	-0.079***	-4.419***	-0.012	-2.163
	(0.020)	(1.136)	(0.047)	(4.858)
Observations	377,555	478,735	113,160	140,655
Primary-owner T4 income	0.153***	10.867***	0.086*	5.162
	(0.024)	(1.505)	(0.050)	(4.705)
Observations	374,695	478,735	117,250	140,655
Spouse T4 from family business	-0.133***	-11.503***	-0.421**	-20.120**
	(0.047)	(1.296)	(0.169)	(7.952)
Observations	215,465	215,660	36,070	36,090
Primary-owner T4 from family business	-0.040	9.729*	-0.217	2.373
	(0.051)	(5.174)	(0.151)	(9.356)
Observations	177,420	177,615	31,015	31,035
Family taxable income	-0.012*	0.928	-0.009	6.831
	(0.007)	(1.875)	(0.019)	(10.374)
Observations	478,260	478,735	140,510	140,655
Family tax liability	0.032*	0.225	0.055	5.059
	(0.017)	(0.755)	(0.039)	(4.471)
Observations	456,090	478,735	136,480	140,655
Dividend receipt				
Spouse receives dividends	-0.168***		-0.076***	
	(0.009)		(0.020)	
Observations	478,735		140,655	
Primary owner receives dividends	0.029***		0.030**	
	(0.007)		(0.014)	
Observations	478,735		140,655	

Notes: TC implementation-period estimates for 2018–2020. The main sample restricts the spouse to shares in one business; the comparison sample includes spouses with shares in multiple businesses. Dollar outcomes are in thousands and receipt is a zero-one indicator. Each outcome has its own sample. Family-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C7: Treatment-Intensity Responses by Spouse Business Ownership

	Single business	Multiple businesses
Spouse dividends	-2.552***	-0.465
	(0.238)	(0.474)
Observations	11,665	2,350
Owner dividends	0.256**	0.049
	(0.130)	(0.409)
Observations	15,900	2,895
Spouse total T4	0.211***	0.641**
	(0.075)	(0.267)
Observations	15,565	2,265
Owner total T4	0.149	0.498**
	(0.097)	(0.243)
Observations	14,910	2,620
Family taxable income	-0.201***	-0.037
	(0.025)	(0.076)
Observations	21,835	3,440
Family tax liability	0.150***	0.350**
	(0.055)	(0.149)
Observations	21,455	3,390
Spouse receives dividends	-0.591***	-0.365***
	(0.038)	(0.082)
Observations	21,835	3,440
Owner receives dividends	-0.043	0.041
	(0.029)	(0.065)
Observations	21,835	3,440

Notes: TI coefficients interact the fixed 2016 log relative net-of-tax incentive with the 2018–2020 indicator, estimated among treated families within each ownership sample. Income and tax outcomes are in logs; dividend-receipt outcomes are indicators. Family-clustered standard errors are in parentheses. Observations are rounded as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C8: Relative Financial Exposure by Spouse Business Ownership with Log Spouse Dividends

	Single business	Multiple businesses
Owner dividends		
First stage	-0.071***	-0.027**
	(0.007)	(0.012)
Second stage	-0.315***	0.041
	(0.061)	(0.191)
Observations	200,680	81,550
First-stage F-statistic	125.40	7.54
Family taxable income		
First stage	-0.080***	-0.030***
	(0.007)	(0.011)
Second stage	0.019	-0.093
	(0.014)	(0.099)
Observations	239,730	87,760
First-stage F-statistic	125.40	7.54
Family tax liability		
First stage	-0.084***	-0.030***
	(0.007)	(0.011)
Second stage	-0.156***	-0.731**
	(0.039)	(0.344)
Observations	232,285	86,100
First-stage F-statistic	125.40	7.54

Notes: First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome has its own estimation sample. Family-clustered standard errors appear in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The endogenous variable is log spouse dividends. TOSI can also affect tax liability directly.

Table C9: Relative Financial Exposure by Spouse Business Ownership with Dividend Receipt

	Single business	Multiple businesses
Owner dividends		
First stage	-0.022***	-0.012***
	(0.001)	(0.002)
Second stage	-0.780***	0.798
	(0.162)	(0.708)
Observations	255,855	93,425
First-stage F-statistic	397.70	28.37
Family taxable income		
First stage	-0.027***	-0.013***
	(0.001)	(0.002)
Second stage	0.222***	-0.004
	(0.034)	(0.167)
Observations	478,150	140,505
First-stage F-statistic	397.70	28.37
Family tax liability		
First stage	-0.028***	-0.013***
	(0.001)	(0.003)
Second stage	-0.528***	-1.623***
	(0.088)	(0.488)
Observations	456,030	136,480
First-stage F-statistic	397.70	28.37

Notes: First stages report the effect of a one-percentage-point increase in the simulated penalty relative to 2016 family income. Dividend receipt is a zero-one indicator. Each outcome has its own estimation sample. Family-clustered standard errors appear in parentheses. Observations and F-statistics are reproduced as released. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The endogenous variable is an indicator for spouse dividend receipt. TOSI can also affect tax liability directly.

D Alternative Dollar-Penalty Instrument

The main financial-exposure specification divides the simulated TOSI penalty by total family income in 2016. This appendix compares those estimates with results using the penalty in thousands of dollars, which compares equal amounts of additional potential tax rather than equal tax costs as shares of baseline income. I first illustrate how the choice of scale affects the dividend and tax-liability comparisons, then report the alternative estimates by income quartile.

D.1 Scaling of Financial Exposure and Tax Responses

Table D1: Income, Tax Liability, and Simulated TOSI Penalty Benchmarks for Treated Families

	P25	Median	P75	P90
Total family income	131,298	203,415	300,142	423,938
Family tax liability	12,737	28,761	54,630	101,150
Simulated TOSI penalty	1	2,004	16,251	26,660
Penalty / 2016 family income (%)	< 0.01	0.99	6.94	13.13

Notes: The table reports separate percentiles among treated families. Total family income and family tax liability are measured in 2016. The simulated penalties use 2016 incomes under post-reform tax rules. Relative penalties are expressed as percentages of 2016 total family income and are calculated for each family before taking percentiles. Pairing the separate percentile values provides illustrative benchmarks rather than the observed income, taxes, and penalties of the same families. Monetary values are in Canadian dollars and are rounded to the nearest dollar. Relative penalties are reported to two decimal places, with positive values below 0.01% shown as < 0.01.

Measuring the penalty relative to income changes the dollar amounts being compared. At the median and ninetieth-percentile income benchmarks in Table D1, a potential penalty equal to 1% of income represents approximately $2,034 and $4,239, respectively. Assuming a common response of -0.05 log points per additional $1,000, these penalties would predict declines of approximately 0.102 and 0.212 log points in spouse dividends. Larger relative-penalty coefficients at higher incomes can therefore arise even when the response to an additional dollar of potential tax is identical. The more similar dollar-penalty estimates support this interpretation, although this illustration does not quantify how much of the estimated gradient reflects scaling.

A separate scaling issue arises because the tax-liability outcome is expressed in logs. Pairing the median income and tax-liability benchmarks, a potential penalty equal to 1% of income represents approximately 7.1% of the initial tax bill. Pairing the corresponding ninetieth-percentile values gives 4.2%. Even if families ultimately paid the same fraction of their potential penalty, the percentage increase in taxes could therefore be larger for families with smaller initial tax bills relative to income. The larger proportional tax increases estimated at moderate incomes remain meaningful, but do not alone establish greater reliance on income splitting or different behavioural responses. Using the dollar penalty as an instrument does not remove this distinction because the tax-liability outcome remains in logs.

D.2 Dividend Responses by Family Income

In the log spouse-dividend first stages, the coefficients for an additional $1,000 are -0.033, -0.047, and -0.051 in Q2–Q4 (Table D2). The income gradient is smaller than under the relative-penalty specification, particularly between Q3 and Q4.

Table D2: Financial-Exposure Estimates for Primary-Owner Dividends by Family-Income Quartile

	Q1	Q2	Q3	Q4
Panel A: Second-stage estimates				
Primary-owner dividend income	-0.367	-0.175	-0.087	-0.370***
	(0.860)	(0.150)	(0.104)	(0.076)
Observations	26,045	52,435	60,555	59,820
Panel B: First-stage estimates for the corresponding sample				
Effect of $1,000 penalty on log spouse dividends	-0.014	-0.033***	-0.047***	-0.051***
	(0.016)	(0.011)	(0.010)	(0.006)
First-stage F-statistic	2.597	15.60	29.38	93.55

Notes: Quartiles are based on family income in 2016. The second stage uses log spouse dividend income, instrumented by the simulated TOSI penalty in thousands of dollars, to estimate the response of log primary-owner dividend income. The first stage reports the effect of the penalty on log spouse dividends. Outcomes are in logs. Standard errors are clustered at the family level and shown in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table D3: Financial-Exposure Dividend-Receipt IV Estimates for Owner Dividends by Income Quartile

	Q1	Q2	Q3	Q4
Primary-owner dividends				
Second stage: effect on log outcome	1.166	-0.054	-0.195	-1.078***
	(5.553)	(0.403)	(0.291)	(0.210)
First stage: effect on spouse dividend receipt	-0.002	-0.015***	-0.014***	-0.012***
	(0.006)	(0.002)	(0.002)	(0.001)
Observations	35,120	66,110	76,855	75,765

Notes: Each row pair comes from a separate IV regression. The instrument is the simulated TOSI penalty in thousands of dollars, and the endogenous variable is an indicator for spouse dividend receipt. The first stage is in probability units per $1,000 penalty; the second-stage outcome is in logs. Quartiles use 2016 family income. Standard errors clustered at the family level appear in parentheses. The Q1 first stage is weak, so its second-stage estimates require caution. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.3 Family Taxable Income and Tax Liability by Family Income

The dividend-receipt IV second-stage coefficients for family taxable income and tax liability remain similar under the dollar and relative instruments (Table D5). Both specifications express tax liability in logs, so this comparison does not remove the role of differences in initial tax bills described above.

Table D4: Financial-Exposure Family Income and Tax Estimates by Quartile

	Q1	Q2	Q3	Q4
Panel A: Second stage				
Family taxable income	0.165	0.091	0.021	0.023**
	(0.144)	(0.064)	(0.027)	(0.011)
Family tax liability	-0.468	-0.292*	-0.192**	-0.019
	(0.523)	(0.165)	(0.083)	(0.023)
Panel B: First stage on log spouse dividends				
Family taxable income sample	-0.034*	-0.039***	-0.049***	-0.055***
	(0.019)	(0.010)	(0.009)	(0.006)
Family tax liability sample	-0.038	-0.038***	-0.050***	-0.055***
	(0.024)	(0.010)	(0.009)	(0.006)

Notes: Quartiles use 2016 family income. Each outcome is estimated in a separate IV regression. Panel B uses the same outcome sample as Panel A and reports the effect of a $1,000 simulated penalty on log spouse dividends. Tax-liability IV estimates may reflect direct tax effects. Standard errors clustered at the family level are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table D5: Financial-Exposure Dividend-Receipt IV Estimates for Family Outcomes by Income Quartile

	Q1	Q2	Q3	Q4
Family taxable income				
Second stage: effect on log outcome	1.213*	0.468***	0.221***	0.139***
	(0.627)	(0.131)	(0.059)	(0.032)
First stage: effect on spouse dividend receipt	-0.012*	-0.016***	-0.018***	-0.014***
	(0.006)	(0.002)	(0.002)	(0.001)
Observations	104,045	132,410	131,055	108,035
Family tax liability				
Second stage: effect on log outcome	-3.700	-0.990***	-0.557***	-0.207***
	(5.501)	(0.333)	(0.152)	(0.060)
First stage: effect on spouse dividend receipt	-0.006	-0.016***	-0.018***	-0.014***
	(0.007)	(0.002)	(0.002)	(0.001)
Observations	86,020	129,740	130,060	107,640

Notes: Each row pair comes from a separate IV regression. The instrument is the simulated TOSI penalty in thousands of dollars, and the endogenous variable is an indicator for spouse dividend receipt. The first stage is in probability units per $1,000 penalty; the second-stage outcome is in logs. Quartiles use 2016 family income. Standard errors clustered at the family level appear in parentheses. The Q1 first stage is weak, so its second-stage estimates require caution. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.